

CS4269

Wang Xiyu A0282500R

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1 Syntax of Propositional Logics

Definition 1.1. *DSyntax decides what are allowed to be written*

1.1 Well-formed Formula (WFF)

A formula ψ is obtained by composing $p \in P$ with $c \in C$, where P is the set of propositions and C is the set of connectors. here we use any set of complete logic, like $C = \{\neg, \vee\}$. This is not a precise definition as counter examples that do not form a valid formula can be formulated based on this, for example $\neg \vee p$ means nothing. To formalize the definition we may attempt to define a language with the following grammar:

$$\Sigma = P \cup C \cup \{(\,,\,)\}$$

$$FORM : \{\psi = p \in P \mid \neg\psi \mid \psi \vee \psi \mid (\psi)\}$$

However this is a recursive definition as the definition of the a formula ψ involves itself. We need to find a declarative way to define this recursive/inductive construction.

Definition 1.2. *FORM is the smallest set of strings over Σ where*

1. $P \subseteq FORM$
2. $\forall \psi \in FORM, \neg\psi \in FORM$
3. $\forall \psi_1, \psi_2 \in FORM, \psi_1 \vee \psi_2 \in FORM$

We can see as a construction rule, we are admitting strings into the set of formula, but we need to somewhat impose restriction on what is not acceptable.

1.2 Closed Set

Let's remove the attributive **the smallest** first and define another set that contains all the set that satisfy these rules listed above:

Definition 1.3. *Closed is a set of strings over Σ where*

1. $P \subseteq Closed$
2. $\forall \psi \in Closed, \neg\psi \in Closed$
3. $\forall \psi_1, \psi_2 \in Closed, \psi_1 \vee \psi_2 \in Closed$

1.2.1 All Closed Set

And we define a set ACS (All-closed sets). We can define FORM via the following constriction:

$$FORM = \bigcap ACS$$

Suppose $BCS = \emptyset$ and from definition

$$U_0 = \bigcap BCS = \{w \in \Sigma^* \mid \forall S \in BCS, w \in S\}$$

However, $BCS = \emptyset$, The statement $\forall S \in BCS, w \in S$ is vacuously true $\forall w \in \Sigma^*$, which means that

$$U_0 = \Sigma^* = \bigcap BCS$$

A contradiction. Therefore we have proved that ACS cannot be empty, thus $\forall P, \exists S$ such that S is a closed set.

1.3 FORM is the minimum of the set of Closed

First let's prove that FORM is closed. It seems to be trivial that the intersection of closed sets is closed but formal proof is still needed. We denote a set from the intersection construction as T_0 .

Lemma 1.4. T_0 is closed

We are proving that the set T_0 we get from the intersection construction still satisfy the rules that define what is closed.

Proof. From rule 1,

$$\begin{aligned} \forall S \in ACS, P \subseteq S \\ \rightarrow \forall p \in P, \forall S \in ACS, p \in S \\ \rightarrow \forall p \in P, p \in \bigcap ACS \\ \rightarrow P \in T_0 \end{aligned}$$

From rule 2,

$$\begin{aligned} \forall S \in ACS, T_0 \subseteq S \\ \text{let } w \in T_0, \forall S \in ACS, w \in S, \neg w \in S \\ \rightarrow \forall w \in T_0, \neg w \in \bigcap ACS \\ \rightarrow \forall w \in T_0, \neg w \in T_0 \end{aligned}$$

From rule 3,

$$\begin{aligned} \forall S \in ACS, T_0 \subseteq S \\ \text{let } w_1, w_2 \in T_0, \forall S \in ACS, w_1 \vee w_2 \in S \\ \rightarrow \forall w_1, w_2 \in T_0, w_1 \vee w_2 \in \bigcap ACS \\ \rightarrow \forall w_1, w_2 \in T_0, w_1 \vee w_2 \in T_0 \end{aligned}$$

□

Lemma 1.5. T_0 is a smallest Closed set

Proof. The proof of minimality is trivial. $T_0 = \bigcap ACS \rightarrow \forall S \in ACS, T_0 \subseteq S$

□

Lemma 1.6. T_0 is **the** smallest Closed set (unique). In other word,

$$\forall T'_0 \in \{T \mid \forall S \in ACS, T \subseteq S\}, T'_0 = T_0$$

Now we can use the generative rule to define formula, which is called **Backus-Naur form**. Note that a context-free grammar is a type of formal language. Backus Naur form is a specification language for this type of grammar. It is used to describe language syntax.

1.4 Subformula

Definition 1.7. ϕ_1 is a subformula of ϕ_2 , denoted by

$$\phi_1 \preceq \phi_2$$

as a relation is

- reflexive: $\forall \phi \in FORM, \phi \preceq \phi$
- transitive: $\forall \phi_1, \phi_2, \phi_3 \in FORM, (\phi_1 \preceq \phi_2) \wedge (\phi_2 \preceq \phi_3) \rightarrow \phi_1 \preceq \phi_3$

Inductive definition:

$$\begin{aligned} \forall \phi \in FORM, \phi \preceq (\neg \phi) \\ \forall \phi_1, \phi_2 \in FORM, \phi_1 \preceq (\phi_1 \vee \phi_2) \\ \phi_2 \preceq (\phi_1 \vee \phi_2) \end{aligned}$$

We might intuitively think that subformula can be defined using the notion of substring, since we have been considering formula as concatenation of proposition and connector symbols. However based on our inductive definition, we can come up with the following counter example to a definition using substring: A formula ϕ_1 is a subformula of ϕ_2 if and only if ϕ_1 is a well-formed formula, and is a substring to ϕ_2

$$p \vee q \not\preceq \neg p \vee q$$

What if we tighten the syntax by strictly enforce the use of parenthese whenever we can

2 Semantics of Propositional Logics

Definition 2.1. *Semantics describe meanings.*

2.1 Truth Assignment/Proposition evaluation

A truth assignment or a valuation of propositions is a function that maps from the set of propositions to true and false.

$$\begin{aligned} v : P \rightarrow \{T, F\} \\ v \in [P \rightarrow \{T, F\}] \end{aligned}$$

Example 2.2. Consider $v = \{p_1 \mapsto T, p_2 \mapsto F\}$, we can derive

$$\begin{aligned} v(p_1) &= T \\ v(p_2) &= F \\ v(\neg p_2) &= T \\ v(p_1 \vee p_2) &= T \text{ etc.} \end{aligned}$$

2.2 Entail/Model

We denote entailment of an evaluation function to formula as

$$v \models \psi$$

We also inductively define the following rule

$$\begin{aligned} \forall p \in P, v \models p &\iff v(p) \\ v \models \psi &\iff v \not\models \neg \psi \\ v \models \psi_1 \vee \psi_2 &\iff (v \models \psi_1) \vee (v \models \psi_2) \end{aligned}$$

2.2.1 Valuation and Model

A valuation v is a model of a formula ϕ , if and only if $v(\phi) = T$.

- $\forall \phi$ over $P, \exists 2^{|P|}$ valuations.
- A formula can have many or one or none models

2.2.2 $\models$ as a relation

Definition 2.3. $\models \subseteq \{2^{|P|} \times FORM\}$ is a binary relation, between truth assignment to formula. We define as follows:

- $\forall p \in P, v \models p \iff v(p) = 1$

- $v \not\models p$ otherwise.

Consider a valuation v as a subset of P such that the propositions in v denoted by $v(p)$ is assigned with true. $v \in \mathcal{P}(P)$

ϕ	ψ	$\neg\phi$	$(\phi \wedge \psi)$	$(\phi \vee \psi)$	$(\phi \rightarrow \psi)$	$(\phi \leftrightarrow \psi)$
0	0	1	0	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	0	0
1	1	0	1	1	1	1

Definition 2.4.

$$p(v) = v(p)$$

A proposition can be viewed as a circuit is just a wire, thus if a proposition is assigned with true by a valuation, the valuation values this proposition to be true

Lemma 2.5. Let $\phi \in FORM, v \in \mathcal{P}(P), v \models \phi \iff \phi(v) = 1$

Proof. Base case: $\forall \phi = p \in P, v \models \phi \iff v(p) = 1 \iff p(v) = 1$
Inductive case:

1.

□

2.2.3 Logical Equivalence

Definition 2.6. Syntax definition of logical equivalence is as follows:

$$\phi_1 \equiv \phi_2 \iff \forall v : P \mapsto \{T, F\} (v \models \phi_1), v \models \phi_2$$

Example 2.7.

$$p \vee (\neg p) \equiv q \vee (\neg q)$$

$$\begin{aligned} \forall v \models (p \vee (\neg p)), \\ v \models p \text{ or } v \models (\neg p) \\ v \models p \text{ or } v \not\models p \\ \forall v : P \mapsto \{T, F\}, \end{aligned}$$

TODO

2.3 Satisfiability of Formula

Definition 2.8. Let $\phi \in FORM, \phi$ is satisfiable

$$\iff \exists v : P \mapsto \{T, F\}, v \models \phi$$

2.3.1 Valid formula (tautology)

Definition 2.9. Tautology: ϕ is a tautology $\iff \forall v, v \models \phi$

Example 2.10.

$$\phi = \neg(p \vee (\neg p))$$

From example 2.4, we have

$$\begin{aligned} \forall v, v \models (p \vee (\neg p)) \\ \rightarrow \forall v, v \not\models \neg(p \vee (\neg p)) \end{aligned}$$

Thus ϕ is NOT satisfiable.

2.4 Satisfiability of a set of formula

Set Satisfaction

Definition 2.11. Let $\Gamma \subseteq FORM$,

$$v \models \Gamma \iff \forall \phi \in \Gamma, v \models \phi$$

Set Satisfiability

Definition 2.12. Let $\Gamma \subseteq FORM$, Γ is satisfiable

$$\iff \exists v, \forall \phi \in \Gamma, v \models \phi$$

Example 2.13. $\Gamma = \emptyset$, Γ is satisfiable.

Proof. Consider $v : P \mapsto \{T, F\}$, $v(p) = T \forall p \in P$, $\forall \phi \in \Gamma, v \models \phi$ is vacuously true. □

Example 2.14. $\Gamma = FORM$, Γ is NOT satisfiable.

Proof. Suppose otherwise,

$$\begin{aligned} \exists v, \forall \phi \in \Gamma, v \models \phi \\ \forall \phi \in \Gamma, (\neg \phi) \in \Gamma \\ v \models (\neg \phi) \end{aligned}$$

A contradiction. □

2.4.1 Relevance

Definition 2.15. Define set $OCCUR(\phi) \subseteq P$, which contains propositions appear in ϕ . Inductively defined as

$$\begin{aligned} OCCUR(\phi) &= \{p\} \iff \phi \equiv p, p \in P \\ OCCUR(\neg \phi) &= OCCUR(\phi) \\ OCCUR(\phi_1 \vee \phi_2) &= OCCUR(\phi_1) \cup OCCUR(\phi_2) \end{aligned}$$

Lemma 2.16. Let $\phi \in FORM$, let v_1, v_2 be valuations over P .

$$\forall p \in OCCUR(\phi), v_1(p) = v_2(p) \rightarrow (v_1 \models \phi \rightarrow v_2 \models \phi)$$

v_1, v_2 can be different valuations over $p' \notin OCCUR(\phi)$

Proof. Let $p \in P$, □

2.4.2 Logical consequence

Semantic logical consequence:

Definition 2.17.

$$\Gamma \models \phi \iff \forall v \models \Gamma, v \models \phi$$

Syntactic logical consequence: (logical derivability)

Definition 2.18. A formula ϕ is a syntactic consequence of Γ if there exists a formal proof (a finite sequence of formulas) ending in ϕ , where each step is either a premise from Γ , an axiom, or follows from previous steps via an inference rule (like Modus Ponens).

Material Implication ($\rightarrow$) vs. Logical Consequence ($\models$)

- Material implication is a connector within a formula, $p \rightarrow q \equiv \neg(p \vee q)$, and can therefore be eval to true or false in a single model.
- Logical consequence is a relation between formulas (meta-logic), it is a statement about all models. $\{p, p \rightarrow q\} \models q$

2.5 Modelling Computations as Propositional Logic

Computation problems can be considered as membership check of an encoded problem instance to a language.

$$L = \{p \mid \langle M, p \rangle\}$$

Here instead of arbitrary encoding, we consider problem encoding with propositional logics only, and solve with SAT solvers by deciding whether the set of formula we form from any instance of the problem is satisfiable. That is

$$Encode : p \mapsto FORM, \forall p, \exists v \models Encode(p) \iff p \in L$$

The rules of such modelling include:

- Propositional logics only
- The final expression must one well formed fomula
- for problem of size n , the size of the formula ϕ must be in $O(poly(n))$

Note that there can be different definition on the size of the

- problem: in unary, binary etc;
- Formula: Whether by the number of nodes in the syntax tree or length of the string **Anything else?**

Example 2.19. Graph coloring problem: given a undirected graph $G = (V, E)$, whether there exists a k coloring of all vertices such that no neighbouring vertices have the same color. Consider the following construction of a proposition set P :

$$P = \{p_{v,i} \mid v \in V, i \in [1, k]\}$$

The valuation of proposition $p_{v,i}$ conceptually represents vertex v is assigned color i .

We can formulate constraints as well formed formula, and use conjunction to connect the formula to model satisfaction of multiple constraints.

Corollary 2.20. $\forall \phi_1, \phi_2 \in FORM, (\phi_1 \wedge \phi_2) \in FORM$

First let us list down the constraints of this problems, including implicit ones in plain words

1. Neighbouring vertices have different colors.
2. All vertices as assigned with exactly one color.

1. ϕ_{nbr}

First lets formula the constraint between any 2 vertices,

$$\forall (\alpha, \beta) \in E, c(v_\alpha) \neq c(v_\beta)$$

And from our proposition set, $c(v_\gamma) = i \iff \exists v, v \models p_{\gamma,i}$. As such we have

$$\phi_{\alpha,\beta,i} = \neg(p_{\alpha,i} \wedge p_{\beta,i})$$

This models vertex α and vertex β are not colored with color i at the same time. For every color this needs to hold, thus we have

$$\phi_{nbr(\alpha,\beta)} = \bigwedge_{i \in [1,k]} \phi_{\alpha,\beta,i}$$

For every pair of neighbouring vertices this also needs to hold. Therefore we have

$$\phi_{nbr} = \bigwedge_{(\alpha,\beta) \in E} \phi_{nbr(\alpha,\beta)}$$

2. $\phi_{\text{exactly-1}}$

To model exactly 1 coloring on each vertex, we can formulate $\phi_{\geq 1}$ and $\phi_{\leq 1}$ first and use conjunction to connect them. Given our propositions, to make sure vertex v is at least assign a color ($\phi_{v,\geq 1}$),

$$\phi_{v,\geq 1} = \bigvee_{i \in [1,k]} p_{v,i}$$

This needs to hold for every vertex, thus we have

$$\phi_{\geq 1} = \bigwedge_{v \in V} \phi_{v,\geq 1}$$

Then we need to model at most 1, (or less than 2) coloring on each vertex. Vertex v is not assignment with both color i and j can be formulated as

$$\phi_{v,i,j,\leq 1} = \neg(p_{v,i} \wedge p_{v,j})$$

This needs to hold for every distinct pair of colors, thus

$$\phi_{v,\leq 1} = \bigwedge_{i \neq j \in [1,k]} \phi_{v,i,j,\leq 1}$$

For all vertices

$$\phi_{\leq 1} = \bigwedge_{v \in V} \phi_{v,\leq 1}$$

Finally we have

$$\phi_{\text{exactly-1}} = \phi_{\leq 1} \wedge \phi_{\geq 1}$$

And the overall formula for an instance of $G = (V, E)$ becomes

$$\phi = \phi_{\text{nbr}} \wedge \phi_{\text{exactly-1}}$$

Similar to reduction, we need also now to show that

$$\exists v : P \mapsto \{T, F\}, v \models \phi \iff \langle G = (V, E), k \rangle \in L_{\text{color}}$$

TODO

Introduce propositional variables that encode atomic choices, then add constraints that enforce the combinatorial structure of the problem.

1. Hamiltonian Path

Instance. Graph $G = (V, E)$ with $|V| = n$.

Variables. For every vertex $v \in V$ and position $i \in \{1, \dots, n\}$:

$$X_{v,i} \quad \text{“vertex } v \text{ is at position } i \text{ in the path.”}$$

Constraints.

1. **Each position has exactly one vertex:**

$$\bigvee_{v \in V} X_{v,i} \quad \text{and pairwise exclusion.}$$

2. **Each vertex appears exactly once:**

$$\bigvee_{i=1}^n X_{v,i} \quad \text{and pairwise exclusion.}$$

3. **Adjacency constraint:** If $(u, v) \notin E$, then for all i :

$$\neg X_{u,i} \vee \neg X_{v,i+1}.$$

Satisfiable iff G has a Hamiltonian path.

2. Clique of Size k

Instance. Graph $G = (V, E)$.

Variables. For each vertex $v \in V$:

X_v “vertex v is selected.”

Constraints.

1. **At least k vertices selected:** Encode using a cardinality constraint.
2. **Selected vertices form a clique:** If $(u, v) \notin E$, then

$$\neg X_u \vee \neg X_v.$$

Satisfiable iff there exists a clique of size $\geq k$.

3. 3-Dimensional Matching

Instance. Sets X, Y, Z and triples $T \subseteq X \times Y \times Z$.

Variables. For each triple $t \in T$:

X_t “triple t is chosen.”

Constraints.

1. **No element used twice:**

If triples t, t' share an element, then

$$\neg X_t \vee \neg X_{t'}.$$

2. **Select exactly k triples:** Encode cardinality constraint.

Satisfiable iff there is a matching of size k .

4. Subset Sum

Instance. Numbers $a_1, \dots, a_n$, target B .

Variables.

X_i “ a_i is selected.”

Constraints. We encode:

$$\sum_{i=1}^n a_i X_i = B.$$

This is reduced to propositional logic by building a binary addition circuit:

- Introduce variables for each bit of partial sums.
- Encode full-adder constraints using propositional clauses.

Satisfiable iff there exists a subset summing to B .

5. Exact Cover

Instance. Universe U , collection $\mathcal{S} = \{S_1, \dots, S_m\}$.

Variables.

X_i “set S_i is selected.”

Constraints. For every element $u \in U$:

1. **At least one covering set:**

$$\bigvee_{u \in S_i} X_i$$

2. **At most one covering set:** If $u \in S_i \cap S_j$ and $i \neq j$:

$$\neg X_i \vee \neg X_j.$$

Satisfiable iff an exact cover exists.

6. Scheduling with Precedence Constraints

Instance. Tasks $T_1, \dots, T_n$, time slots $1, \dots, m$.

Variables.

$$X_{i,t} \quad \text{“task } i \text{ scheduled at time } t\text{.”}$$

Constraints.

1. Each task scheduled exactly once.
2. At most one task per time slot.
3. If T_i must precede T_j :

$$X_{i,t} \rightarrow \bigvee_{t' > t} X_{j,t'}.$$

2.6 Encoding a Full Adder in Pure Propositional Logic

A full adder takes three input bits:

$$x, y, c_{in}$$

and produces two outputs:

$$s \quad (\text{sum bit}), \quad c_{out} \quad (\text{carry bit})$$

Semantically:

$$s = x \oplus y \oplus c_{in}$$

$$c_{out} = (x \wedge y) \vee (x \wedge c_{in}) \vee (y \wedge c_{in})$$

We now encode this purely in propositional logic using only $\wedge, \vee, \neg$ (or $\rightarrow, \perp$ if preferred).

1. Encoding the Sum Bit

The sum is true iff an odd number of inputs are true.

Truth table characterization:

$$s \leftrightarrow (x \wedge \neg y \wedge \neg c_{in}) \vee (\neg x \wedge y \wedge \neg c_{in}) \vee (\neg x \wedge \neg y \wedge c_{in}) \vee (x \wedge y \wedge c_{in})$$

Thus we encode:

$$(s \rightarrow \Phi_s) \wedge (\Phi_s \rightarrow s)$$

where Φ_s is the disjunction above.

2. Encoding the Carry Bit

Carry is true iff at least two inputs are true:

$$c_{out} \leftrightarrow (x \wedge y) \vee (x \wedge c_{in}) \vee (y \wedge c_{in})$$

Again encode as:

$$(c_{out} \rightarrow \Phi_c) \wedge (\Phi_c \rightarrow c_{out})$$

3. CNF Encoding (SAT-friendly)

To encode for SAT, we introduce clauses.

Carry Constraints

$$\neg c_{out} \vee x \vee y$$

$$\neg c_{out} \vee x \vee c_{in}$$

$$\neg c_{out} \vee y \vee c_{in}$$

$$\neg x \vee \neg y \vee c_{out}$$

$$\neg x \vee \neg c_{in} \vee c_{out}$$

$$\neg y \vee \neg c_{in} \vee c_{out}$$

Sum Constraints (XOR Form) We can encode $s = x \oplus y \oplus c_{in}$ using clauses:

$$(x \vee y \vee c_{in} \vee \neg s)$$

$$(\neg x \vee \neg y \vee c_{in} \vee \neg s)$$

$$(\neg x \vee y \vee \neg c_{in} \vee \neg s)$$

$$(x \vee \neg y \vee \neg c_{in} \vee \neg s)$$

$$(\neg x \vee \neg y \vee \neg c_{in} \vee s)$$

$$(x \vee y \vee \neg c_{in} \vee s)$$

$$(x \vee \neg y \vee c_{in} \vee s)$$

$$(\neg x \vee y \vee c_{in} \vee s)$$

These clauses exactly encode the XOR behavior.

4. Circuit-Style Encoding (Cleaner)

Instead of expanding XOR directly, introduce auxiliary variables:

$$t = x \oplus y$$

$$s = t \oplus c_{in}$$

Each XOR can be encoded with 4 clauses:

$$(t \leftrightarrow x \oplus y)$$

which in CNF becomes:

$$(\neg x \vee \neg y \vee \neg t) \wedge (x \vee y \vee \neg t) \wedge (x \vee \neg y \vee t) \wedge (\neg x \vee y \vee t)$$

This keeps formula size linear.

5. Why This Matters

Full adders are the atomic building blocks of:

- Binary addition circuits
- Subset Sum encodings
- Integer programming to SAT reductions
- Cook–Levin tableau arithmetic

Any arithmetic constraint can be reduced to propositional logic by composing full adders.

Conceptual Insight

Arithmetic $\longrightarrow$ Circuits $\longrightarrow$ Propositional Logic

This translation is what makes SAT complete for NP: even numeric computation can be flattened into Boolean structure.

General Pattern

Almost all NP problems can be modelled by:

1. Encoding choices with propositional variables.
2. Enforcing structural constraints with local clauses.
3. Encoding counting via cardinality circuits if needed.

This modelling perspective is precisely what makes SAT solvers universal engines for combinatorial search.

3 Compactness Theorem for Propositional Logics

Definition 3.1. *Axiom of Choice: formally*

$$\forall X, \emptyset \in X \vee \exists f, \text{dom}(f) = X \wedge \forall A \in X, f(A) \in A$$

A choice function (or selection) f defined on a collection of nonempty sets X , such that for every set A in the collection X , $f(A)$ is an element in A . That is, Axiom—For any set X of nonempty sets, there exists a choice function f that is defined on X and maps each set of X to an element of that set.

Theorem 3.2. *If P is finite, $FORM$ over P is finite.*

Theorem 3.3. *If P is countable, $FORM$ over P is countable.*

Proof. Given $\forall \phi \in FORM$, ϕ is finitely long, this theorem is another way of stating: The language formed from countable alphabet $P \cup C$ with the inductive construction rules, is countable. We can prove a stronger theorem first. $\square$

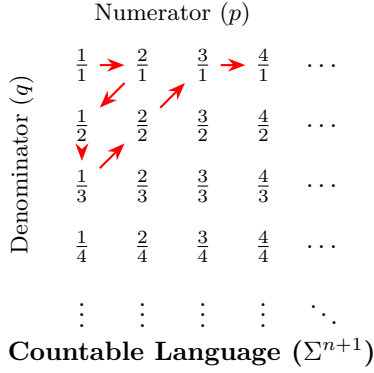
Theorem 3.4. *All languages formed from a countable alphabet is countable.*

Proof. Consider $L = \Sigma^*$, we will prove that Σ is countable $\rightarrow L$ is countable. We use the same diagonal proof for the countability of $\mathbb{Q}$ to show an enumeration of all strings formable from a countable alphabet. To formalize, we use induction to prove that $\forall i \geq 0, \Sigma^i$ is countable

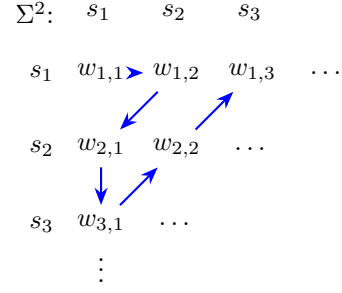
Base case: $\Sigma^1 = \Sigma$, countable as defined.

Inductive hypothesis: Σ^n is countable: we enumerate all strings in Σ^n , and use the same diagonal proof to show the concatenation of one symbol from the countable alphabet to every string Σ^{n+1} is also countable. Thus we can give an enumeration of all Σ^i and use the diagonal argument again to show the existence of an

Rational Numbers ($\mathbb{Q}^+$)



Countable Language (Σ^2)



Countable Language (Σ^*)

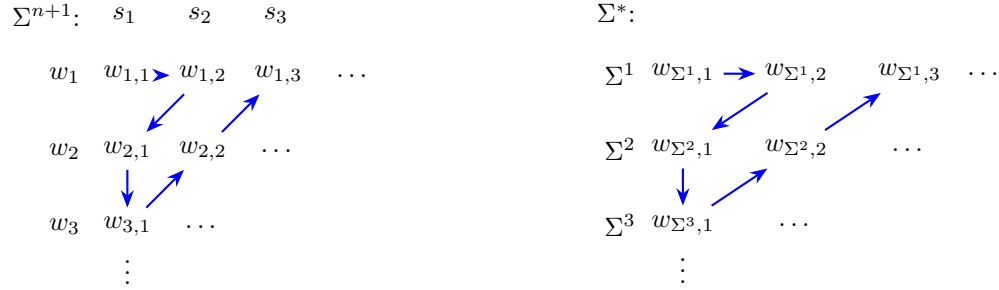


Figure 1: Comparison of diagonal enumeration for numbers vs. strings.

enumeration of Σ^* .

Alternatively, Given

$$\Sigma^* = \bigcup_{i \geq 0} \Sigma^i$$

From the Axiom of Choice, which implies the union of countably many countable set is countable, Σ^* is countable. Since $\forall L$ over $\Sigma, L \subseteq \Sigma^*, \Sigma^*$ is countable, all languages over countable Σ is countable. $\square$

Definition 3.5 (Finitely Satisfiable). *A set of formulas Γ is said to be **finitely satisfiable** if and only if for all finite subsets $\Gamma_0 \subseteq \Gamma$, Γ_0 is satisfiable.*

Theorem 3.6 (Compactness Theorem).

$$\Gamma \text{ is satisfiable} \iff \Gamma \text{ is finitely satisfiable}$$

Proof. Consider a special case where Γ is “ T -good”, defined as:

$$\forall p \in P, (p \in \Gamma \vee \neg p \in \Gamma)$$

Claim: If Γ is T -good, then Γ is finitely satisfiable $\implies \Gamma$ is satisfiable.

Suppose Γ is T -good and finitely satisfiable. We define a candidate valuation v_L such that for all $p \in P$:

$$v_L(p) = \begin{cases} T & \text{if } p \in \Gamma \\ F & \text{if } \neg p \in \Gamma \end{cases}$$

First, we prove that for every $p \in P$, **exactly one** of $\{p, \neg p\}$ is in Γ . By the T -good property, at least one is in Γ . Suppose both $\{p, \neg p\} \subseteq \Gamma$. Consider the finite subset:

$$\Gamma' = \{p, \neg p\}$$

Since Γ' is a finite subset of a finitely satisfiable set, it must be satisfiable. However, $\neg \exists v$ such that $v \models \{p, \neg p\}$, which is a contradiction. Thus, v_L is well-defined.

Now we show $v_L \models \Gamma$. Suppose $\exists \phi \in \Gamma$ such that $v_L \not\models \phi$. Define the following finite sets based on the atoms occurring in ϕ :

$$\begin{aligned}\Gamma_\phi^+ &= \{p \mid p \in \text{OCCUR}(\phi) \wedge v_L(p) = T\} \\ \Gamma_\phi^- &= \{\neg p \mid p \in \text{OCCUR}(\phi) \wedge v_L(p) = F\} \\ \Gamma_\phi &= \{\phi\} \cup \Gamma_\phi^+ \cup \Gamma_\phi^-\end{aligned}$$

Since Γ_ϕ is a finite subset of Γ , it must be satisfiable by some valuation v . For this v :

- $\forall p$ such that $p \in \Gamma_\phi^+$, $v(p) = T$.
- $\forall \neg p$ such that $\neg p \in \Gamma_\phi^-$, $v(\neg p) = T \implies v(p) = F$.

By construction, v and v_L agree on all atoms in $\text{OCCUR}(\phi)$. By the **Relevance Lemma**, $v(\phi) = v_L(\phi)$. Since $v \models \Gamma_\phi$, we have $v \models \phi$. This implies $v_L \models \phi$, contradicting our assumption $v_L \not\models \phi$. Therefore, $\forall \phi \in \Gamma, v_L \models \phi$, which implies $v_L \models \Gamma$, Γ is satisfiable.

Now we will generalize this to all Γ which may not be ‘T-good’, by showing that we can construct a ‘T-good’ finitely satisfiable set Δ , from any finitely satisfiable Γ , and therefore show that $\Gamma SAT \iff \Delta SAT$.

We will only prove the case for countable P now. Let an enumeration of $p \in P$ be $\{p_1, p_2, \dots\}$

Define finite subset Γ_i of Γ as such:

$$\begin{aligned}\Gamma_0 &= \Gamma \\ \Gamma_{i+1} &= \begin{cases} \Gamma_i \cup \{p_i\} & \text{if } \Gamma_i \cup \{p_i\} \text{ is finitely satisfiable} \\ \Gamma_i \cup \{\neg(p_i)\} & \text{Otherwise} \end{cases}\end{aligned}$$

Here we need to prove that $\forall i$, between $\Gamma_i \cup \{p_i\}$ and $\Gamma_i \cup \{\neg(p_i)\}$ at least one case is also finitely satisfiable.

Theorem 3.7. *Let Γ be finitely satisfiable set, and ϕ be a formula, then $\Gamma \cup \{\phi\}$ is finitely satisfiable or $\Gamma \cup \{\neg(\phi)\}$ is finitely satisfiable.*

Proof. We prove the contrapositive of this statement: if $\Gamma \cup \{\phi\}$ and $\Gamma \cup \{\neg(\phi)\}$ are both not finitely satisfiable, Γ is not finitely satisfiable. Given both are not finitely satisfiable, there exist finite subsets $\Gamma_1, \Gamma_2 \subseteq \Gamma$ such that $\Gamma_1 \cup \{\phi\}$ and $\Gamma_2 \cup \{\neg(\phi)\}$ are not satisfiable. Therefore for all valuations, it must not model either Γ_1 or $\{\phi\}$, and not model either Γ_2 or $\{\neg(\phi)\}$. Denote the set of valuations that models formula ψ as $\text{models}(\psi)$, the set of valuations that models set of formula Σ as $\text{models}(\Sigma)$

$$\begin{aligned}\text{models}(\Gamma_1) \cap \text{models}(\phi) &= \emptyset \\ &\rightarrow \forall v \in \text{models}(\Gamma_1), v \not\models \phi \rightarrow v \models (\neg\phi) \\ &\rightarrow \text{models}(\Gamma_1) \subseteq \text{models}((\neg\phi)) \\ \text{models}(\Gamma_2) \cap \text{models}((\neg\phi)) &= \emptyset \\ &\rightarrow \forall v \in \text{models}(\Gamma_2), v \not\models (\neg\phi) \rightarrow v \models (\phi) \\ &\rightarrow \text{models}(\Gamma_2) \subseteq \text{models}(\phi)\end{aligned}$$

Now consider the set of valuations that models $\Gamma_1 \cup \Gamma_2$

$$\begin{aligned}\text{models}(\Gamma_1 \cup \Gamma_2) &= \text{models}(\Gamma_1) \cap \text{models}(\Gamma_2) \\ (\text{models}(\Gamma_1) \cap \text{models}(\Gamma_2)) &\subseteq (\text{models}(\phi) \cap \text{models}((\neg\phi))) = \emptyset\end{aligned}$$

Since $\Gamma_1 \cup \Gamma_2$ is a finite subset of Γ that is not satisfiable, Γ is not finitely satisfiable. □

Now prove that

Lemma 3.8. *$\forall i, \Gamma_i$ is finitely satisfiable.*

Proof. Base case:

$$\Gamma_0 = \Gamma \text{ is finitely satisfiable by definition}$$

Inductive hypothesis: Suppose

$$\Gamma_n \text{ is finitely satisfiable}$$

By Lemma 3.7, $\Gamma_n \cup \{p_n\}$ and $\Gamma_n \cup \{\neg p_n\}$ there must be at least one that is finitely satisfiable. $\rightarrow \Gamma_{n+1}$ is therefore finitely satisfiable. $\square$

Next we will show that

Lemma 3.9. $\Gamma_U = \bigcup_{i \geq 0} \Gamma_i$ is finitely satisfiable.

Proof. Consider finite set $\Gamma' \subseteq \Gamma_U$, since Γ' is finite, and $\forall k \geq 0, \Gamma_k \subseteq \Gamma_{k+1}$ by definition, $\exists j \geq 0, \Gamma' \subseteq \Gamma_j = \bigcup_{i=0}^j \Gamma_i$. By Lemma 3.8, Γ_j is satisfiable $\rightarrow \Gamma'$ is finitely satisfiable. $\square$

Lemma 3.10. Γ_U is satisfiable.

Proof. From the construction, $\forall i \geq 0$ at least one of p_i and $(\neg p_i)$ is in $\bigcup_{i \geq 0} \Gamma_i$. Since $\bigcup_{i \geq 0} \Gamma_i$ is finitely satisfiable, $\forall i \geq 0, p_i$ and $(\neg p_i)$ cannot both be in $\bigcup_{i \geq 0} \Gamma_i$. Otherwise $\{p_i\} \cup \{(\neg p_1)\} \subseteq \bigcup_{i \geq 0} \Gamma_i$ is not satisfiable, contradiction. $\square$

Now we have proven that the construction proposed from a finitely satisfiable formula set Γ to $\bigcup_{i \geq 0} \Gamma_i$ is also finitely satisfiable. Now to complete the proof of the compactness theorem, we need to prove that $\exists v, v \models \Gamma$.

Proof. From Lemma 3.10, $\forall i \geq 0, (p_i \in \Gamma_U) \oplus ((\neg p_i) \in \Gamma_U), \exists v, v \models \Gamma_U$

$$v(p) : P \mapsto \{T, F\} = \begin{cases} T, & \text{if } p \in \Gamma_U \\ F, & \text{if } (\neg p) \in \Gamma_U \end{cases}$$

Define the literal set $Literals = \{p_i, (\neg p_i) \mid i \geq 0\}$, and function

$$\tau(p) : P \mapsto Literals = \begin{cases} p, & \text{if } p \in \Gamma_U \\ (\neg p), & \text{if } (\neg p) \in \Gamma_U \end{cases}$$

Lemma 3.11.

$$v \models \Gamma_U \rightarrow v \models \tau(\Gamma)$$

Let $\phi \in \Gamma$, Define literal set $X = \{\tau(p) \mid p \in OCCUR(\phi)\}$ where X is a set containing single proposition literals. By definition we have

$$X \subseteq \Gamma_U \rightarrow v \models X \rightarrow \forall \psi \in X, v(\psi) = T$$

and $OCCUR(X) = OCCUR(\phi)$ Since both X is finite, $X \cup \{\phi\}$ is a finite subset to Γ_U , thus satisfiable. $\rightarrow \exists v', v' \models X \cup \{\phi\} \rightarrow v' \models X, v' \models \phi \rightarrow \forall \psi \in X, v'(\psi) = T$.

$$\forall p \in OCCUR(\phi), v(p) = v'(p), v' \models \phi \rightarrow v \models \phi$$

Therefore, the valuation that models enlarged construction Γ_U also models Γ , thus Γ is satisfiable. $\square$

$\square$

3.1 Application of the Compactness theorem for proposition logics

We have already shown that we can use propositional logic to model computational problems, the compactness theorem allows us to extend problem modelling to instances of infinite size. Recall the k-colorable problem, where we used propositional logic to model any finite size instance.

Theorem 3.12. A graph is k-colorable iff every finite subgraph is k-colorable.

Definition 3.13 (Infinite graph). $G = (V, E)$ is an infinite graph if V is infinite.

Definition 3.14 (Subgraph). $G' = (V', E')$ is a subgraph to G if $V' \subseteq V, E' \subseteq E$

Definition 3.15 (Induced subgraph). $G' = (V', E')$ is an induced subgraph to G if G' is a subgraph to G , and $\{u, v\} \in E' \iff \{u, v\} \in E \wedge u, v \in V'$

Proof. Since we have formulated any finite instance of k -colorable graph as propositional logic formula, we can define infinite set

$$\Gamma_G = \{\phi_{G'}^{k\text{-colorable}} \mid G' \text{ is finite subgraph of } G\}$$

We will first prove that

Lemma 3.16. G is finitely k -colorable $\rightarrow \Gamma_G$ is finitely satisfiable.

Proof. We will reuse the proposition definition from last chapter, where $|V|$ and thus P is countable

$$P = \{p_{v,i} \mid v \in V, i \in [1, k]\}$$

For each instance on $G = (V, E)$ and k colors, we formulate as such to enforce constraints on edges (No same color on adjacent vertices), and vertices (exactly 1 coloring on each vertex).

$$\phi = \phi^{nbr} \wedge \phi^{exactly-1\text{-color}}$$

Consider the infinite graph $G = (V, E)$

$$\begin{aligned} \Gamma_G &= \Gamma_G^{nbr} \cup \Gamma_G^{exactly-1\text{-color}} \\ \Gamma_G^{exactly-1\text{-color}} &= \{\phi_v^{exactly-1\text{-color}} \mid v \in V\} \\ \phi_v^{exactly-1\text{-color}} &= (\bigvee_{i \in [1, k]} p_{v,i}) \wedge (\bigwedge_{i \neq j \in [1, k]} (p_{v,i} \rightarrow p_{v,j})) \\ \Gamma_G^{nbr} &= \{\phi_{\{u,v\}}^{nbr} \mid \{u, v\} \in E\} \\ \phi_{\{u,v\}}^{nbr} &= \bigwedge_{i \in [1, k]} (p_{u,i} \rightarrow p_{v,i}) \end{aligned}$$

Suppose the contrary, where $G = (V, E)$ is finitely k -colorable, Γ_G is not finitely satisfiable, $\exists$ finite subset $\Gamma_0 \subseteq \Gamma_G$, Γ_0 is not satisfiable. Let $G_0 = (V_0, E_0)$ be a finite subgraph of G , such that

$$\begin{aligned} V_0 &= \{v \mid v \in V, \phi_v^{exactly-1\text{-color}} \in \Gamma_0\} \cup \{u \mid u \in V, \exists u', \phi_{\{u,u'\}}^{nbr} \in \Gamma_0\} \\ E_0 &= \{\{u, v\} \mid \{u, v\} \in E, u, v \in V_0\} \end{aligned}$$

Here we are formulating an instance of finite k -colorable induced subgraph of G from finite subset Γ_0 . Now we need to show that G_0 is not k -colorable to reach the contradiction, with another contradiction. Suppose G_0 is k -colorable, let C_0 be a k -coloring of G_0 , define valuation

$$v_0(p_{u,i}) = \begin{cases} T, & \text{if } C_0(u) = i \\ F, & \text{if } C_0(u) \neq i \end{cases}$$

Let

$$\Gamma'_0 = \{\phi_v^{exactly-1\text{-color}} \mid v \in V_0\} \cup \{\phi_{\{u,v\}}^{nbr} \mid \{u, v\} \in E_0\}$$

Consider

$$\Gamma_0 \subseteq \Gamma'_0$$

Since Γ'_0 is the same construction from k -colorability to formula, G_0 is k -colorable, we have

$$v_0 \models \Gamma'_0$$

By relevance lemma

$$v_0 \models \Gamma_0$$

Contradicting to our initial assumption that Γ_0 is NOT satisfiable. □

By Compactness theorem, Γ_G is finitely satisfiable $\rightarrow \Gamma_G$ is satisfiable.

Lemma 3.17. Γ_G is satisfiable $\rightarrow G$ is k -colorable

Proof. Γ_G is satisfiable $\rightarrow \exists v, v \models \Gamma_G$. We can construct a coloring C from v similarly to how we constructed v_0 from C_0 from previous proof:

$$C(v) = i, \text{ if } v(p_{v,i}) = T$$

By construction the coloring is valid, detailed technicality is omitted. □

□

4 Proof System

4.1 Hilbert Style Proof System

Four axiom schema of the hilberts style proof system. Axiom schema means that the ϕ, ψ, ρ in these schema are placeholders that can be replaced by any valid formula

1. Axiom 1: Simplification

$$\frac{}{p \rightarrow (q \rightarrow p)}$$

2. Axiom 2: Distribution

$$\frac{}{(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))}$$

3. Axiom 3: Double Negation Elimination (DNE)

$$\frac{}{((p \rightarrow \perp) \rightarrow \perp) \rightarrow p}$$

4. Rule of Inference: Modus Ponens

$$\frac{p \rightarrow q \quad p}{q}$$

The horizontal lines are judgement lines, which means that whats below can be inferred from whats above. The first 3 rules has no assumptions or premises, meaning they are ways we can introduce arbitrary claims or assumption into our proofs.

Definition 4.1 (Proof). A proof of the consequence of ϕ from Γ is a finite sequence of formula $c_1, c_2, \dots, c_k$ where $\forall i \leq k, c_i \in \Gamma$ or, c_i is obtained using one of the 4 axioms:

- $\exists j, l < i, c_j \equiv c_l \rightarrow c_i$
- $\exists \phi, \psi, c_i \equiv \phi \rightarrow (\psi \rightarrow \phi)$
- $\exists \phi, \psi, \rho, c_i \equiv [\phi \rightarrow (\psi \rightarrow \rho)] \rightarrow [(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \rho)]$
- $\exists \phi, c_i \equiv ((\phi \rightarrow \perp) \rightarrow \perp) \rightarrow \phi$

Definition 4.2 (Derivable). ϕ is derivable from Γ if there is a proof in HSPS of " ϕ is a consequence of Γ "

$$\Gamma \vdash \phi$$

Example 4.3.

$$\Gamma = \{p, p \rightarrow q\}, \quad \phi = q$$

Proof. We can construct proof sequence:

$$\begin{array}{ll}
c_1 = p & (p \in \Gamma) \\
c_2 = p \rightarrow q & (p \rightarrow q \in \Gamma) \\
c_3 = q & (\text{Modus Ponens on } c_1, c_2)
\end{array}$$

□

Sometimes HSPS requires us to creatively introduce new hypothesis to assist in the proof.

Example 4.4.

$$\Gamma = \{\perp\}, \phi = p$$

Intuitively this does not sound provable, just from $\perp$, but axioms that require no premises allows us to introduce helper hypothesis. In this case $(p \rightarrow \perp)$

Proof.

$$\begin{array}{ll}
c_1 = \perp & (\perp \in \Gamma) \\
c_2 = \perp \rightarrow [(p \rightarrow \perp) \rightarrow \perp] & (\text{Axiom 1: Simplification}) \\
c_3 = (p \rightarrow \perp) \rightarrow \perp & (\text{Modus Ponens on } \perp, (p \rightarrow \perp)) \\
c_4 = [(p \rightarrow \perp) \rightarrow \perp] \rightarrow p & (\text{Axiom 3: Double negation}) \\
c_5 = p & (\text{Modus Ponens on } c_3 \equiv [(p \rightarrow \perp) \rightarrow \perp], c_4 \equiv [(p \rightarrow \perp) \rightarrow \perp] \rightarrow p)
\end{array}$$

□

As seen from this example, HSPS can get convoluted and unintuitive for relatively straightforward consequences.

1. Self-Implication: $\vdash p \rightarrow p$

Idea. We must simulate identity using only distribution.

1. $p \rightarrow ((p \rightarrow p) \rightarrow p)$ (Axiom 1)
2. $p \rightarrow (p \rightarrow p)$ (Axiom 1)
3. $(p \rightarrow ((p \rightarrow p) \rightarrow p)) \rightarrow ((p \rightarrow (p \rightarrow p)) \rightarrow (p \rightarrow p))$ (Axiom 2)
4. $(p \rightarrow (p \rightarrow p)) \rightarrow (p \rightarrow p)$ (MP 1,3)
5. $p \rightarrow p$ (MP 2,4)

Even the trivial tautology requires a nontrivial derivation.

2. Contraposition: $\vdash (p \rightarrow q) \rightarrow ((q \rightarrow \perp) \rightarrow (p \rightarrow \perp))$

Idea. Assume $p \rightarrow q$ and $q \rightarrow \perp$; derive $p \rightarrow \perp$ via Axiom 2.

1. $(p \rightarrow (q \rightarrow \perp)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow \perp))$ (Axiom 2)
2. $(q \rightarrow \perp) \rightarrow (p \rightarrow (q \rightarrow \perp))$ (Axiom 1)
3. $(p \rightarrow q) \rightarrow ((q \rightarrow \perp) \rightarrow (p \rightarrow \perp))$

The last line follows by two applications of Modus Ponens using (1) and (2).
This shows implication reversal must be encoded indirectly.

3. Peirce's Law: $\vdash ((p \rightarrow q) \rightarrow p) \rightarrow p$

This formula characterizes classical logic.

Sketch.

1. $((p \rightarrow q) \rightarrow p) \rightarrow (((p \rightarrow q) \rightarrow p) \rightarrow ((p \rightarrow q) \rightarrow p))$ (Ax1)
2. Use Axiom 2 repeatedly to simulate assumption chaining.
3. Derive $((p \rightarrow q) \rightarrow p) \rightarrow \perp \rightarrow \perp$.
4. Apply Axiom 3 (DNE) to conclude $((p \rightarrow q) \rightarrow p) \rightarrow p$.

The essential trick: Peirce's law is obtained by embedding the formula into a double-negation context and applying Axiom 3.

This is far from syntactically obvious.

4. Explosion: $\vdash \perp \rightarrow p$

In this system $\perp$ is primitive via $p \rightarrow \perp$.

Idea. Show that from $\perp$ we can derive anything.

1. $p \rightarrow (\perp \rightarrow p)$ (Ax1)
2. $\perp \rightarrow (p \rightarrow \perp)$ (Ax1)
3. $(\perp \rightarrow (p \rightarrow \perp)) \rightarrow ((\perp \rightarrow p) \rightarrow (\perp \rightarrow \perp))$ (Ax2)
4. Derive $\perp \rightarrow \perp$ using the $p \rightarrow p$ pattern.
5. Conclude $\perp \rightarrow p$.

Notice how even explosion must be simulated through distribution.

5. Double Negation Introduction: $\vdash p \rightarrow ((p \rightarrow \perp) \rightarrow \perp)$

Although Axiom 3 gives elimination, introduction must be derived.

1. $(p \rightarrow \perp) \rightarrow (p \rightarrow (p \rightarrow \perp))$ (Ax1)
2. Use Axiom 2 to rearrange implication nesting.
3. Conclude $p \rightarrow ((p \rightarrow \perp) \rightarrow \perp)$.

This illustrates asymmetry between DNE and its converse.

4.2 Metatheorems

4.2.1 Monotonicity theorem

Theorem 4.5.

$$\Gamma \vdash A, \Gamma \subseteq \Delta \implies \Delta \vdash A$$

Proof. Let the proof of A from Γ be the sequence $\langle \phi_1, \phi_2, \dots, \phi_n \rangle, \phi_n = A$

Base case: $n = 1$

- Case 1: $\phi_1 \in \Gamma \implies \phi_1 \in \Delta \implies \Delta \vdash \phi_1$
- Case 2: ϕ_1 is from one of the axiom schema, $\Delta \vdash \phi_1$

Inductive step: $\forall i < n, \Gamma \vdash \phi_i, \Delta \vdash \phi_i$

- Case 1, 2: $\phi_{i+1} \in \Gamma$ or ϕ_{i+1} is from one of the axiom schema, trivial
- Case 3: ϕ_{i+1} is from Modus Ponens, $\exists \phi_j$ such that

$$\frac{\phi_j \rightarrow \phi_{i+1} \quad \phi_j}{\phi_{i+1}}$$

$\Gamma \vdash \phi_j, \Gamma \vdash \phi_j \rightarrow \phi_{i+1}$ by inductive hypothesis, $\Delta \vdash \phi_j, \Delta \vdash \phi_j \rightarrow \phi_{i+1}$, by MP, $\Delta \vdash \phi_{i+1}$

□

4.2.2 Deduction

Theorem 4.6.

$$\Gamma \cup \{A\} \vdash B \iff \Gamma \vdash A \rightarrow B$$

Proof. ($\implies$)

We will use structural induction over the length of derivation from $\Gamma \cup \{A\}$ to B , $\Gamma \cup \{A\} \vdash B \implies \exists \langle \phi_1, \phi_2, \dots, \phi_n \rangle, \phi_n = B$.

Base case: $n = 1, \phi_n = B$

- Case 1: B is an axiom $\Gamma \vdash B$. By simplification axiom, $\Gamma \vdash B \rightarrow (A \rightarrow B)$, by MP, $\Gamma \vdash A \rightarrow B$.
- Case 2: $B = A$, we will use the simplification and distribution axioms:

- $p \equiv A$
- $q \equiv (A \rightarrow A)$
- $r \equiv A$
- $(A \rightarrow ((A \rightarrow A) \rightarrow A)) \rightarrow ((A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A))$ (Distribution)
- $A \rightarrow ((A \rightarrow A) \rightarrow A)$ (Simplification)
- $(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$ (MP)
- $A \rightarrow (A \rightarrow A)$ (Simplification)
- $(A \rightarrow A)$ (MP)

$$\Gamma \vdash A \rightarrow A \implies \Gamma \vdash A \rightarrow B$$

Inductive step: Suppose $\forall i < n, \Gamma \cup \{A\} \vdash \phi_i \implies \Gamma \vdash A \rightarrow \phi_i$

- Case 1, 2, 3: ϕ_{i+1} is in Γ or ϕ_{i+1} is from one of the axioms, trivial; If $\phi_{i+1} = A$, the same as Case 2 in base case
- Case 4: ϕ_{i+1} is obtained from MP, where $\exists j \leq i$,

$$\begin{aligned} \Gamma \cup \{A\} \vdash \phi_j &\implies \Gamma \vdash A \rightarrow \phi_j \\ \Gamma \cup \{A\} \vdash \phi_j \rightarrow \phi_{i+1} &\implies \Gamma \vdash A \rightarrow (\phi_j \rightarrow \phi_{i+1}) \end{aligned}$$

$$\frac{\phi_j \rightarrow \phi_{i+1} \quad \phi_j}{\phi_{i+1}}$$

$$\begin{aligned} \Gamma \vdash (A \rightarrow \phi_j) \rightarrow (A \rightarrow \phi_{i+1}) &\text{by distribution} \\ \Gamma \vdash A \rightarrow \phi_{i+1} &\text{by MP} \end{aligned}$$

□

Theorem 4.7 (Soundness of HSPS). *HSPS is sound.*

$$\Gamma \vdash \phi \implies \Gamma \models \phi$$

Proof. Define property $H(i) : \forall j \leq i, \Gamma \models c_j$.

Base case: $i = 1$

Suppose $\Gamma \vdash \phi, \exists \pi = c_1, c_2, \dots, c_k, c_k = \phi$. By definition, $c_1 \in \Gamma$ or

- $\exists \phi, \psi, c_1 \equiv \phi \rightarrow (\psi \rightarrow \phi)$
- $\exists \phi, \psi, \rho, c_1 \equiv [\phi \rightarrow (\psi \rightarrow \rho)] \rightarrow [(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \rho)]$
- $\exists \phi, c_1 \equiv ((\phi \rightarrow \perp) \rightarrow \perp) \rightarrow \phi$

Note that Modus Ponens is not possible to yield c_1 as we need at least 1 previous formula.

If $c_1 \in \Gamma, \Gamma \models c_1$; Else, if c_1 comes from any other axiom, c_1 is valid, i.e. $\forall v, v \models c_1$, therefore $\Gamma \models c_1$

Inductive step:

Suppose $H(i), \forall j < i, \Gamma \models c_j, \forall v \models \Gamma, v \models c_j$.

- Case 1, 2: $c_{i+1} \in \Gamma$, or c_{i+1} is from an axiom, $\Gamma \models c_{i+1}, H(i+1)$
- Case 3: c_{i+1} comes from MP: $\exists j < i, \Gamma \vdash c_j, \Gamma \vdash c_j \rightarrow c_{i+1}$. By inductive hypothesis, $\forall v \models \Gamma, v \models c_j, v \models c_j \rightarrow c_{i+1}$

$$\begin{aligned} v(c_j) &= T \\ v(c_j \rightarrow c_{i+1}) &= T \equiv v(\neg c_j \vee c_{i+1}) = T \\ v(\neg c_j) &= T \text{ OR } v(c_{i+1}) = T \\ v(c_j) = T &\implies v(\neg c_j) = F \implies v(c_{i+1}) = T \\ &\implies \forall v \models \Gamma, v \models c_{i+1} \implies \Gamma \models c_{i+1} \end{aligned}$$

□

Theorem 4.8 (Completeness of HSPS). *HSPS is complete.*

$$\Gamma \models \phi \implies \Gamma \vdash \phi$$

4.3 Weak Completeness and Completeness of HSPS

We work with the Hilbert-Style Propositional System (HSPS):

- Axiom 1: $p \rightarrow (q \rightarrow p)$
- Axiom 2: $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$
- Axiom 3: $((p \rightarrow \perp) \rightarrow \perp) \rightarrow p$
- Rule: Modus Ponens

We write $\vdash \varphi$ for provability and $\models \varphi$ for semantic validity.

1. Weak Completeness

Theorem 4.9 (Weak Completeness). *If $\models \varphi$, then $\vdash \varphi$.*

Idea. We prove the contrapositive: if $\not\vdash \varphi$, then $\not\models \varphi$. That is, if φ is not provable, then there exists a valuation falsifying it.

Step 1: Consistency and Maximal Consistency. A set Γ is *consistent* if $\Gamma \not\vdash \perp$.

If $\not\vdash \varphi$, then $\{\neg\varphi\}$ is consistent.

Using a standard Lindenbaum construction:

- Extend any consistent set Γ
- to a *maximal consistent set* Δ
- such that for every formula ψ :

$$\psi \in \Delta \quad \text{or} \quad \neg\psi \in \Delta.$$

This construction relies only on classical logic and the deduction theorem (derivable in HSPS).

Step 2: Canonical Valuation. Define a valuation v_Δ by:

$$v_\Delta(p) = \begin{cases} \text{true} & \text{if } p \in \Delta \\ \text{false} & \text{otherwise} \end{cases}$$

We prove by structural induction:

$$\psi \in \Delta \quad \Longleftrightarrow \quad v_\Delta \models \psi.$$

The key steps use:

- Maximality of Δ
- Closure under Modus Ponens
- Axiom 3 for classical negation

Step 3: Falsifying φ . Since $\neg\varphi \in \Delta$, we obtain:

$$v_\Delta \models \neg\varphi, \quad \text{hence} \quad v_\Delta \not\models \varphi.$$

Therefore φ is not valid.

Conclusion.

$$\models \varphi \implies \vdash \varphi.$$

□

2. Strong Completeness (Full Completeness)

Theorem 4.10 (Completeness). *If $\Gamma \models \varphi$, then $\Gamma \vdash \varphi$.*

Reduction to Weak Completeness. Suppose $\Gamma \models \varphi$.

Then:

$$\models \bigwedge \Gamma \rightarrow \varphi.$$

If Γ is finite, we directly apply weak completeness:

$$\vdash (\bigwedge \Gamma) \rightarrow \varphi.$$

Repeated applications of Modus Ponens yield:

$$\Gamma \vdash \varphi.$$

Infinite Case. If Γ is infinite:

- If $\Gamma \models \varphi$,
- then some finite subset $\Gamma_0 \subseteq \Gamma$ satisfies $\Gamma_0 \models \varphi$ (compactness of propositional logic).

Apply the finite case to Γ_0 , then use monotonicity of derivability:

$$\Gamma \vdash \varphi.$$

3. Conceptual Structure

Weak completeness:

$$\models \varphi \Rightarrow \vdash \varphi.$$

Strong completeness:

$$\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi.$$

The essential engine is:

1. Maximal consistent sets
2. Canonical valuation
3. Truth lemma

Axiom 3 ensures classicality, making the canonical model work.

4. Final Picture

Soundness:

$$\vdash \varphi \Rightarrow \models \varphi.$$

Completeness:

$$\models \varphi \Rightarrow \vdash \varphi.$$

Hence:

$$\vdash \varphi \iff \models \varphi.$$

HSPS is therefore sound and complete for classical propositional logic. ■

4.4 Resolution Proof System

Definition 4.11 (Clause). For each proposition p define literals p and $\neg p$. A clause C is finite set of literals over the proposition set P .

$$v \models C = \{\ell_1, \ell_2, \dots, \ell_k\} \iff \exists j, v \models \ell_j$$

As such, a clause C is satisfiable if $\exists v, v \models C$

Note that from this definition, an empty clause (an empty set of literals) is not satisfiable, as intuitively, a valuation models a clause if there exists at least one literal in the clause such that $v \models \ell$.

Definition 4.12 (Conjunctive normal form (CNF)). A formula ϕ is said to be in Conjunctive Normal Form is

$$\phi \equiv \bigwedge_{i=1}^k C_i$$

Definition 4.13 (Satisfiability of a set of clauses).

$$\Gamma = \{C_1, C_2, \dots\}$$

- Γ is satisfiable if $\exists v, \forall C \in \Gamma, v \models C$
- Γ is unsatisfiable if $\forall v, \exists C \in \Gamma, v \not\models C$

By this definition,

- $\Gamma = \emptyset$ is vacuously satisfiable
- $\Gamma = \{\emptyset\}$ is not satisfiable.
- Any set of clauses that contains the empty set is unsatisfiable

Theorem 4.14. *Each well-formed formula can be transformed to a logically equivalent CNF*

Proof. **Base case:** $\phi \in P$

trivial

Inductive Case 1: $\neg$

Suppose there exists a CNF $\bigwedge_i \bigvee_j \ell_{i,j}$ logically equivalent to ϕ .

$$\begin{aligned} \neg\phi &= \neg((\ell_{1,1} \vee \ell_{1,2} \vee \dots) \wedge (\ell_{2,1} \vee \ell_{2,2} \vee \dots) \wedge \dots) \\ &= \neg(\ell_{1,1} \vee \ell_{1,2} \vee \dots) \vee \neg(\ell_{2,1} \vee \ell_{2,2} \vee \dots) \vee \dots \\ &= (\neg\ell_{1,1} \wedge \neg\ell_{1,2} \wedge \dots) \vee (\neg\ell_{2,1} \wedge \neg\ell_{2,2} \wedge \dots) \vee \dots \\ &= (\neg\ell_{1,1} \vee \neg\ell_{2,1} \vee \dots) \wedge () \end{aligned}$$

Inductive Case 2: $\wedge$

Suppose $\phi \equiv \bigwedge_{i \in I} C_i$ and $\psi \equiv \bigwedge_{j \in J} D_j$ are CNFs.

$$\begin{aligned} \phi \wedge \psi &\equiv \left(\bigwedge_{i \in I} C_i \right) \wedge \left(\bigwedge_{j \in J} D_j \right) \\ &\equiv \bigwedge_{k \in I \cup J} E_k \end{aligned}$$

where E_k is a clause from the set of clauses of ϕ or ψ . Since the conjunction of two conjunctions is simply a larger conjunction of the original clauses, the result is by definition a CNF.

Inductive Case 3: $\vee$

Suppose $\phi \equiv \bigwedge_{i \in I} C_i$ and $\psi \equiv \bigwedge_{j \in J} D_j$ are CNFs. By applying the Distributive Law:

$$\begin{aligned} \phi \vee \psi &\equiv \left(\bigwedge_{i \in I} C_i \right) \vee \left(\bigwedge_{j \in J} D_j \right) \\ &\equiv \bigwedge_{i \in I} \left(C_i \vee \left(\bigwedge_{j \in J} D_j \right) \right) \\ &\equiv \bigwedge_{i \in I} \left(\bigwedge_{j \in J} (C_i \vee D_j) \right) \\ &\equiv \bigwedge_{i \in I, j \in J} (C_i \vee D_j) \end{aligned}$$

Since each C_i and D_j are disjunctions of literals, their union $C_i \vee D_j$ is also a disjunction of literals, and thus a valid clause. The resulting expression is a conjunction of these clauses, satisfying the definition of a CNF.

Conclusion

By the principle of induction on the complexity of the formula, we have shown that if the sub-formulas of ϕ have equivalent CNFs, then $\neg\phi$, $\phi \wedge \psi$, and $\phi \vee \psi$ also have equivalent CNFs. Since the set of connectives $\{\neg, \wedge, \vee\}$ is functionally complete, every well-formed formula can be transformed into a logically equivalent CNF. We can bring all negations to literal level, and supply an additional negation to each literal to negate the whole formula. $\square$

The Resolution Rule

The system relies on a single rule of inference. Given two clauses C' and D' such that $C' = C \oplus \{p\}$ and $D' = D \oplus \{\neg p\}$ (where $\oplus$ denotes a disjoint union), the **resolvent** with respect to p is defined as the clause:

$$Res(C', D') = C \cup D$$

Resolution Proofs

A resolution proof of Γ is a finite sequence of clauses $\pi = C_1, C_2, \dots, C_n$ where the final clause C_n is the empty clause $\square$. For every clause C_i in the sequence, one of the following must be true:

1. C_i is an original clause from the set Γ ($C_i \in \Gamma$).
2. C_i is the resolvent of two previous clauses C_j and C_k (where $j, k < i$) with respect to some proposition p .

Soundness and Completeness

Theorem 4.15 (Soundness). *The resolution proof system is sound:*

$$\Gamma \vdash_{RES} \square \implies \Gamma \text{ is unsatisfiable}$$

Proof. Suppose there exists a resolution derivation $C_1, C_2, \dots, C_k$ where $C_k = \square$. We proceed by induction on the length of the derivation k .

Base Case: $k = 1$

If the empty clause $\square$ is in the derivation at step 1, then $\square \in \Gamma$. Since an empty clause is by definition unsatisfiable (it is an empty disjunction), Γ is unsatisfiable.

Inductive Step

Assume any clause derived in $i \leq k$ steps is a semantic consequence of Γ . Consider C_{k+1} derived from C_a and C_b ($a, b \leq k$) using the resolution rule on literal p :

$$C_a = \{p\} \cup C'_a, \quad C_b = \{\neg p\} \cup C'_b, \quad C_{k+1} = C'_a \cup C'_b$$

By the Inductive Hypothesis, $\Gamma \models C_a$ and $\Gamma \models C_b$. Let v be a valuation satisfying Γ .

- If $v(p) = F$, then since $v(C_a) = T$, it must be that $v(C'_a) = T$.
- If $v(p) = T$, then since $v(C_b) = T$, it must be that $v(C'_b) = T$.

In either case, $v(C'_a \cup C'_b) = T$. Thus $\Gamma \models C_{k+1}$. If $C_{k+1} = \square$, then $\Gamma \models \perp$, meaning Γ has no model and is unsatisfiable. $\square$

Theorem 4.16 (Completeness). *The resolution proof system is complete:*

$$\Gamma \text{ is unsatisfiable} \implies \Gamma \vdash_{RES} \square$$

Proof. We prove this by induction on the number of distinct propositional variables n in Γ .

Base Case: $n = 0$

If Γ is unsatisfiable and has 0 variables, it must contain the empty clause $\square$. Thus $\Gamma \vdash_{RES} \square$.

Inductive Step

Pick a variable p and define two sets:

- $\Gamma_{p=T}$: The set of clauses obtained by setting p to True (remove clauses containing p , remove $\neg p$ from remaining).
- $\Gamma_{p=F}$: The set of clauses obtained by setting p to False (remove clauses containing $\neg p$, remove p from remaining).

If Γ is unsatisfiable, then both $\Gamma_{p=T}$ and $\Gamma_{p=F}$ are unsatisfiable. By the Inductive Hypothesis, there exist resolution proofs for $\square$ from both sets. By re-inserting p and $\neg p$, these proofs yield $\Gamma \vdash_{RES} \{p\}$ and $\Gamma \vdash_{RES} \{\neg p\}$. Resolving these two results yields $\square$. $\square$

5 Complexity

5.1 Turing Machine

Definition 5.1 (Turing Machine).

$$\mathcal{M} = \langle Q, \Sigma, \Gamma, \delta, q_0, q_{Acc}, q_{Rej}, k \rangle$$

- Q : the set of states
- Σ : finite input alphabet
- Γ : finite worktap alphabet, $\square \in \Gamma$
- δ : the set of transitions
- q_0 : starting state, $q_0 \in Q$
- q_{Acc} : accepting state, $q_{Acc} \in Q$
- q_{Rej} : rejecting state, $q_{Rej} \in Q$ (Optional)
- k : the number of worktaps (Optional)

A TM has 1 read-only input tape, may have 1 write only output tape, and k work tape. We can show that any finitely many k tapes is equivalent to 1 worktap TM. [TODO]

Transifiton function

$$\delta = \langle (Q \times \Sigma \times \Gamma^k) \times (Q \times \{L, R, S\} \times \Gamma^k \times \{L, R, S\}^k \times \Gamma \times \{L, R, S\}) \rangle$$

A state transition

$$\tau = \langle q, \sigma, \gamma_1, \gamma_2, \dots, \gamma_k, q', d, \gamma'_1, \gamma'_2, \dots, \gamma'_k, d_1, d_2, \dots, d_k, \gamma', d_o \rangle$$

$$\begin{aligned} \tau = & \langle q, \sigma, \\ & \gamma_1, \gamma_2, \dots, \gamma_k, \\ & q', d, \\ & \gamma'_1, \gamma'_2, \dots, \gamma'_k, \\ & d_1, d_2, \dots, d_k, \\ & \gamma', d_o \rangle \end{aligned}$$

Represents a transition from state q , when the head reading input symbol σ , and worktap symbol γ_i on worktape i , transition into state q' , write worktape symbol γ'_i , move the input head in the direction d , and workhead i in direction d_i , writing γ' on the output tape and move output head in direction d_o

Non-deterministic TM

Non deterministic TM has non-deterministic transition functions

$$\delta \subseteq \langle (Q \times \Sigma \times \Gamma^k) \times (Q \times \{L, R, S\} \times \Gamma^k \times \{L, R, S\}^k \times \Gamma \times \{L, R, S\}) \rangle$$

5.2 Semantics of TM

Definition 5.2 (Configuration). A configuration C_t of a Turing machine at step t captures the overall state of the TM. Now we only consider TM with no rejecting states and output tape.

$$\begin{aligned} C = & \langle q, \\ & w_{in}^1 \nabla w_{in}^2, \\ & w_{w1}^1 \nabla w_{w1}^2 \\ & w_{w2}^1 \nabla w_{w2}^2 \\ & \dots \\ & w_{wk}^1 \nabla w_{wk}^2 \rangle \end{aligned}$$

Where $w^1 \nabla w^2 = w$, the symbol read by the head at each configuration is the first symbol after ∇ . Note

that ∇ is not a Symbol but just an indication of the head position.

The initial configuration will starts at the starting state q_0 , input head and work heads all at the beginning, and all worktapes empty.

$$C_0 = \langle q_0, \nabla w_{in}, \underbrace{\nabla, \nabla, \dots, \nabla}_k \rangle$$

Configuration

$$C_{t+1} = \langle q, \\ w_{in}^1 \nabla w_{in}^2, \\ w_{w1}^1 \nabla w_{w1}^2, \\ w_{w2}^1 \nabla w_{w2}^2, \\ \dots \\ w_{wk}^1 \nabla w_{wk}^2 \rangle$$

is a valid successor of

$$C_t = \langle q', \\ w_{in}^{1'} \nabla w_{in}^{2'}, \\ w_{w1}^{1'} \nabla w_{w1}^{2'}, \\ w_{w2}^{1'} \nabla w_{w2}^{2'}, \\ \dots \\ w_{wk}^{1'} \nabla w_{wk}^{2'} \rangle$$

$$\exists \tau = \langle p, \sigma, \gamma_1, \gamma_2, \dots, \gamma_k, p', d, \gamma'_1, \gamma'_2, \dots, \gamma'_k, d_1, d_2, \dots, d_k, \gamma', d_o \rangle$$

such that:

- σ is the first symbol after the input head: $w_{in}^2[0]$
- γ_i is the first symbol after work head i : $w_{wi}^2[0]$
- - if $d = S$: $w'_{in} = w_{in}$
 $|w_{in}^1| - |w_{in}^{1'}| = 0$
 - if $d = L$: $w'_{in} = w_{in}^1[: -1] \nabla w_{in}^1[-1] \cdot w_{in}^2$
 $|w_{in}^1| - |w_{in}^{1'}| = 1$
 - if $d = L$: $w'_{in} = w_{in}^1 \cdot w_{in}^2[0] \nabla w_{in}^2[1 :]$
 $|w_{in}^1| - |w_{in}^{1'}| = -1$
- $\forall i \in [1, k]$,
 - if $d_i = S$: $w'_{wi} = w_{wi}$
 $|w_{wi}^1| - |w_{wi}^{1'}| = 0$
 - if $d_i = L$: $w'_{wi} = w_{wi}^1[: -1] \nabla w_{wi}^1[-1] \cdot w_{wi}^2$
 $|w_{wi}^1| - |w_{wi}^{1'}| = 1$
 - if $d_i = L$: $w'_{wi} = w_{wi}^1 \cdot w_{wi}^2[0] \nabla w_{wi}^2[1 :]$
 $|w_{wi}^1| - |w_{wi}^{1'}| = -1$

Definition 5.3. $\forall w \in \Sigma^*$, $\mathcal{M}$ yields some valid sequences of configurations, which is called a run $\pi_w = C_0, C_1, \dots$ where

- C_0 is the initial configuration
- $\forall t \geq 1, C_t$ is a valid successor of C_{t-1}

A run is accepting if

1. π is finite, and

2. π ends with a configuration C_n , where $C_n.q = q_{Acc}$

A run is rejecting if

- π is infinite, or
- π ends with a configuration C_n , where $C_n.q \neq q_{Acc}$

Definition 5.4. $w \in \Sigma^*$ is accepted by $\mathcal{M}$ if $\exists \pi_w, \pi_w$ is accepting.

Now we can bridge to the concept of languages,

Definition 5.5. The set of strings accepted by $\mathcal{M}$ is the language described by $\mathcal{M}$, ie

$$L(\mathcal{M}) = \{w \in \Sigma^* \mid w \text{ is accepted by } \mathcal{M}\}$$

Now we will focus on decidable languages, and their recognizing non deterministic TM.

Definition 5.6 (Terminating Turing Machine). $\mathcal{M}$ is terminating if $\forall w \in \Sigma^*, \forall \pi_w, \pi_w$ is finite.

5.3 Running time and Space

Definition 5.7.

$$T : \mathbb{N} \mapsto \mathbb{N}$$

The running time of $\mathcal{M}$ is bounded by T if

$$\forall w \in \Sigma^*, \forall \pi_w, |\pi_w| \leq T(|w|)$$

Definition 5.8 (Space usage). The space usage of at configuration C_i is

$$space(C_i) = |w_{out}| + \sum_{j=1}^k |w_{wj}|$$

Definition 5.9.

$$S : \mathbb{N} \mapsto \mathbb{N}$$

The space usage of $\mathcal{M}$ is bounded by S if

$$\forall w \in \Sigma^*, \forall \pi_w, \forall C \in \pi_w, space(C) \leq S(|w|)$$

Definition 5.10. L is computable by $\mathcal{M}$ if $L = L(\mathcal{M})$

Definition 5.11. L is computable in time T if $\exists \mathcal{M}$

1. $L = L(\mathcal{M})$ and
2. the running time of $\mathcal{M}$ is bounded by T

Definition 5.12. L is computable in space S if $\exists \mathcal{M}$

1. $L = L(\mathcal{M})$ and
2. the space usage of $\mathcal{M}$ is bounded by S

Theorem 5.13 (Church-Turing Thesis). *Any model of computation is only as powerful as the Turing machine.*

$\forall L, \exists N$ in a new model of computation, $L(N) = L \implies \exists \mathcal{M} \in TM, L(\mathcal{M}) = L$

5.4 Complexity classes

Definition 5.14 (*NTIME*).

$$NTIME(T(\cdot)) = \{L \mid L \text{ is computable in time } T(\cdot)\}$$

Definition 5.15 (*DTIME*).

$$DTIME(T(\cdot)) = \{L \mid \exists \text{ deterministic } \mathcal{M}, L(\mathcal{M}) = L, \mathcal{M} \text{ running time is bounded by } T(\cdot)\}$$

Definition 5.16 (*NSPACE*).

$$NSPACE(S(\cdot)) = \{L \mid L \text{ is computable in space } S(\cdot)\}$$

Definition 5.17 (*DSPACE*).

$$DSPACE(S(\cdot)) = \{L \mid \exists \text{ deterministic } \mathcal{M}, L(\mathcal{M}) = L, \mathcal{M} \text{ space usage is bounded by } S(\cdot)\}$$

$$\begin{aligned} P &= PTIME = \bigcup_{k \geq 0} DTIME(f_k), & f_k(n) &= n^k \\ PSPACE &= \bigcup_{k \geq 0} DSPACE(f_k), & f_k(n) &= n^k \\ NP &= NPTIME = \bigcup_{k \geq 0} NTIME(f_k), & f_k(n) &= n^k \\ NPSPACE &= \bigcup_{k \geq 0} NSPACE(f_k), & f_k(n) &= n^k \end{aligned}$$

Theorem 5.18.

$$PTIME \subseteq NPTIME$$

Theorem 5.19.

$$PSPACE \subseteq NPSPACE$$

Theorem 5.20.

$$NPSPACE \subseteq PSPACE$$

Proof.

□

Corollary 5.21.

$$PSPACE = NPSPACE$$

5.5 Hardness

Definition 5.22 (Hardness of Turing machine). $\mathcal{M}_1 \leq \mathcal{M}_2$ if all runs of $\mathcal{M}_1$ are smaller than $\mathcal{M}_2$

5.6 Reduction

Theorem 5.23 (Constant speed up theorem). *L is computable by $f \implies L$ is computable by f' , where*

$$f'(n) = f\left(\frac{n}{c}\right)$$

Theorem 5.24. *For all non-deterministic turing machines $\mathcal{M}_1$ with, there exists a deterministic turing machine $\mathcal{M}_2$, such that $L(\mathcal{M}_1) = L(\mathcal{M}_2)$*

Lemma 5.25. *For all deterministic turing machine with $\mathcal{M}_1$ with k worktapes, there exists a deterministic turing machine $\mathcal{M}_2$ with 1 worktape such that $L(\mathcal{M}_1) = L(\mathcal{M}_2)$. Same for non deterministic TM.*

Consider deterministic TM $\mathcal{M}$ with an output tape, it induces function

$$f_{\mathcal{M}} : \Sigma^* \mapsto \Gamma^*$$

Such that

$$f_{\mathcal{M}} = \begin{cases} \text{undefined, if } \mathcal{M} \text{ does not terminate;} \\ u, \text{ if } \mathcal{M} \text{ terminates, such that in the final configuration, output tape contains } u \end{cases}$$

Corollary 5.26. *$f_{\mathcal{M}}$ is total if $\mathcal{M}$ is terminating*

Theorem 5.27. *A partial function $f : \Sigma^* \mapsto \Gamma^*$ is computable is $\exists \mathcal{M}$ that induces $f_{\mathcal{M}}, f = f_{\mathcal{M}}$*

Theorem 5.28. *A total function is computable in time $T : \mathbb{N} \mapsto \mathbb{N}$ if there exists $\mathcal{M}$ such that $f_{\mathcal{M}} = f$, and $\mathcal{M}$ is running time bounded by T*

5.7 Cook–Levin Theorem

Theorem 5.29 (Cook–Levin). *SAT is NP-complete. In particular, for every nondeterministic Turing machine $\mathcal{M}$ running in polynomial time, there exists a polynomial-time reduction mapping any input w to a Boolean formula $\Phi_{\mathcal{M},w}$ such that*

$$w \in L(\mathcal{M}) \iff \Phi_{\mathcal{M},w} \text{ is satisfiable.}$$

Proof Outline.

1. SAT $\in$ NP.
2. Every language in NP reduces to SAT.

We focus on (2) under the Turing machine semantics defined above.

Step 1: SAT $\in$ NP

Given a Boolean formula φ , guess a valuation and evaluate φ in time polynomial in $|\varphi|$. Hence SAT $\in$ NP.

Step 2: Reduction from NP to SAT

Let $L \in$ NP. Then there exists a nondeterministic Turing machine $\mathcal{M}$ and a polynomial $p(n)$ such that for every input w :

$$w \in L \iff \mathcal{M} \text{ has an accepting run on } w \text{ of length } \leq p(|w|).$$

Fix w with $|w| = n$, and let $T = p(n)$. We construct a Boolean formula $\Phi_{\mathcal{M},w}$ expressing:

“There exists an accepting run of $\mathcal{M}$ on w of length $\leq T$.”

Encoding Configurations

Since $\mathcal{M}$ runs for at most T steps:

- The input head moves at most T cells.
- Each work tape head moves at most T cells.

Thus each tape region used is bounded by $O(T)$ cells.

We represent the entire computation as a *tableau*:

$$\text{Time } 0, 1, \dots, T \quad \times \quad \text{Tape positions } -T, \dots, T$$

Each row corresponds to a configuration.

Propositional Variables

We introduce variables:

- $Q_{t,q}$: machine is in state q at time t .
- $H_{t,x}^i$: head of tape i is at position x at time t .
- $S_{t,x,\gamma}^i$: tape i at position x contains symbol γ at time t .

Here $i = 0$ denotes the input tape and $i = 1, \dots, k$ the work tapes.

The number of variables is polynomial in T .

Constraints

The formula $\Phi_{\mathcal{M},w}$ is a conjunction of four parts.

(A) Exactly-One Constraints For every time t :

- Exactly one state holds:

$$\bigvee_{q \in Q} Q_{t,q} \quad \wedge \quad \bigwedge_{q \neq q'} (\neg Q_{t,q} \vee \neg Q_{t,q'})$$

- Exactly one head position per tape.
- Exactly one symbol per tape cell.

(B) Initial Configuration We enforce:

- Q_{0,q_0} .
- Input tape contains w .
- Work tapes contain blanks.
- All heads at starting position.

All of these are expressed as unit clauses.

(C) Transition Constraints For every $t < T$ and every possible local configuration, we encode the transition relation.

Fix a transition

$$\tau = \langle q, \sigma, \gamma_1, \dots, \gamma_k, q', d, \gamma'_1, \dots, \gamma'_k, d_1, \dots, d_k \rangle.$$

For each position x , we enforce:

If at time t :

- state is q ,
- input head at x reading σ ,
- work head i at x_i reading γ_i ,

then at time $t + 1$:

- state is q' ,
- written symbols updated,
- heads moved according to d, d_i ,
- all other cells unchanged.

Each such condition is encoded by clauses of the form:

$$(\text{current configuration}) \rightarrow (\text{next configuration})$$

which is translated into CNF using Tseitin transformation.

(D) Acceptance We require that for some $t \leq T$:

$$Q_{t,q_{Acc}}$$

This is encoded as:

$$\bigvee_{t=0}^T Q_{t,q_{Acc}}.$$

Correctness

Soundness If $\mathcal{M}$ has an accepting run of length $\leq T$, assign variables according to that run. All constraints (A)–(D) are satisfied. Hence $\Phi_{\mathcal{M},w}$ is satisfiable.

Completeness If $\Phi_{\mathcal{M},w}$ is satisfiable, the satisfying assignment defines:

- A unique state per time,
- A unique head position per time,
- A unique symbol per cell,

and transition constraints ensure each configuration is a valid successor of the previous one. Thus we extract a valid accepting run of $\mathcal{M}$ on w . Hence $w \in L(\mathcal{M})$.

Size and Complexity

- Time bound $T = p(n)$.
- Tableau size $O(T^2)$.
- Number of variables and clauses polynomial in T .

Therefore $\Phi_{\mathcal{M},w}$ can be constructed in polynomial time.

Conclusion

We have shown:

$$L \in \text{NP} \implies L \leq_p \text{SAT}.$$

Since $\text{SAT} \in \text{NP}$, we conclude:

SAT is NP-complete.

□

6 Syntax of First Order Logics

Let consider the limitation of propositional logics. Propositional logics intuitively represents choices, and each proposition has no internal structure, it is simply a boolean value to be evaluated. And since formula needs to be finitely long, we cannot express problems of infinite size. Thus we introduce first order logic, which has constant, variables and quantifiers, allowing us to express more nuanced ideas.

6.1 Vocabulary of FOL

1. Signature: $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R})$
 - Constant symbols: $\mathcal{C}$
 - Function symbols: $\mathcal{F}$
 - Relation symbols: $\mathcal{R}$
2. Connector: $C = \{\neg, \vee\}$
3. Variables: V
4. Quantifiers: $\{\forall, \exists\}$
5. Equality: $=$ as a special relation

Definition 6.1. *Term* Similar to how we define formula in propositional logic, a term can also be recursively defined as

$$t := c \in \mathcal{C} \mid x \in V \mid f(\underbrace{t, \dots}_{\text{arity}(f)}), f \in \mathcal{F}$$

Here we use term to represent the idea of an "object" in FOL.

Definition 6.2. *Formula*

$$\phi := t = t \mid R(\underbrace{t, \dots}_{\text{arity}(R)}), R \in \mathcal{R} \mid \phi \vee \phi \mid \neg \phi \mid \exists x \in V, \phi \mid \forall x \in V, \phi$$

6.2 Modelling with FOL

Example 6.3. *Number line*

$$\Sigma = (\{0\}, \{succ\}, \{<\})$$

Under this signature, we can write formula:

- $\phi_1 = \forall x \exists y, y = succ(x)$
- $x = 0$
- $0 = 1$

These are all syntactically valid formula.

7 Semantics of First Order Logics

In propositional logics, semantics are conveyed by valuations, which assigns true or false to each proposition. In FOL, we need to define the universe and interpret the symbols to unpack the semantics.

Definition 7.1. *Structure*

$$\mathfrak{A} = (U, I)$$

A structure is a tuple of U , an non-empty universe set, and I , the interpretation of symbols.

Definition 7.2. *Interpretation* An interpretation is a mapper from symbols to elements in the universe.

- For every constant $c \in \mathcal{C}$, $I(c) \in U$
- For every function symbol $f \in \mathcal{F}$, $I(f) \in [U^k \mapsto U]$, $k = \text{arity}(f)$
- For every relation symbol $R \in \mathcal{R}$, $I(R) \subseteq U^k$, $k = \text{arity}(R)$

Definition 7.3. *Assignment* Given a structure $\mathfrak{A} = (U, I)$

$$\alpha \in [V \mapsto U]$$

Is an assignment of each variable to elements in the universe.

Definition 7.4. *Valuation of terms*

$$\begin{aligned} \text{Val}_{\mathfrak{A}, \alpha}(c) &= I(c) \in U \\ \text{Val}_{\mathfrak{A}, \alpha}(x) &= \alpha(x) \in U \\ \text{Val}_{\mathfrak{A}, \alpha}(f(t, \dots)) &= I(f)(\text{Val}_{\mathfrak{A}, \alpha}(t), \dots) \in U \end{aligned}$$

Essentially Valuation is an assignment of elements in a universe to terms, the objects in FOL

Definition 7.5. *Substitution*

$$\alpha[x \mapsto u](z) = \begin{cases} u, & \text{if } z = x \\ \alpha(z), & \text{otherwise} \end{cases}$$

Definition 7.6. *Valuation of Formula*

Base case: $\phi \equiv t_1 = t_2$

$$\mathfrak{A}, \alpha \models \phi \iff \text{Val}_{\mathfrak{A}, \alpha}(t_1) = \text{Val}_{\mathfrak{A}, \alpha}(t_2)$$

Inductive cases:

$$1. \phi \equiv R(\underbrace{t, \dots}_{\text{arity}(R)})$$

$$\mathfrak{A}, \alpha \models \phi \iff (\underbrace{\text{Val}_{\mathfrak{A}, \alpha}(t), \dots}_{\text{arity}(R)}) \in I(R)$$

$$2. \phi \equiv \neg \varphi$$

$$\mathfrak{A}, \alpha \models \phi \iff \mathfrak{A}, \alpha \not\models \varphi$$

$$3. \phi \equiv \phi_1 \vee \phi_2$$

$$\mathfrak{A}, \alpha \models \phi \iff \mathfrak{A}, \alpha \models \phi_1 \text{ OR } \mathfrak{A}, \alpha \models \phi_2$$

$$4. \phi \equiv \exists x, \varphi$$

$$\mathfrak{A}, \alpha \models \phi \iff \text{there is } u \in U, \mathfrak{A}, \alpha[x \mapsto u] \models \varphi$$

$$5. \phi \equiv \forall x, \varphi$$

$$\mathfrak{A}, \alpha \models \phi \iff \text{for all } u \in U, \mathfrak{A}, \alpha[x \mapsto u] \models \varphi$$

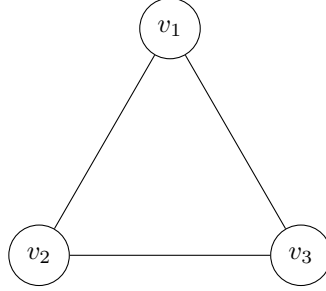
Example 7.7. *graphs* Consider signature

$$\Sigma = (\{s, t, u\}, \emptyset, \{E\})$$

We can come up with a structure

$$\begin{aligned}
\mathfrak{A} &= (U_G, I_G) \\
U_G &= \{v_1, v_2, v_3\} \\
I_G(s) &= v_1 \\
I_G(t) &= v_2 \\
I_G(u) &= v_3 \\
I_G(E) &= \{(v_1, v_2), (v_2, v_3), (v_3, v_1)\}
\end{aligned}$$

The graph represented by $\mathfrak{A}$ is:



Definition 7.8. *Variables and Free variables* Variables in a term is defined as:

$$\begin{aligned}
Var(x) &= x \\
Var(c) &= \emptyset \\
Var(f(t_1, \dots, t_k)) &= \bigcup_{i=1}^k Var(t_i)
\end{aligned}$$

Free variables in a formula can be defined as:

$$\begin{aligned}
FVar(\forall x, \phi) &= FVar(\phi) \setminus \{x\} \\
FVar(\exists x, \phi) &= FVar(\phi) \setminus \{x\}
\end{aligned}$$

In words, free variables are variables appear in a formula that are not bounded by quantifiers

Definition 7.9. *Satisfaction of FOL formula*

$$\phi \text{ is satisfiable} \iff \exists (\mathfrak{A}, \alpha), \mathfrak{A}, \alpha \models \phi$$

Definition 7.10. *Logical Consequence*

$$\Gamma \models \phi \iff \forall (\mathfrak{A}, \alpha), \mathfrak{A}, \alpha \models \Gamma \implies \mathfrak{A}, \alpha \models \phi$$

Lemma 7.11 (Relevance Lemma for FOL). *Let ϕ be a FOL formula, let $\mathfrak{A}$ be a structure and let α_1, α_2 be variable assignments such that for all $x \in FVar(\phi)$,*

$$\alpha_1(x) = \alpha_2(x).$$

Then

$$\mathfrak{A}, \alpha_1 \models \phi \iff \mathfrak{A}, \alpha_2 \models \phi.$$

Proof. By symmetry, we only need to prove if for all $x \in FVar(\phi)$, $\alpha_1(x) = \alpha_2(x)$ then

$$\mathfrak{A}, \alpha_1 \models \phi \implies \mathfrak{A}, \alpha_2 \models \phi.$$

Lemma 7.12 (Substitution lemma).

$$\mathfrak{A}, \alpha \models \phi[x \mapsto t] \iff \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi$$

Lemma 7.13 (Substitution lemma).

$$\forall x \in \text{Var}(t), \alpha_1(x) = \alpha_2(x) \implies \text{Val}_{\mathfrak{A}, \alpha_1}(t) = \text{Val}_{\mathfrak{A}, \alpha_2}(t)$$

Lemma 7.14 (Substitution lemma).

$$\mathfrak{A} \models \psi[x \mapsto t] \iff \mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}}(t)] \models \psi$$

Lemma 7.15.

$$\text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(t)$$

Proof. **Base case:**

1. $t = c \in C$:

$$\begin{aligned} \text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) &= \text{Val}_{\mathfrak{A}, \alpha}(c) = I(c) \\ \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(t) &= \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(c) = I(c) \end{aligned}$$

2. $t = y \in V$:

Case 1: $x = y$

$$\begin{aligned} &\implies y[x \mapsto t'] = t' \\ \text{Val}_{\mathfrak{A}, \alpha}(y[x \mapsto t']) &= \text{Val}_{\mathfrak{A}, \alpha}(t') \\ \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(y) &= \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')](y) = \text{Val}_{\mathfrak{A}, \alpha}(t') \end{aligned}$$

Case 2: $x \neq y$

$$\begin{aligned} &\implies y[x \mapsto t'] = y \\ \text{Val}_{\mathfrak{A}, \alpha}(y[x \mapsto t']) &= \text{Val}_{\mathfrak{A}, \alpha}(y) = \alpha(y) \\ \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(y) &= \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')](y) = \alpha(y) \end{aligned}$$

Inductive step: $t = f(t_1, \dots)$

$$\begin{aligned} t[x \mapsto t'] &= f(t_1[x \mapsto t'], t_2[x \mapsto t'], \dots) \\ \text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) &= \text{Val}_{\mathfrak{A}, \alpha}(f(t_1[x \mapsto t'], t_2[x \mapsto t'], \dots)) = \\ &= I(f)(\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t']), \text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t']), \dots) \end{aligned}$$

The other side

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(f(t_1, t_2, \dots)) = I(f)(\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t']), \text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t']), \dots)$$

□

Proof. We prove the substitution lemma for formulae by structural induction on ψ .

Base case: $\psi \equiv t_1 = t_2$

$$\psi[x \mapsto t] \equiv t_1[x \mapsto t] = t_2[x \mapsto t]$$

Thus

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff \text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t]).$$

By the substitution lemma for terms,

$$\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_1)$$

and

$$\text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_2).$$

Hence

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_1) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_2),$$

that is,

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models t_1 = t_2.$$

Inductive steps

1. $\psi \equiv R(t_1, \dots, t_n)$.

$$\psi[x \mapsto t] \equiv R(t_1[x \mapsto t], \dots, t_n[x \mapsto t]).$$

Then

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff (Val_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]), \dots, Val_{\mathfrak{A}, \alpha}(t_n[x \mapsto t])) \in I(R).$$

By the substitution lemma for terms, for each i ,

$$\begin{aligned} Val_{\mathfrak{A}, \alpha}(t_i[x \mapsto t]) &= Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_i). \\ (Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_1), \dots, Val_{\mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)]}(t_n)) &\in I(R), \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\models R(t_1, \dots, t_n). \end{aligned}$$

2. $\psi \equiv \neg\phi$.

$$\begin{aligned} \mathfrak{A}, \alpha \models (\neg\phi)[x \mapsto t] &\iff \mathfrak{A}, \alpha \models \neg(\phi[x \mapsto t]) \\ &\iff \mathfrak{A}, \alpha \not\models \phi[x \mapsto t]. \end{aligned}$$

By IH,

$$\begin{aligned} \mathfrak{A}, \alpha \models \phi[x \mapsto t] &\iff \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi. \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\not\models \phi, \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] &\models \neg\phi. \end{aligned}$$

3. $\psi \equiv \phi_1 \vee \phi_2$.

$$\begin{aligned} \mathfrak{A}, \alpha \models (\phi_1 \vee \phi_2)[x \mapsto t] &\iff \mathfrak{A}, \alpha \models \phi_1[x \mapsto t] \vee \phi_2[x \mapsto t]. \\ \mathfrak{A}, \alpha \models \phi_1[x \mapsto t] \quad \text{or} \quad \mathfrak{A}, \alpha \models \phi_2[x \mapsto t]. \end{aligned}$$

By IH

$$\begin{aligned} \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi_1 \quad \text{or} \quad \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi_2. \\ \mathfrak{A}, \alpha[x \mapsto Val_{\mathfrak{A}, \alpha}(t)] \models \phi_1 \vee \phi_2. \end{aligned}$$

□

□

Lemma 7.16 (Analogous relevance lemma on structure). *If $\mathfrak{A}$ and $\mathfrak{B}$ agree on all parameters occur in ϕ , then $\mathfrak{A} \models \phi \iff \mathfrak{B} \models \phi$*
If

$$\begin{aligned} \forall c \in C, Val_{\mathfrak{A}, \alpha}(c) &= Val_{\mathfrak{B}, \alpha}(c) \\ \forall x \in V, Val_{\mathfrak{A}, \alpha}(x) &= Val_{\mathfrak{B}, \alpha}(x) \\ \forall f(t, \dots), Val_{\mathfrak{A}, \alpha}(f(t, \dots)) &= Val_{\mathfrak{B}, \alpha}(f(t, \dots)) \\ \forall R(t, \dots), R(t, \dots) \in I_{\mathfrak{A}}(R) &\iff R(t, \dots) \in I_{\mathfrak{B}}(R) \\ \mathfrak{A} \models \phi &\iff \mathfrak{B} \models \phi \end{aligned}$$

8 Compactness of First Order Logics

Definition 8.1. *Compactness Theorem for First Order Logics*

$$\Gamma \text{ is satisfiable} \iff \Gamma \text{ is finitely satisfiable}$$

$$\Gamma \text{ is satisfiable} \iff \forall \Gamma' \subseteq_{fin} \Gamma, \exists (\mathfrak{A}, \alpha), \mathfrak{A}, \alpha \models \Gamma'$$

Theorem 8.2. *Structures are not countable, and the uncountability is uncapped, as U and I can be arbitrarily defined*

Definition 8.3. *Prenex Normal Form* ϕ is in *Prenex Normal Form* if all quantifiers are in the beginning of the formula.

$$\phi \equiv Q_1 x_1 Q_2 x_2 \dots \psi$$

Where ψ is the matrix of ϕ , which is a quantifier-free formula.

8.1 Prenex Normal Form

1. Remove implication and biconditional.

Replace

$$\begin{aligned} A \rightarrow B &\equiv \neg A \vee B \\ A \leftrightarrow B &\equiv (A \rightarrow B) \wedge (B \rightarrow A) \end{aligned}$$

2. Push negations inward (Negation Normal Form).

Apply De Morgan's laws and quantifier negation:

$$\begin{aligned} \neg(A \wedge B) &\equiv \neg A \vee \neg B \\ \neg(A \vee B) &\equiv \neg A \wedge \neg B \\ \neg \forall x \varphi &\equiv \exists x \neg \varphi \\ \neg \exists x \varphi &\equiv \forall x \neg \varphi \end{aligned}$$

After this step, negations appear only directly in front of atomic formulas.

3. Standardize variables apart: α -renaming

Rename bound variables so that each quantifier binds a unique variable and no variable is bound by more than one quantifier.

4. Move quantifiers outward.

Move quantifiers across logical connectives when the variable does not occur free in the other formula. For example,

$$\begin{aligned} (\forall x A) \wedge B &\equiv \forall x (A \wedge B) \quad \text{if } x \notin \text{FV}(B) \\ (\exists x A) \vee B &\equiv \exists x (A \vee B) \quad \text{if } x \notin \text{FV}(B) \end{aligned}$$

Continue moving quantifiers outward until all quantifiers appear at the front of the formula.

5. Obtain Prenex Normal Form.

The formula now has the form

$$Q_1 x_1 Q_2 x_2 \dots Q_n x_n \varphi$$

where each $Q_i \in \{\forall, \exists\}$ and φ is quantifier-free.

Example 8.4.

$$\phi \equiv P(x) \wedge \forall z (R(x, z) \vee \exists y Q(y))$$

1. move out quantifiers:

$$\begin{aligned} \phi &\equiv P(x) \wedge \forall z \exists y (R(x, z) \vee Q(y)) \\ \phi &\equiv \forall z \exists y [P(x) \wedge (R(x, z) \vee Q(y))] \end{aligned}$$

2. PNF obtained, $\varphi \equiv P(x) \wedge (R(x, z) \vee Q(y))$

Example 8.5.

$$\phi \equiv \forall x (\neg P(x) \vee \exists y (Q(x, y) \wedge \forall z R(y, z))) \vee \exists x S(x)$$

1. α -renaming

$$\phi \equiv \forall x (\neg P(x) \vee \exists y (Q(x, y) \wedge \forall z R(y, z))) \vee \exists u S(u)$$

2.

Lemma 8.6. ψ , the PNF of ϕ is logically equivalent to ϕ

$$\forall (\mathfrak{A}, \alpha), \mathfrak{A}, \alpha \models \psi \iff \mathfrak{A}, \alpha \models \phi$$

Proof.

□

Definition 8.7. *Universally quantified formula ϕ is universally quantified if all quantifiers in ϕ are $\forall$. Analogous to Existential quantified formula*

Definition 8.8. *Sentence ϕ is a sentence if there is no free variables.*

$$FVar(\phi) = \emptyset$$

To determine satisfiability, we can ignore existential quantifiers.

Theorem 8.9. *Let φ be a formula and $FVar(\varphi) = \{x_1, x_2, \dots, x_k\}$ φ is satisfiable iff*

$$\varphi' \equiv \exists x_1 \exists x_2, \dots, \exists x_k \varphi \text{ is satisfiable}$$

φ' is a sentence

Definition 8.10. *Skolemization Transform PNF into universally quantified PNF.*

$\forall \phi, \exists \phi'$, such that

1. ϕ and ϕ' are equisatisfiable. (Not necessarily equivalent!)
2. ϕ' is universally quantified PNF

8.2 Skolemization

1. Convert the formula to prenex normal form.

Transform the formula so that all quantifiers appear at the front:

$$Q_1 x_1 Q_2 x_2 \dots Q_n x_n \varphi$$

where φ is quantifier-free.

2. Process quantifiers from left to right.

For each existential quantifier $\exists x$, determine the universal variables that precede it in the prefix.

3. Replace existential variables with Skolem terms.

- If $\exists x$ has no preceding universal quantifiers, replace x with a new Skolem constant c .

$$\exists x P(x) \Rightarrow P(c)$$

- If $\exists x$ is preceded by universal variables $u_1, \dots, u_k$, replace x with a new Skolem function $f(u_1, \dots, u_k)$.

$$\forall u_1 \dots \forall u_k \exists x P(x, u_1, \dots, u_k) \Rightarrow P(f(u_1, \dots, u_k), u_1, \dots, u_k)$$

4. Remove the existential quantifier.

After substitution with the Skolem term, delete the corresponding existential quantifier from the prefix.

5. Repeat until no existential quantifiers remain.

The resulting formula contains only universal quantifiers.

6. Optionally drop universal quantifiers.

In many applications (e.g., resolution), the remaining universal quantifiers are omitted, since variables are implicitly assumed to be universally quantified.

Example 8.11.

$$\phi \equiv \exists x P(x) \wedge \forall z (R(x, z) \vee \exists y Q(y))$$

1. convert to PNF

$$\phi \equiv \exists x \forall z \exists y (P(x) \wedge (R(x, z) \vee Q(y)))$$

2. if the leftmost quantifier is existential, replace the variable with a new constant c

$$\phi' \equiv \forall z \exists y (P(c_x) \wedge (R(c_x, z) \vee Q(y)))$$

$$c_x \notin \Sigma$$

At this stage, ϕ and ϕ' are equisatisfiable. No longer equivalent

Proof. ϕ is satisfiable $\implies \exists \mathfrak{A}, \alpha \models \phi$. We can construct a new signature and structure from it as such

$$\Sigma' = \Sigma \cup \{c_x\}, \quad I'(c_x) = \alpha(x)$$

We can show that $\mathfrak{A}', \alpha \models \phi'$

The other way can be done similarly □

3. if the next existential quantifier is not the outer most, add a new function symbol to replace each variable quantified by the existential quantifier with the function call using all previous universally quantified variables

$$\phi'' \equiv \forall z (P(c_x) \wedge (R(c_x, z) \vee Q(f_y(z))))$$

We can show that ϕ' and ϕ'' are equisatisfiable

Theorem 8.12. Skolemization preserves satisfiability.

Proof. Consider skolemization as a recursive algorithm described as follows:

Algorithm 1: Skolemization on Prenex Formula

Input: Signature $\Sigma = (C, F, R)$;

Formula $\phi = (\pi, \psi)$ where π is a quantifier prefix and ψ is quantifier-free;

List u of universally quantified variables in scope (initially empty).

Output: Skolemized formula (π', ψ') equisatisfiable with ϕ .

if π is empty **then**

return $(\emptyset, \psi)$

$(q, x) \leftarrow \text{head}(\pi)$;

$\pi_{\text{rest}} \leftarrow \text{tail}(\pi)$

if $q = \exists$ **then**

if u is empty **then**

 introduce fresh constant c_x ;

$\Sigma.C \leftarrow \Sigma.C \uplus \{c_x\}$;

return $\text{Skol}(\Sigma, (\pi_{\text{rest}}, \psi[x \mapsto c_x]), u)$

else

 introduce fresh function f_x of arity $|u|$;

$\Sigma.F \leftarrow \Sigma.F \uplus \{f_x\}$;

 let $t = f_x(u_1, \dots, u_k)$ where $u = [u_1, \dots, u_k]$;

return $\text{Skol}(\Sigma, (\pi_{\text{rest}}, \psi[x \mapsto t]), u)$

else

 // $q = \forall$

$u' \leftarrow u \cdot x$ // append x to u

$(\pi', \psi') \leftarrow \text{Skol}(\Sigma, (\pi_{\text{rest}}, \psi), u')$;

return $((\forall x) \cdot \pi', \psi')$

We claim that the invariant is the satisfiability of input and output is the same. Let $\text{Skol}(\Sigma, (\pi, \psi), u)$ be the recursive algorithm defined. We prove by induction on the length of the quantifier prefix $n = |\pi|$ that the algorithm preserves equisatisfiability.

Inductive Hypothesis (IH)

For any prefix π of length n , any signature Σ , and any list of universal variables $u = [u_1, \dots, u_k]$, a formula $\phi = \forall u(\pi\psi)$ is satisfiable in a Σ -structure $\mathfrak{A}$ if and only if there exists a larger $\mathfrak{A}'$ than $\mathfrak{A}$ to Σ' (containing new symbols) such that $\mathfrak{A}'$ satisfies the output of $\text{Skol}(\Sigma, (\pi, \psi), u)$.

$$\mathfrak{A} \models \phi = \forall u(\pi\psi) \iff \mathfrak{A}' \models \text{Skol}(\Sigma, (\pi, \psi), u)$$

Case 1: Base Case ($n = 0$)

If π is empty, the algorithm returns $(\emptyset, \psi)$, trivially equisatisfiable

Case 2: Universal Case ($q = \forall$)

Let $\pi = \forall x \cdot \pi_{\text{rest}}$. The algorithm sets $u' = u \cdot x$ and recurses

We will show that:

$$\mathfrak{A} \models \forall u \forall x (\pi_{\text{rest}} \psi) \iff \mathfrak{A}' \models \forall u \forall x (\text{Skol}(\pi_{\text{rest}}, \psi, u'))$$

- $(\Rightarrow)$: Suppose $\mathfrak{A} \models \forall u \forall x (\pi_{\text{rest}} \psi) \implies \mathfrak{A} \models \forall (u \cup \{x\}) (\pi_{\text{rest}} \psi)$. There exists an expansion $\mathfrak{A}'$ such that $\mathfrak{A}' \models \forall (u \cup \{x\}) \text{Skol}(\pi_{\text{rest}}, \psi, u')$. rearrange the quantifiers gives $\mathfrak{A}' \models \forall u \forall x \text{Skol}(\dots)$.
- $(\Leftarrow)$: Suppose $\mathfrak{A}' \models \text{Skol}(\pi_{\text{rest}}, \psi, u')$. Since $\mathfrak{A}'$ is an expansion of $\mathfrak{A}$, and the universal quantifier $\forall x$ does not introduce new symbols in this step, same logic

Case 3: Existential Case ($q = \exists$)

□

Theorem 8.13 (Herbrand's Theorem). *Informally, if a first-order formula ϕ is satisfiable, then its satisfiability is witnessed by a special kind of structure called a Herbrand structure.*

Definition 8.14 (Herbrand structure). *Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{P})$ be a signature. A structure*

$$\mathfrak{A} = (U, I)$$

is a Herbrand structure if

1. *The universe is the Herbrand universe*

$$U = T_{\mathcal{C} \cup \mathcal{F}},$$

the set of all ground terms constructed from the constant symbols $\mathcal{C}$ and function symbols $\mathcal{F}$.

2. *The interpretation I is fixed syntactically:*

- *For every constant symbol $c \in \mathcal{C}$,*

$$I(c) = c$$

- *For every function symbol $f \in \mathcal{F}$ of arity n ,*

$$I(f)(t_1, \dots, t_n) = f(t_1, \dots, t_n)$$

Definition 8.15. *Ground term*

$$gt := c \mid f(gt, \dots)$$

Basically terms without variables

Definition 8.16. *Ground Atom (Let's ignore = first)*

$$ga := R(gt, \dots)$$

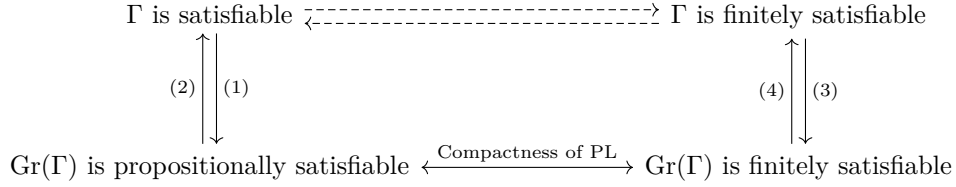
Definition 8.17 (Ground Formula).

$$gf := gt = gt \mid R(gt, \dots) \mid \neg gf \mid gf \vee gf$$

No variables, no quantifiers

Now that we have developed the machinery needed to prove Compactness for first-order logic, we outline the overall proof strategy. From the definition of grounding, it is clear that the goal is to connect first-order logic with propositional logic. By grounding, we eliminate variables and quantifiers, and treat each ground atom as a propositional atom. In this way, we translate the satisfiability problem for a first-order theory

into a corresponding satisfiability problem in propositional logic. This suggests the following proof plan: establish a correspondence between satisfiability and finite satisfiability for Γ and for its grounding $\text{Gr}(\Gamma)$, then apply Compactness of propositional logic to $\text{Gr}(\Gamma)$ in order to deduce Compactness for first-order logic.



Definition 8.18 (Ground instantiation). ϕ is a universally quantified sentence in PNF, let ψ be a ground formula, with the same signature. ψ is a ground instantiation of ϕ , or $\text{Gr}(\phi) = \psi$ if:

$$\phi = \forall x_1 \forall x_2, \dots, \forall x_n \cdot \gamma$$

There are n ground terms $gt_1, gt_2, \dots, gt_n$, such that

$$\psi = \gamma[x_1 \mapsto gt_1][x_2 \mapsto gt_2] \dots [x_n \mapsto gt_n]$$

Definition 8.19 (Ground instantiation of a set).

$$\text{Gr}(\Gamma) = \bigcup_{\phi \in \Gamma} \text{Gr}(\phi)$$

Lemma 8.20 (Ground instantiation of a formula is countable).

Corollary 8.21 (Ground instantiation of a set is countable).

Definition 8.22 (Ground valuation).

$$gv : ga_\Sigma \mapsto \{true, false\}$$

$$gv \models_{gr} R(t, \dots) \iff gv(R(t, \dots)) = true$$

$$gv \models_{gr} \neg gf \iff gv \not\models_{gr} gf$$

$$gv \models_{gr} gf_1 \vee gf_2 \iff gv \models_{gr} gf_1 \text{ OR } gv \models_{gr} gf_2$$

Definition 8.23 (Propositionally satisfiable). A ground formula gf is propositionally satisfiable if

$$\exists \tau \models_{gr} gf$$

Definition 8.24 (Propositionally satisfiable set). A set of ground formula Γ is propositionally satisfiable if there exist a ground atomic valuation τ such that

$$\forall \phi \in \Gamma, \tau \models_{Gr} \phi$$

8.3 Proof of the compactness theorem for FOL

Let's first consider the case without equality, the definition of ground formula becomes:

$$gf := R(ga, \dots) \mid \neg gf \mid gf \vee gf$$

8.3.1 Γ is satisfiable $\implies Gr(\Gamma)$ is propositionally satisfiable

Lemma 8.25 (Substitution Lemma for Ground Terms). *Let $\mathfrak{A} = (U, I)$ be a structure, ψ a formula, x a variable, and t a ground term. Then for any assignment α ,*

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff \mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}}(t)] \models \psi.$$

Proof. We have proven a more general case of substitution for FOL formula and terms □

Lemma 8.26 (Ground Instantiation for Universal Prefix). *Let $\mathfrak{A} = (U, I)$ be a structure and*

$$\phi = \forall x_1 \cdots \forall x_n \psi.$$

If $\mathfrak{A} \models \phi$, then for all ground terms $t_1, \dots, t_n$,

$$\mathfrak{A} \models \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n].$$

Proof. Assume $\mathfrak{A} \models \forall x_1 \cdots \forall x_n \psi$.

Let $t_1, \dots, t_n$ be arbitrary ground terms. Define

$$u_i = \text{Val}_{\mathfrak{A}}(t_i) \in U \quad (1 \leq i \leq n).$$

By the semantics of universal quantifiers,

$$\mathfrak{A} \models \forall x_1 \cdots \forall x_n \psi \implies \mathfrak{A}, \alpha[x_1 \mapsto u_1] \cdots [x_n \mapsto u_n] \models \psi$$

for every assignment α .

By the substitution lemma for ground terms,

$$\mathfrak{A}, \alpha \models \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n] \iff \mathfrak{A}, \alpha[x_1 \mapsto u_1] \cdots [x_n \mapsto u_n] \models \psi.$$

Hence,

$$\mathfrak{A}, \alpha \models \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n].$$

Since α was arbitrary,

$$\mathfrak{A} \models \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n].$$

□

Lemma 8.27 (Ground Semantic Equivalence). *Let $\mathfrak{A} = (U, I)$ and τ be defined as above. Then for every ground formula χ ,*

$$\mathfrak{A} \models \chi \iff \tau \models_{Gr} \chi.$$

Proof. Trivial by definition □

Lemma 8.28 (Ground Semantic Equivalence). *Let $\mathfrak{A} = (U, I)$ and τ be defined as above. Then for every ground formula χ ,*

$$\mathfrak{A} \models \chi \iff \tau \models_{Gr} \chi.$$

Proof. We prove the claim by structural induction on the ground formula χ .

Base case: χ is a ground atom. Then

$$\mathfrak{A} \models \chi \iff \tau \models_{Gr} \chi$$

by the Atomic Agreement Lemma.

Inductive case 1: $\chi = \neg \chi_1$ for some ground formula χ_1 .

By the semantics of negation,

$$\mathfrak{A} \models \neg \chi_1 \iff \mathfrak{A} \not\models \chi_1.$$

By the induction hypothesis,

$$\mathfrak{A} \not\models \chi_1 \iff \tau \not\models_{Gr} \chi_1.$$

Again by the semantics of negation,

$$\tau \not\models_{Gr} \chi_1 \iff \tau \models_{Gr} \neg \chi_1.$$

Hence

$$\mathfrak{A} \models \neg \chi_1 \iff \tau \models_{Gr} \neg \chi_1.$$

Inductive case 2: $\chi = \chi_1 \vee \chi_2$ for some ground formulas χ_1, χ_2 .

By the semantics of disjunction,

$$\mathfrak{A} \models \chi_1 \vee \chi_2 \iff (\mathfrak{A} \models \chi_1 \text{ or } \mathfrak{A} \models \chi_2).$$

By the induction hypothesis,

$$(\mathfrak{A} \models \chi_1 \text{ or } \mathfrak{A} \models \chi_2) \iff (\tau \models_{Gr} \chi_1 \text{ or } \tau \models_{Gr} \chi_2).$$

Again by the semantics of disjunction,

$$(\tau \models_{Gr} \chi_1 \text{ or } \tau \models_{Gr} \chi_2) \iff \tau \models_{Gr} \chi_1 \vee \chi_2.$$

Hence

$$\mathfrak{A} \models \chi_1 \vee \chi_2 \iff \tau \models_{Gr} \chi_1 \vee \chi_2.$$

Therefore, for every ground formula χ ,

$$\mathfrak{A} \models \chi \iff \tau \models_{Gr} \chi.$$

□

Lemma 8.29 (Grounding Preserves Truth). *Let Γ be a set of sentences (in Skolem form). If*

$$\mathfrak{A} \models \Gamma,$$

then

$$\mathfrak{A} \models \text{Ground}(\Gamma).$$

Lemma 8.30 (Transfer to Ground Valuation). *Let $\mathfrak{A} = (U, I)$ and τ be defined as above. If*

$$\mathfrak{A} \models \text{Ground}(\Gamma),$$

then

$$\tau \models_{Gr} \text{Ground}(\Gamma).$$

Proof. Γ is satisfiable $\implies$ there exists a structure $\mathfrak{A}$ such that

$$\mathfrak{A} \models \Gamma$$

Define ground valuation on atoms

$$\tau : \text{ga}_\Sigma \mapsto \{\text{true}, \text{false}\}$$

and for all ground atoms over Σ , define

$$\tau(a) = \text{true} \iff \mathfrak{A} \models a$$

Now we will show that: $\mathfrak{A} \models \phi \implies \tau \models_{Gr} \phi$

Define ground valuation τ as such:

$$\tau(R(t_1, \dots, t_n)) = \begin{cases} \text{true} & \text{if } (Val_{\mathfrak{A}}(t_1), \dots, Val_{\mathfrak{A}}(t_n)) \in I_{\mathfrak{A}}(R) \\ \text{false} & \text{otherwise} \end{cases}$$

Base case: ϕ is ground atom

By definition, trivial

Inductive steps: induction over the structure of formula(not gf)

- $\varphi = \neg\phi: \mathfrak{A} \models \neg\phi \implies \mathfrak{A} \not\models \phi.$
- $\varphi = \phi_1 \vee \phi_2$

□

8.4 Special FOL fragments

8.4.1 Quantifier-free FOL

Definition 8.31 (quantifier free formula).

$$\phi := t = t \mid R(t, \dots) \mid \neg\phi \mid \phi \vee \phi$$

All variables are free

Definition 8.32 (QT-free SAT). $QT\text{-}F\text{ SAT} = \{\phi \mid \phi \text{ is quantifier free, } \phi \text{ is satisfiable}\}$

Definition 8.33 (Small Model Property). A fragment $\mathbb{F}$ of logic is said to have the **Small Model Property** (SMP) if there exists a computable function $b_{\mathbb{F}} : \mathbb{F} \rightarrow \mathbb{N}$ such that for every formula $\phi \in \mathbb{F}$:

$$\text{If } \phi \text{ is satisfiable, then } \exists \mathfrak{A} \models \phi \text{ such that } |U_{\mathfrak{A}}| \leq b_{\mathbb{F}}(\phi)$$

We are interested in QT-free formula because it satisfy the SMP

Theorem 8.34. If $\mathbb{F}$ satisfy SM, the satiefiability of $\mathbb{F}$ is decidable.

Corollary 8.35. $FORM_{FOL}$ does not have SMP

counter example. Consider the folowing formula:

$$\phi \equiv \forall x, S(x) \neq 0 \wedge (x \neq y \rightarrow S(x) \neq S(y))$$

This basically describes a structure that has no right endpoint, and no 2 unique elements have the same successor. This requires an infinite (unbounded) structure, thus $FORM_{FOL}$ does not possess the SMP □

Proof. Notie that, for quantifier free formula, the set of ground insitantiations is finite.

$$\phi \in QT - freeSAT \iff skol(\phi) \in QT - freeSAT \wedge \phi \text{ is grounded}$$

As skolemization preserves satisfiability

□

8.5 Homomorphism and Isomorphism

Definition 8.36 (Homomorphism). Let $\Sigma = (C, F, R)$ be a First order signature. $\mathfrak{A}_1 = (U_1, I_1)$ and $\mathfrak{A}_2 = (U_2, I_2)$ be Σ -structures. A function

$$h : U_1 \mapsto U_2$$

is a homomorphism from $\mathfrak{A}_1$ to $\mathfrak{A}_2$ if:

1. $\forall c \in C, h(I_1(c)) = I_2(c)$
2. $\forall f^k \in F, \forall (u_1, \dots, u_k) \in U_1^k, h(I_1(f)(u_1, \dots, u_k)) = I_2(f)(h(u_1), \dots, h(u_k))$
3. $\forall R^k \in R, \forall (u_1, \dots, u_k) \in U_1^k, (u_1, \dots, u_k) \in I_1(R) \iff (h(u_1), \dots, h(u_k)) \in I_2(R)$

Definition 8.37 (Isomorphism). Let $\Sigma = (C, F, R)$ be a First order signature. $\mathfrak{A}_1 = (U_1, I_1)$ and

$\mathfrak{A}_2 = (U_2, I_2)$ be Σ -structures. A function

$$h : U_1 \mapsto U_2$$

is an isomorphism from $\mathfrak{A}_1$ to $\mathfrak{A}_2$ if:

1. h is a homomorphism from $\mathfrak{A}_1$ to $\mathfrak{A}_2$
2. h is a bijection

denoted as

$$\mathfrak{A}_1 \cong \mathfrak{A}_2$$

Theorem 8.38 (Isomorphism Theorem).

$$\mathfrak{A}_1, \alpha \models \phi \iff \mathfrak{A}_2, (h \circ \alpha) \models \phi$$

8.6 Theory

Definition 8.39 (Theory). A theory T is a set of sentences.

Example 8.40 (Partial order).

$$T_{po} = \begin{cases} \forall x \forall y \forall z (x < y \wedge y < z \rightarrow x < z) & (\text{transitivity}) \\ \forall x \neg (x < x) & (\text{irreflexivity}) \end{cases}$$

Definition 8.41 (T-satisfiable). Let T be a theory over Σ , a formula ϕ over Σ is T -satisfiable if

$$\exists \mathfrak{A}, \alpha, \mathfrak{A}, \alpha \models \phi$$

AND

$$\forall \psi \in T, \mathfrak{A} \models \psi$$

Definition 8.42 (T-validity). Let T be a theory over Σ , a formula ϕ over Σ is T -valid if

$$\forall \mathfrak{A}, \text{ if } \forall \psi \in T, \mathfrak{A} \models \psi \rightarrow \mathfrak{A} \models \phi$$

8.6.1 Common clauses used to define theories

- $\sigma_1 \equiv \forall x \neg (x < x) \wedge \forall x \forall y \forall z (x < y \wedge y < z \rightarrow x < z)$ (partial order)
- $\sigma_2 \equiv \forall x \forall y (x < y \vee x = y \vee y < x)$ (trichotomy)
- $\sigma_3 \equiv \forall x \forall y (x < y \rightarrow \exists z (x < z < y))$ (dense)
- $\sigma_4 \equiv \forall x \exists y \exists z (y < x < z)$ (no endpoints)
- $\varphi_1 \equiv \forall x \forall y \forall z ((x + y) + z = x + (y + z))$ (associativity)
- $\varphi_2 \equiv \forall x ((x + 0 = x = 0 + x) \wedge (x + (-x) = (-x) + x = 0))$ (additive identity & add. inverse axioms)
- $\varphi_3 \equiv \forall x \forall y (x + y = y + x)$ (commutativity)
- $\varphi_4 \equiv \forall x \forall y \forall z (x < y \rightarrow ((x + z) < (y + z)))$ (order-compatibility axiom for addition)
- $\psi_n \equiv \forall x (nx = 0 \rightarrow x = 0)$ (torsion-freeness)

Theories formed over above sentences include:

- $\mathfrak{A} \models \sigma_1 \iff \mathfrak{A}$ is a partial order
- $\mathfrak{A} \models \{\sigma_1, \sigma_2\} \iff \mathfrak{A}$ is a linear/total order

- $\mathfrak{A} \models \{\sigma_1, \sigma_2, \sigma_3, \sigma_4\} \iff \mathfrak{A}$ is a dense linear order without endpoints
- $\mathfrak{A} \models \{\varphi_1, \varphi_2\} \iff \mathfrak{A}$ is a group
- $\mathfrak{A} \models \{\varphi_1, \varphi_2, \varphi_3\} \iff \mathfrak{A}$ is an abelian group
- $\mathfrak{A} \models \{\varphi_1, \varphi_2, \varphi_3, \varphi_4, \sigma_1, \sigma_2\} \iff \mathfrak{A}$ is an ordered abelian group
- $\mathfrak{A} \models \{\varphi_1, \varphi_2, \varphi_3, \psi_n\}_{n \in \mathbb{N}} \iff \mathfrak{A}$ is a torsion-free abelian group

8.6.2 Monotonicity under theory inclusion

Lemma 8.43. *If $T_1 \subseteq T_2$, ϕ is T_1 -valid $\implies \phi$ is T_2 -valid. Or*

$$T_1 \subseteq T_2, T_1 \models \phi \implies T_2 \models \phi$$

Lemma 8.44. *If $T_1 \subseteq T_2$, ϕ is T_2 -sat $\implies \phi$ is T_1 -sat. Or*

$$T_1 \subseteq T_2, T_2 \cup \{\phi\} \text{ is satisfiable} \implies T_1 \cup \{\phi\} \text{ is satisfiable}$$

$$T_1 \subseteq T_2, T_1 \cup \{\phi\} \text{ is unsatisfiable} \implies T_2 \cup \{\phi\} \text{ is unsatisfiable}$$

8.6.3 Theory of a structure

Instead of a set of sentences, we can also define a theory with a structure.

Definition 8.45. *Let $\mathfrak{A}$ be a structure,*

$$Th(\mathfrak{A}) = \{\phi \mid \phi \text{ is a sentence, } \mathfrak{A} \models \phi\}$$

Definition 8.46 (Class of structures). *Let $\mathfrak{C}$ be a collection of structures,*

$$Th(\mathfrak{C}) = \bigcap_{\mathfrak{A} \in \mathfrak{C}} Th(\mathfrak{A})$$

Definition 8.47.

$$Th(\Gamma) = \{\phi \mid \phi \text{ is a sentence, } \Gamma \models \phi\}$$

Theorem 8.48.

$$Th(Th(\Gamma)) \subseteq Th(\Gamma)$$

Proof. Suppose otherwise, $\exists \phi \in Th(Th(\Gamma)), \phi \notin Th(\Gamma)$.

$$\phi \in Th(Th(\Gamma)) \implies Th(\Gamma) \models \phi$$

$$\phi \notin Th(\Gamma) \implies \Gamma \not\models \phi \implies \exists \mathfrak{A}, \mathfrak{A} \models \Gamma, \mathfrak{A} \not\models \phi$$

$$\mathfrak{A} \models Th(\Gamma), \text{ contradiction}$$

□

Theorem 8.49.

$$Th(\Gamma) \text{ is inconsistent} \implies Th(\Gamma) \models \text{all sentences}$$

Definition 8.50 (Consistency of a theory). *Theory T is consistent if*

$$\forall \phi, \neg(\{\phi\} \subseteq T \wedge \{\neg\phi\} \subseteq T)$$

Definition 8.51 (Completeness of a theory). *Theory T is complete if*

$$\forall \phi, \{\phi, \neg\phi\} \cap T \neq \emptyset$$

Lemma 8.52.

$Th(\mathfrak{A})$ is consistent

Proof. Suppose otherwise, $\phi, \neg\phi \in Th(\mathfrak{A})$. $\mathfrak{A} \models \phi \oplus \mathfrak{A} \models \neg\phi$, contradiction. $\square$

Lemma 8.53.

$Th(\mathfrak{A})$ is complete

Proof. Let ϕ be a sentence,

- if $\mathfrak{A} \models \phi, \phi \in Th(\mathfrak{A})$
- if $\mathfrak{A} \not\models \phi, \mathfrak{A} \models \neg\phi, \neg\phi \in Th(\mathfrak{A})$

$\square$

Lemma 8.54.

$Th(\mathfrak{C})$ is consistent

Proof. Suppose otherwise, $\exists \phi, \phi, \neg\phi \in Th(\mathfrak{C})$

$$\forall \mathfrak{A} \in \mathfrak{C}, \mathfrak{A} \models \phi \oplus \mathfrak{A} \models \neg\phi$$

contradiction $\square$

Lemma 8.55. *$Th(\mathfrak{C})$ is not complete.*

Proof. A theory T is complete if for every sentence ψ , either $T \models \psi$ or $T \models \neg\psi$. To show $Th(\mathfrak{C})$ is incomplete, we can find a sentence ϕ such that $Th(\mathfrak{C}) \not\models \phi$ and $Th(\mathfrak{C}) \not\models \neg\phi$.

Let $\phi \equiv \exists x \exists y, R(x, y)$. Consider the following two structures in $\mathfrak{C}$:

1. Structure $\mathfrak{A} = (U_{\mathfrak{A}}, I_{\mathfrak{A}})$:

- $U_{\mathfrak{A}} = \{0, 1\}$
- $I_{\mathfrak{A}}(R) = \{(0, 1)\}$

In this structure, $R(0, 1)$ holds, therefore $\mathfrak{A} \models \phi$. This implies $Th(\mathfrak{C}) \not\models \neg\phi$.

2. Structure $\mathfrak{B} = (U_{\mathfrak{B}}, I_{\mathfrak{B}})$:

- $U_{\mathfrak{B}} = \{0, 1\}$
- $I_{\mathfrak{B}}(R) = \emptyset$

In this structure, no pair (x, y) satisfies R , therefore $\mathfrak{B} \models \neg\phi$. This implies $Th(\mathfrak{C}) \not\models \phi$.

Since there exists a structure in $\mathfrak{C}$ that satisfies ϕ and another that satisfies $\neg\phi$, the theory $Th(\mathfrak{C})$ contains neither. Thus, $Th(\mathfrak{C})$ is not complete. $\square$

Theorem 8.56. *Let T be a theory, $T \in RE$. If T is consistent and complete, $T \in REC$*

Proof. Let E be an enumerator for T . To decide if $\psi \in T$:

1. Start E to produce sentences $\sigma_1, \sigma_2, \sigma_3, \dots$
2. For each σ_i , check if $\sigma_i = \psi$ or $\sigma_i = \neg\psi$.
3. By completeness, one of these must eventually appear.
4. If ψ appears, accept. If $\neg\psi$ appears, reject.

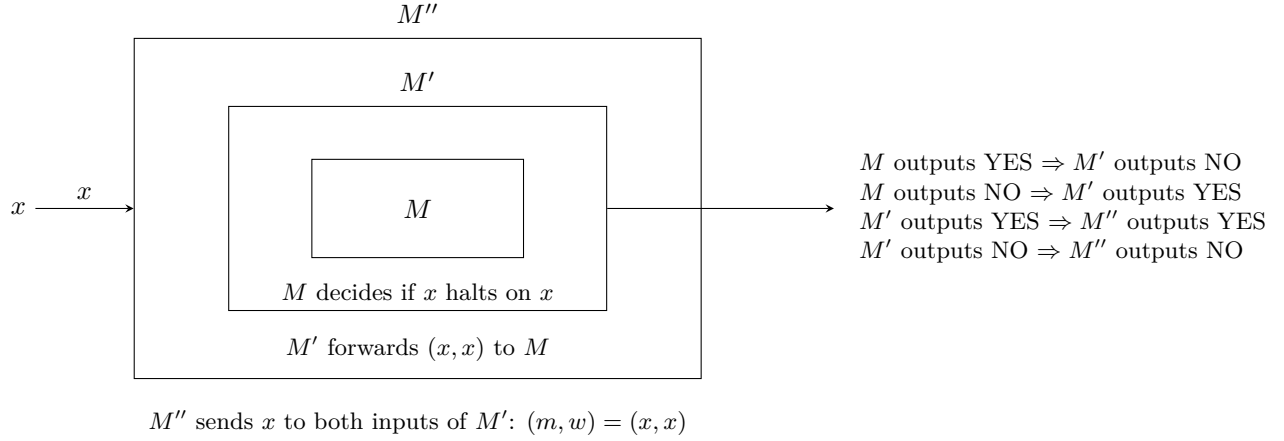
Since the algorithm always halts, $T \in REC$. $\square$

9 Computability

9.1 Halting problem

Theorem 9.1. *The halting problem is undecidable*

Proof.



Now consider $M''(M'')$.

Case 1: $M''(M'') = NO$

□

Definition 9.2 (Recursively Enumerable). $L \subseteq \Sigma^*$ is recursively enumerable (RE), if there is a Turing machine $\mathcal{M}$, such that $L = L(\mathcal{M})$. Other equivalent terms for an RE language are:

- Semi-decidable
- Turing recognizable

Definition 9.3 (Recursive). $L \subseteq \Sigma^*$ is a recursive (REC) language if there is a Turing machine $\mathcal{M}$, such that $L = L(\mathcal{M})$, and $\mathcal{M}$ is terminating. Recursive languages are also called decidable

Theorem 9.4.

$$L_H = \{\langle \mathcal{M}, w \rangle \mid w \text{ terminates on } \mathcal{M}\} \in RE$$

Proof. We can construct a TM $\mathcal{M}_{Halt}$ which is the turing machine that returns YES if its input machine terminates on input string.

```

 $\mathcal{M}_{Halt}(\mathcal{M}, w)$ :
   $\mathcal{M}(w)$ 
  return YES
  
```

This suffice to show that $L_H \in RE$

$L_H \notin REC$

Similar to how we prove undecidability of the Halting problem

□

Theorem 9.5 (Membership).

$$MP = \{\langle \mathcal{M}, w \rangle \mid w \text{ is accepted by } \mathcal{M}\} \in RE$$

Proof. We can construct TM

```

 $\mathcal{M}_{UTM}(\mathcal{M}, w):$ 
   $b = \mathcal{M}(w)$ 
  return  $b$ 

```

□

Quick Reference: Reduction Recipes

If you are stuck on the construction logic, these are the standard "wrappers" for M' :

1. **To show X is "Large" (e.g., Non-Empty):** Construct $M'(x)$ to simulate $M(w)$. If it halts, **Accept**. *Result:* $L(M')$ is either Σ^* (if M halts) or $\emptyset$ (if it loops).
2. **To show X is "Small" (e.g., Empty or Finite):** To reduce $\overline{L_{Halt}}$ to L_{Empty} , use the same construction as above. *Logic:* M loops on $w \iff L(M') = \emptyset \iff \langle M' \rangle \in L_{Empty}$.

Goal: Show X is...	Hierarchy	Source (A)	Requirement
Undecidable	$\notin \Delta_1$	L_{Halt}	$A \leq_m X$
Not in RE	$\notin \Sigma_1$	$\overline{L_{Halt}}$	$A \leq_m X$
Not in co-RE	$\notin \Pi_1$	L_{Halt}	$A \leq_m X$
Not in RE $\cup$ co-RE	$\notin \Sigma_1 \cup \Pi_1$	L_{Total}	$A \leq_m X$

Table 1: Strategic mapping for complexity proofs.

3. **To involve all strings (e.g., Totality):** Construct $M'(x)$ to simulate $M(w)$ for exactly $|x|$ steps. This is often used for more complex Π_2 proofs.

The "Direction" Rule

Always remember: Hard $\leq$ New.

- You are "borrowing" the difficulty of the known language to prove the unknown one is just as bad.
- If the reduction f maps "Yes" instances to "Yes" instances and "No" to "No", the complexity flows from the source to the target.

Theorem 9.6.

$$L \in REC \iff \bar{L} \in REC$$

Proof. $L \in REC \implies \exists \mathcal{M}, L = L(\mathcal{M}), \mathcal{M}$ is terminating, we can construct another TM

```

 $\mathcal{M}'(w):$ 
   $b = \mathcal{M}(w)$ 
  return  $\neg b$ 

```

It is easy to show that $L(\mathcal{M}') = \bar{L}$, and $\mathcal{M}'$ is terminating.

□

Theorem 9.7.

$$L \in RE \wedge \bar{L} \in RE \implies L \in REC$$

Proof. Consider then TM for L and $\bar{L}$, $\mathcal{M}_1$ and $\mathcal{M}_2$, we can construct a new TM

```

 $\mathcal{M}(w):$ 
  for  $i > 0:$ 
    run the  $i^{th}$  step of  $\mathcal{M}_1$ 
    if  $\mathcal{M}_1$  reaches  $q_{\mathcal{M}_1, Acc}$ 
      return YES

```

```

run the  $i^{th}$  step of  $\mathcal{M}_2$ 
if  $\mathcal{M}_2$  reaches  $q_{\mathcal{M}_2, Acc}$ 
    return NO

```

Since $\forall w, \forall L \subseteq \Sigma^*, w \in L \vee w \in \bar{L}$, w must be accepted by either $\mathcal{M}_1$ or $\mathcal{M}_2$, therefore $\mathcal{M}$ is terminating. $\square$

9.2 Reduction

Definition 9.8. $L_1 \leq_m L_2$ if there is a computable function f , such that

$$\begin{aligned} w \in L_1 &\implies f(w) \in L_2 \\ w \notin L_1 &\implies f(w) \notin L_2 \end{aligned}$$

Theorem 9.9.

$$L_1 \leq_m L_2, L_2 \in REC \implies L_1 \in REC$$

Theorem 9.10.

$$L_1 \leq_m L_2, L_2 \in RE \implies L_1 \in RE$$

Proof. $\square$

Theorem 9.11.

$$L_1 \leq_m L_2, L_1 \in REC \not\implies L_2 \in REC$$

Proof. Consider the following recursive language:

$$L_1 = \{\langle \mathcal{M}, w \rangle \mid w \text{ terminates on } \mathcal{M} \text{ in 10 steps}\}$$

and

$$L_H = \{\langle \mathcal{M}, w \rangle \mid w \text{ terminates on } \mathcal{M}\}$$

We have shown that $L_H \in RE \setminus REC$. We can construct the following terminating TM to define a computable function f

```

 $\mathcal{M}_f(\langle \mathcal{M}, w \rangle)$ :
  run  $\mathcal{M}_1$  on  $w$  for 10 steps
  if output is YES:
    return  $\langle \mathcal{M}, w \rangle$ 
  else:
    return  $\langle NH, w \rangle$ 

```

Where NH is the encoding of a TM that loops forever on all input.

It is easy to show that

$$\begin{aligned} \langle \mathcal{M}, w \rangle \in L_1 &\implies f(\langle \mathcal{M}, w \rangle) \in L_H \\ \langle \mathcal{M}, w \rangle \notin L_1 &\implies f(\langle \mathcal{M}, w \rangle) \notin L_H \end{aligned}$$

But $L_H \notin REC$ as we have previously shown. Contradiction $\square$

9.3 RE-hardness

There are languages beyond the RE class. For example

Theorem 9.12.

$$\bar{L}_H \notin RE$$

Proof. Suppose otherwise, $\bar{L}_H \in RE, L_H \in RE \implies L_H \in REC$, contradiction $\square$

Definition 9.13 (RE-hard). L is RE-hard, if

$$\forall L' \in RE, L' \leq_m L$$

Example 9.14.

$$MP = \{\langle M, w \rangle \mid w \in L(M)\}$$

$$L_{Halt} \leq_m MP$$

Proof. We construct a computable function $f(\langle M, w \rangle) = \langle M', w \rangle$. The machine M' is defined as follows:

$M'(x)$:

Simulate M on input w
Accept // Only reached if $M(w)$ halts

Clearly, M halts on $w \iff M'$ accepts x (for any x). Thus $\langle M, w \rangle \in L_{Halt} \iff \langle M', w \rangle \in MP$. $\square$

Similarly reduction can show some languages are not in RE, if they are at least as hard as a known not in RE language

Example 9.15. $\bar{MP} \notin RE$

Proof by Reduction: $\bar{K} \leq_m \bar{MP}$

Given $\bar{K} = \{\langle M \rangle \mid \langle M \rangle \notin L(M)\}$ is not in RE. Define a computable transformation $f(\langle M \rangle) = \langle M', \epsilon \rangle$, where M' is:

$M'(x)$:

Simulate M on input $\langle M \rangle$
Accept // Only reached if M accepts $\langle M \rangle$

1. If $\langle M \rangle \in \bar{K}$, then M does not accept $\langle M \rangle$. M' will loop/reject on the simulation, so $L(M') = \emptyset$. Thus, $\epsilon \notin L(M') \implies \langle M', \epsilon \rangle \in \bar{MP}$.
2. If $\langle M \rangle \notin \bar{K}$, then M accepts $\langle M \rangle$. M' will reach the "Accept" command for any input. Thus, $\epsilon \in L(M') \implies \langle M', \epsilon \rangle \notin \bar{MP}$.

Conclusion: $\langle M \rangle \in \bar{K} \iff f(\langle M \rangle) \in \bar{MP}$.

9.4 Validity of FOL

Definition 9.16.

$$VALID : \{\varphi \mid \varphi \text{ is valid} \}$$

Theorem 9.17. $VALID \in RE$

Theorem 9.18 (Church-Turing Theorem). $VALID$ is RE-hard

10 Second Order Logics

10.1 Syntax of Second Order Logics

in second order logics, terms are defined the same as in FOL.

$$\Sigma = (C, F, R)$$

$$V = FOV \cup SOV$$

where FOV is the set containing variables in FOL, and SOV contains the relation variables, for example

$$SOV = \{X^2, Y^1, \dots\}$$

where the superscripts denote the arity of these relation variable.

Definition 10.1.

$$\varphi := t = t \mid R^k(t_1, \dots, t_k) \mid X^k(t_1, \dots, t_k) \mid \neg\varphi \mid \varphi \vee \varphi \mid \exists x, \varphi \mid \forall x, \varphi \mid \exists X, \varphi \mid \forall X, \varphi$$

Example 10.2.

$$\phi \equiv \forall x, \exists X^1, X(x)$$

This formula means for all x in the universe, there exist such a unary relation, such that x is in the relation. This formula is valid as we can always interpret $X = U$.

Semantics of SOL

Similar to FOL, the semantics of SOL is defined by a structure and an assignment (with free variables)

$$\mathfrak{A} = (U, I)$$

Definition 10.3. for FOL variable:

$$\alpha(x) \in U$$

for SOL variables:

$$\alpha(X^k) \subseteq U^k$$

Similar to FOL, model relation is defined recursively:

$$\begin{aligned} \mathfrak{A}, \alpha \models X^k(t_1, \dots, t_k) &\iff Val_{\mathfrak{A}, \alpha}(t_1), \dots, Val_{\mathfrak{A}, \alpha}(t_k) \in \alpha(X) \\ \mathfrak{A}, \alpha \models \exists X^k, \phi &\iff \text{there is a set of } k\text{-ary tuples of } U, S \subseteq U^k \\ &\text{such that } \mathfrak{A}, \alpha[X \mapsto S] \models \phi \\ \mathfrak{A}, \alpha \models \forall X^k, \phi &\iff \text{for all sets of } k\text{-ary tuples of } U, S \subseteq U^k \\ &\text{such that } \mathfrak{A}, \alpha[X \mapsto S] \models \phi \end{aligned}$$

10.2 Expressiveness of SOL

Definition 10.4. Let ϕ be a sentence (no free variables), we can define the set of structures that models ϕ as

$$\llbracket \phi \rrbracket = \{\mathfrak{A} \mid \mathfrak{A} \models \phi\}$$

We have proven by compactness of FOL, it is impossible to express the set of all finite/infinite structures with FOL formula. However we can do so in SOL.

10.2.1 Finite structures

Theorem 10.5. There exist a SOL formula ϕ_{fin} such that

$$\llbracket \phi_{fin} \rrbracket = \{\mathfrak{A} \mid |U_{\mathfrak{A}}| < \infty\}$$

We need to first find some unique property of finite sets

Theorem 10.6. Let S be a set, S is finite $\iff \forall f : S \mapsto S$ that are injective, f is also surjective

Definition 10.7 (structure classes defined by non-sentence). For formula ϕ , where $FVar(\phi) = \{\bar{x}\}$

$$\llbracket \phi(\bar{x}) \rrbracket = \{(\mathfrak{A}, \bar{e} \in U^k) \mid \mathfrak{A}, \alpha[\bar{x} \mapsto \bar{e}] \models \phi\}$$

We can first use SOL to define function-like relations.

Definition 10.8. A binary relation $S^2 \subseteq U^2$ is function-like, if

1. $\forall u \in U, \exists v, (u, v) \in S$
2. $\forall u, v_1, v_2, (u, v_1) \in S \wedge (u, v_2) \in S \rightarrow (v_1 = v_2)$

This is a literal translation of all elements in domain has a value to be mapped to, and no element maps to 2 different elements

These constraints are be expressed with SOL with free variable:

$$\begin{aligned}\phi_{func}(X^2) &\equiv \forall x \exists y X(x, y) \wedge \forall x \forall y \forall z (X(x, y) \wedge X(x, z) \rightarrow (y = z)) \\ \phi_{injection}(X^2) &\equiv \phi_{func}(X^2) \wedge \forall x \forall y \forall z (X(x, y) \wedge X(y, z) \rightarrow (x = y)) \\ \phi_{surjection}(X^2) &\equiv \phi_{func}(X^2) \wedge \forall y \exists x X(x, y)\end{aligned}$$

Therefore, we can use a SOL formula to express finite structures:

$$\phi_{fin} \equiv \forall X^2 (\phi_{injection}(X^2) \rightarrow \phi_{surjection}(X^2))$$

Follow the same logic as how we proved that it is impossible to express such concept in FOL due to compactness, we can prove that in SOL, compactness does not hold.

10.2.2 Reachability

Consider the following signature:

$$\Sigma = (\{s, t\}, \emptyset, \{E^2\})$$

$$\mathfrak{C}_{s \rightsquigarrow t} = \{\mathfrak{A} \mid I(s) \text{ has a path using } I(E) \text{ to } T(t)\}$$

We have shown this is not expressable in FOL. To express this concept, we need to characterize reachability as a property, which is a binary relation that is

1. reflexive
2. transitive

$$\begin{aligned}\phi_{reflexive}(X^2) &\equiv \forall x X(x, x) \\ \phi_{transitive}(X^2) &\equiv \forall x \forall y \forall z X(x, y) \wedge X(y, z) \rightarrow X(x, z) \\ \llbracket \phi_{s \rightsquigarrow t} \rrbracket &\equiv \left\{ (\mathfrak{A} = (U, I), R \subseteq U^2) \mid \begin{array}{l} \exists w_0, \dots, w_k \in U, \text{ such that:} \\ (s, w_0) \in I(E), \\ \forall i \in \{0, \dots, k-1\}, (w_i, w_{i+1}) \in I(E), \\ (w_k, t) \in I(E) \end{array} \right\}\end{aligned}$$

However we can artificially inflate the set to make it still satisfy the semantics of the formula, but contain more pairs that are not connected. We can use the same technique used to define the set of WFF, which is the **smallest** set.

Definition 10.9. The set of reachable pairs $\mathfrak{R}$ is the **smallest** set of pairs that are

1. $E \subseteq S$
2. transitive and reflexive

First we define the concept of closure

$$\phi_{closed}(X^2) \equiv \phi_{reflexive}(X^2) \wedge \phi_{transitive}(X^2) \wedge E \subseteq X$$

the concept of E subsetting X can also be written in SOL as a shorthand

$$\subseteq (X^2) \equiv \forall x \forall y E(x, y) \rightarrow X(x, y)$$

Then we can formally define the relation reachable:

$$\phi_{reach}(X^2) \equiv \phi_{closed}(X^2) \wedge \forall Y^2 (\phi_{closed}(Y^2) \rightarrow X^2 \supseteq Y^2)$$

10.2.3 Cyclic graph

A cyclic graph is a graph that contains at least one cycle.

Irreflexive reachability can be defined similarly as reachability, but without the reflexive clause.

$$\begin{aligned}\phi_{irreflexive_closed} &= \phi_{transitive}(X^2) \wedge E \subseteq X \\ \phi_{irreflexive_reach}(X^2) &\equiv \phi_{irreflexive_closed} \wedge \forall Y^2 (\phi_{irreflexive_closed}(Y^2) \rightarrow X^2 \supseteq Y^2) \\ \phi_{cyclic} &\equiv \exists x \exists X^2 (\phi_{irreflexive_reach}(X^2) \wedge X(x, x))\end{aligned}$$

Alternatively we can use reachability and additional constraints to enforce reflexivity:

$$\exists x \exists y (\exists X^2 \phi_{reach}(X^2) \wedge E(x, y) \wedge E(y, x))$$

10.2.4 Even

All structures that have finite and even number of elements in the universe

$$\begin{aligned}\phi_{disjoint}(X_1, X_2) &\equiv \forall x X_1(x) \leftrightarrow \neg(X_2(x)) \\ \phi_{exhaustive}(X_1, X_2) &\equiv \forall x X_1(x) \vee X_2(x)\end{aligned}$$

Before we defined a formula to describe relations that are injection-function-like, now we define a formula that describe, there exists a binary relation, between the sets defined by 2 relations.

$$\begin{aligned}\phi_{injection}(X, Y) &\equiv \exists F \left[(\forall x : X(x) \rightarrow \exists y : (Y(y) \wedge F(x, y))) \wedge \right. \\ &\quad (\forall x, y_1, y_2 : X(x) \wedge Y(y_1) \wedge Y(y_2) \wedge F(x, y_1) \wedge F(x, y_2) \rightarrow y_1 = y_2) \wedge \\ &\quad \left. (\forall x_1, x_2, y : X(x_1) \wedge X(x_2) \wedge Y(y) \wedge F(x_1, y) \wedge F(x_2, y) \rightarrow x_1 = x_2) \right] \\ \phi_{surjection}(X, Y) &\equiv \exists F \left[(\forall x : X(x) \rightarrow \exists y : (Y(y) \wedge F(x, y))) \wedge \right. \\ &\quad (\forall x, y_1, y_2 : X(x) \wedge Y(y_1) \wedge Y(y_2) \wedge F(x, y_1) \wedge F(x, y_2) \rightarrow y_1 = y_2) \wedge \\ &\quad \left. (\forall y : Y(y) \rightarrow \exists x : (X(x) \wedge F(x, y))) \right]\end{aligned}$$

The first and second lines define totality and function-like property.

$$\begin{aligned}\phi_{bijection}(X_1, X_2) &\equiv \phi_{injection}(X_1, X_2) \wedge \phi_{surjection}(X_1, X_2) \\ \phi_{even} &\equiv \exists X_1^1 \exists X_2^1 (\phi_{disjoint}(X_1, X_2) \wedge \phi_{bijection}(X_1, X_2) \wedge \phi_{exhaustive}(X_1, X_2))\end{aligned}$$

10.2.5 Bipartite graph

$$\begin{aligned}\phi_{bipartite} &\equiv \exists X_1^1, X_2^1 \left[\phi_{disjoint}(X_1, X_2) \wedge \phi_{exhaustive}(X_1, X_2) \wedge \right. \\ &\quad \left. \forall x, y (E(x, y) \rightarrow \neg(X_1(x) \wedge X_1(y)) \wedge \neg(X_2(x) \wedge X_2(y))) \right]\end{aligned}$$

We can observe that n-partite graphs are exactly thos graphs that are n-colorable, they are identical structures

10.2.6 Connectedness

A graph is connected if every pair of vertices has a path between them. Since you have already defined $\phi_{reach}(X^2)$, we can use it to define a connected graph.

$$\phi_{connected} \equiv \exists X^2 (\phi_{reach}(X^2) \wedge \forall u \forall v X(u, v))$$

Alternatively, we can express it by stating that there is no non-trivial partition of the graph that "cuts" all edges:

$$\phi_{connected} \equiv \forall X^1 ((\exists x X(x) \wedge \exists y \neg X(y)) \rightarrow \exists u \exists v (X(u) \wedge \neg X(v) \wedge E(u, v)))$$

10.2.7 Well-Ordering

In FOL, we can define a total order, but we cannot express that a relation is a well-order (every non-empty set has a least element) because we cannot quantify over sets. In SOL, this is straightforward: Let $\phi_{total}(R^2)$ be the FOL formula for a total order.

$$\phi_{well_order}(R^2) \equiv \phi_{total}(R^2) \wedge \forall X^1 (\exists x X(x) \rightarrow \exists m (X(m) \wedge \forall y (X(y) \rightarrow R(m, y))))$$

10.2.8 Hamiltonian Graphs

A graph is Hamiltonian if it contains a cycle that visits every vertex exactly once. While this property is non-expressible in FOL, it is easily captured in SOL. By Fagin's Theorem, this existential second-order formula (Σ_1^1) also serves as a proof that the problem is in NP. We define a binary relation H^2 to act as a successor relation. The formula $\phi_{\text{hamiltonian}}$ is defined as:

$$\phi_{\text{hamiltonian}} \equiv \exists H^2 \left[\underbrace{\phi_{\text{inj}}(H^2) \wedge \phi_{\text{surj}}(H^2)}_{\text{Permutation}} \wedge \underbrace{\forall x \forall y (H(x, y) \rightarrow E(x, y))}_{\text{Sub-graph of } E} \right. \\ \left. \wedge \underbrace{\forall X^1 ((\exists x X(x) \wedge \forall u \forall v (X(u) \wedge H(u, v) \rightarrow X(v))) \rightarrow \forall x X(x))}_{\text{Connectivity (Induction Axiom)}} \right]$$

Logical Breakdown: Permutation: $\phi_{\text{inj}} \wedge \phi_{\text{surj}}$ ensures that every vertex has exactly one incoming edge and one outgoing edge in H . This forces H to be a collection of one or more disjoint cycles. Sub-graph: Ensures that H only uses edges that actually exist in the original graph E . Induction Axiom: This is the "secret sauce." It states that any set X closed under the successor relation H must be the entire universe U . This prevents the graph from being split into multiple smaller disjoint cycles (sub-cycles), forcing a single cycle that covers all vertices.

10.2.9 Well-Foundedness

A relation E is well-founded if there are no infinite descending chains $x_1 \leftarrow x_2 \leftarrow x_3 \dots$. This is equivalent to saying every non-empty set has an E -minimal element.

$$\phi_{\text{well_founded}} \equiv \forall X^1 (\exists x X(x) \rightarrow \exists m (X(m) \wedge \forall y (X(y) \rightarrow \neg E(y, m))))$$

10.2.10 Graph Isomorphism

Another powerful use of SOL is expressing whether two graphs are isomorphic. Given two graphs $\mathfrak{A} = (U_A, E_A)$ and $\mathfrak{B} = (U_B, E_B)$, they are isomorphic if there exists a bijection $f : U_A \rightarrow U_B$ that preserves the edge relation.

$$\phi_{\text{iso}} \equiv \exists F^2 \left[\phi_{\text{bijection}}(F^2) \wedge \forall u \forall v (E_A(u, v) \leftrightarrow \exists u' \exists v' (F(u, u') \wedge F(v, v') \wedge E_B(u', v'))) \right]$$

HW1

Definition 10.10 (Generic Closure). Let U be a universe of elements and let O be a set of functions whose domain is U and codomain is $\mathcal{P}(U)$. The arity of a function $f \in O$ is denoted by $\text{arity}(f)$. A function f of arity r has type

$$f : U^r \rightarrow \mathcal{P}(U).$$

Let $I \subseteq U$ be an initial set. A set $S \subseteq U$ is said to be (I, O) -closed if:

1. $I \subseteq S$.
2. For every $f \in O$ with $\text{arity}(f) = r$ and for all $e_1, \dots, e_r \in U$, if $e_1, \dots, e_r \in S$, then $f(e_1, \dots, e_r) \subseteq S$.

Lemma 10.11. Let U be a universe, O a set of operators, and $I \subseteq U$ an initial set. There exists a smallest (I, O) -closed set.

Define $\mathcal{P}(U)$ as a collection of all ordering of all elements in the power set $\mathcal{P}(U)$

Q1

Claim: The smallest (I, O) -closed set can be constructed by taking the intersection of all (I, O) -closed set, ie.

$$AIOCS = \{S \subseteq U \mid S \text{ is } (I, O)\text{-closed}\}$$

$$T_0 = \bigcap AIOCS$$

Existence

First we prove the existence of such a set.

Proof. Suppose otherwise,

$$T_0 = \bigcap AIOCS = \emptyset$$

$$T_0 = \{x \in \mathcal{P}(U) \mid \forall S \in AIOCS, x \in S\}$$

$$T_0 = \emptyset \rightarrow x \in S \text{ is vacuously true } \forall x \in \mathcal{P}(U) \rightarrow T_0 = \mathcal{P}(U)$$

contradiction. □

Closed

Now we prove that T_0 is (I, O) -closed.

Proof. Property 1:

$$\forall S \in AIOCS, I \subseteq S$$

$$\rightarrow \forall i \in I, \forall S \in AIOCS, i \in S$$

$$\rightarrow \forall i \in I, i \in \bigcap AIOCS$$

$$\rightarrow \forall i \in I, i \in T_0$$

Property 2:

$$\forall S \in AIOCS, T_0 \subseteq S$$

$$\text{Let } e_1, e_2, \dots, e_r \in T_0, \forall S \in AIOCS, e_1, e_2, \dots, e_r \in S$$

$$\rightarrow \exists f \in O, f(e_1, e_2, \dots, e_r) \in S$$

$$\rightarrow \forall e_1, e_2, \dots, e_r \in T_0, \exists f \in O, f(e_1, e_2, \dots, e_r) \in \bigcap AIOCS$$

$$\rightarrow f(e_1, e_2, \dots, e_r) \in T_0$$

□

Uniqueness

Claim:

$$\forall T'_0 \in \{T \in \mathcal{P}(U) \mid \forall S \in AIOCS, T \subseteq S\}, T'_0 = T_0$$

Proof. Suppose otherwise, $\exists T'_0 \in AIOCS, T'_0 \neq T_0$.

$$\begin{aligned} T_0, T'_0 &\in AIOCS \rightarrow \forall S \in AIOCS, T_0 \subseteq S, T'_0 \subseteq S \\ &\rightarrow T_0 \subseteq T'_0, T'_0 \subseteq T_0 \\ &\rightarrow T_0 = T'_0 \end{aligned}$$

□

Q2

For each of the following strings, prove or disprove that it is a well-formed formula:

1. $w_1 = \epsilon$ (the empty string).
2. $w_2 = (\neg p)$ where $p \in P$.
3. $w_3 = (p\neg)$ where $p \in P$.

2.1

$$w_1 = \epsilon \notin FORM$$

Claim: $\forall w \in FORM, |w| > 0$ I'd like to use definition 2.19 and lemma 2.21 which I proved later without using the conclusion from this question.

$$FORM = INDFORM = \bigcup_{i \geq 0} FORM_i$$

Proof. Base case:

$$\begin{aligned} FORM_0 &= P \\ \forall \phi \in FORM_0, \phi &= p \in P \\ &\rightarrow \forall \phi \in FORM_i, |\phi| > 0 \end{aligned}$$

Inductive Hypothesis: Suppose

$$\forall i \geq 0, \forall \phi \in FORM_i, |\phi| > 0$$

$$\begin{aligned} FORM_{i+1} &= FORM_i \cup \{(\neg\phi) \mid \phi \in FORM_i\} \cup \{(\phi_1 \vee \phi_2) \mid \phi_1, \phi_2 \in FORM_i\} \\ \forall \psi \in FORM_{i+1}, \exists \phi \in FORM_i, |\psi| &= |\phi| \text{ or } |\psi| = |\phi| + 3 \text{ or} \\ \exists \phi_1, \phi_2 \in FORM_i, |\psi| &= |\phi_1| + |\phi_2| + 3 \end{aligned}$$

□

2.2

$$p \in P \rightarrow p \in FORM \rightarrow (\neg p) \in FORM$$

2.3

$$(p\neg) \notin FORM$$

Proof. Suppose otherwise, $(p\neg) \in FORM$

$$\begin{aligned} \forall \phi \in FORM, (\neg\phi) &\in FORM \\ &\rightarrow \epsilon \in FORM \end{aligned}$$

Contradiction from 2.1

□

Q3

Fix $U = \mathbb{R}$. For each of the following sets S , define a finite initial set $I \subseteq \mathbb{R}$ and a set of operators O with $|O| < 10$ such that the smallest (I, O) -closed set is S .

1. $S = \{0, 1, 2, 3, \dots\}$.
2. $S = \{1, 3, 5, 7, \dots\}$.
3. $S = \mathbb{Q}$.
4. $S = \{\frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \frac{5}{3}, \frac{7}{3}, \dots\}$.

Answer:

1. $\mathbb{N} : I = \{0\}, O = \{f(x) = x + 1\}$
2. $Odd : I = \{1\}, O = \{f(x) = x + 2\}$
3. $\mathbb{Q} : I = \{0\}, O = \{f(x) = -x, f(x) = x + 1, f(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ \emptyset & \text{if } y = 0 \end{cases}\}$
4. $\{\frac{x}{3} \mid x \in \mathbb{N}\} \setminus \mathbb{N} : I = \{\frac{1}{3}, \frac{2}{3}\}, O = \{f(x) = x + 1\}$

Finite Directed Graphs

Definition 10.12 (Graph Reachability). *Let $G = (V, E)$ be a directed graph. A path from s to t is a sequence of vertices $\pi = u_1, \dots, u_k$ such that:*

1. $u_1 = s$,
2. $(u_i, u_{i+1}) \in E$ for all $1 \leq i < k$,
3. $u_k = t$.

The length of π is $k - 1$. We say that t is reachable from s if $s = t$ or such a path exists. The set of ancestors of t is defined as

$$Reach_t = \{v \in V \mid t \text{ is reachable from } v\}.$$

Q4

Let $G = (V, E)$ be a directed graph and let $t \in V$. Fix the universe $U = V$.

1. Define an initial set I (with $|I| \leq 2$) and a unary operator f such that the smallest $(I, \{f\})$ -closed set is the set of vertices reachable from t .
2. Let $d \in \mathbb{N} \setminus \{0\}$ and $r < d$. Define I and a unary operator f such that the smallest $(I, \{f\})$ -closed set is the set of vertices reachable from s via paths of length ℓ with $\ell \equiv r \pmod{d}$.

4.1

$$I = \{t\}, O = \{f(v) = succ(v)\}$$

4.2

$$I = \{s\} \cup \{v \in V \mid v \in succ^r(s)\}, O = \{f(v) = succ^d(v)\}$$

Q5

Let $G = (V, E)$ be a directed graph and fix $U = V \times V$. Define an initial set I with $|I| \leq |V|$ and a binary operator f such that the smallest $(I, \{f\})$ -closed set is

$$AllReachPairs = \{(u, v) \mid v \text{ is reachable from } u\}.$$

5.1

$$I = \{(v, v) \mid v \in V\}, O = \{f((v_1, v_2), (v_3, v_4)) \mapsto \begin{cases} \{(v_1, v_4)\} & \text{if } v_2 = v_3 \\ \emptyset & \text{if } v_2 \neq v_3 \end{cases}\}$$

Inductive Definitions

Definition 10.13 (Inductive Well-Formed Formulae). *Let P be a set of propositions and $C = \{\neg, \vee\}$. Define sets $FORM_i$ as follows:*

$$FORM_0 = P,$$

$$FORM_{i+1} = FORM_i \cup \{(\neg w) \mid w \in FORM_i\} \cup \{(w_1 \vee w_2) \mid w_1, w_2 \in FORM_i\}.$$

Define

$$INDFORM = \bigcup_{i \geq 0} FORM_i.$$

Q6

1. Show that $FORM_i \subseteq FORM_{i+1}$ for all i .
2. Prove or disprove that $FORM_i = FORM_{i+1}$ for some i .
3. Prove or disprove that $FORM_i$ is (P, C) -closed for all i .
4. Prove or disprove that $FORM_i$ is (P, C) -closed for some i .

6.1

Base case: $FORM_0 \subseteq FORM_1$ as

$$FORM_1 = FORM_0 \cup \{(\neg w) \mid w \in FORM_0\} \cup \{(w_1 \vee w_2) \mid w_1, w_2 \in FORM_0\}$$

Wait this is just trivial

6.2

Claim:

$$\exists i \geq 0, FORM_i = FORM_{i+1}$$

It is obvious that $\forall i \geq 0, FORM_i$ is finite. Suppose

6.3

Claim:

$$\forall i \geq 0, FORM_i \text{ is } (P, C)\text{-closed}$$

Proof. Suppose otherwise, consider $FORM_0$,

$$FORM_0 = P$$

$$\forall p \in P, (\neg p) \in FORM_0$$

Contradiction

□

6.4

Claim:

$$\exists i \geq 0, FORM_i \text{ is (P,C)-closed}$$

Lemma 10.14.

$$\forall i \geq 0, \forall \phi \in FORM_i, |\phi| \leq 2^{i+3} - 3$$

The longest formula in $FORM_i$ has size $2^{i+3} - 3$. Parentheses are counted in string length

Proof. Base case:

$$\begin{aligned} FORM_0 &= P \\ \forall \phi \in FORM_0, |\phi| &= 1 \end{aligned}$$

Inductive hypothesis:

$$\forall i \geq 0, \forall \phi \in FORM_i, |\phi| \leq 2^{i+3} - 3$$

Consider all new strings formed from the longest string $\phi \in FORM_i$,

$$\begin{aligned} |\phi| &= 2^{i+3} - 3 \\ |(\neg\phi)| &= 2^{i+3} \\ |(\phi \vee \phi)| &= 2 \times (2^{i+3} - 3) + 3 = 2^{(i+1)+3} - 3 \end{aligned}$$

□

Proof. Suppose $\exists i \geq 0, FORM_i$ is (P,C)-closed, and from lemma 2.20 the longest formula in $FORM_i$ has size $|\phi_{max}| = 2^{i+3} - 3$. $\rightarrow \forall \phi \in FORM_i, (\neg\phi) \in FORM_i \rightarrow \phi'_{max} = (\neg\phi_{max}) \in FORM_i, |\phi'_{max}| > 2^{i+3} - 3$, a contradiction.

□

Lemma 10.15. $FORM = INDFORM$.

Q7

Prove the lemma by showing:

1. $INDFORM$ is (P, C) -closed.
2. For every (P, C) -closed set S , $INDFORM \subseteq S$.

7.1

Claim: $INDFORM$ is (P,C)-closed

Proof. **Claim:** $P \subseteq INDFORM$

First condition of (P,C)-closed set:

$$FORM_0 \subseteq INDFORM \rightarrow P \subseteq INDFORM$$

Claim: $\forall w \in INDFORM, (\neg w) \in INDFORM$

$$\begin{aligned} INDFORM &= \bigcup_{i \geq 0} FORM_i \\ &\rightarrow \forall i \geq 0, FORM_i, FORM_{i+1} \in INDFORM \\ &\quad \forall w \in INDFORM, \exists j \geq 0, w \in FORM_j \\ &\rightarrow (\neg w) \in FORM_{j+1} \rightarrow (\neg w) \in INDFORM \end{aligned}$$

Claim: $\forall w_1, w_2 \in \text{INDFORM}, (w_1 \vee w_2) \in \text{INDFORM}$

Case 1: $\exists i \geq 0, w_1, w_2 \in \text{FORM}_i$

$$w_1, w_2 \in \text{FORM}_i \rightarrow (w_1 \vee w_2) \in \text{FORM}_{i+1} \rightarrow (w_1 \vee w_2) \in \text{INDFORM}$$

Case 2: $\nexists i \geq 0, w_1, w_2 \in \text{FORM}_i$. WLOG, suppose $w_1 \in \text{FORM}_i, w_2 \in \text{FORM}_j, i < j$

$$\forall w \in \text{FORM}_n, w \in \text{FORM}_{n+1} \text{ By induction } w_1 \in \text{FORM}_i \rightarrow w_1, w_2 \in \text{FORM}_j$$

Back to case 1. □

7.2

$$\forall (\text{P,C})\text{-closed set } S, \text{INDFORM} \subseteq S$$

To prove this, we only need to prove that $\text{INDFORM} \subseteq S_{\min} = \bigcap_S \text{ is } (\text{P,C})\text{-closed } S$.

Proof. Base case:

$$\bigcup_{i=0}^0 \text{FORM}_i = \text{FORM}_0 = P \subseteq S_{\min}$$

Inductive hypothesis: Suppose

$$\bigcup_{i=0}^n \text{FORM}_i \subseteq S_{\min}$$

$$\bigcup_{i=0}^n \text{FORM}_i \subseteq S_{\min} \rightarrow \text{FORM}_n \in S_{\min}$$

$$S_{\min} \text{ is } (\text{P,C})\text{-closed} \rightarrow \forall i \in [1, n], \forall \phi \in \text{FORM}_n, (\neg \phi) \in S_{\min},$$

$$\forall i \in [0, n], \forall \phi_1, \phi_2 \in \text{FORM}_i, (\phi_1 \vee \phi_2) \in S_{\min}$$

$$\rightarrow \bigcup_{i=0}^n \text{FORM}_i \cup \{(\neg \phi) \mid \forall \phi \in \bigcup_{i=0}^n \text{FORM}_i\} \cup \{(\phi_1 \vee \phi_2) \mid \forall \phi_1, \phi_2 \in \bigcup_{i=0}^n \text{FORM}_i\} \subseteq S_{\min}$$

$$\rightarrow \bigcup_{i=0}^{n+1} \text{FORM}_i \subseteq S_{\min}$$

□

Definition 10.16 (Alternative Inductive Definition). Define $\text{AltFORM}_0 = P$ and for $i \geq 0$:

$$\begin{aligned} \text{AltFORM}_{i+1} = & \{(\neg w) \mid w \in \text{AltFORM}_i\} \\ & \cup \{(w_1 \vee w_2) \mid w_1 \in \text{AltFORM}_i, \exists j \leq i, w_2 \in \text{AltFORM}_j\} \\ & \cup \{(w_1 \vee w_2) \mid w_2 \in \text{AltFORM}_i, \exists j \leq i, w_1 \in \text{AltFORM}_j\}. \end{aligned}$$

Define

$$\text{AltINDFORM} = \bigcup_{i \geq 0} \text{AltFORM}_i.$$

Question 8

1. Show that $\text{FORM}_i = \bigcup_{j=0}^i \text{AltFORM}_j$ for all i .
2. Show that $\text{INDFORM} = \text{AltINDFORM}$.

8.1

Lemma 10.17.

Proof. Base case:

$$FORM_0 = \bigcup_{j=0}^0 AltFORM_j$$

$$FORM_0 = P = AltFORM_0$$

Inductive hypothesis: Suppose

$$\forall j \in [0, i], FORM_i = \bigcup_{k=0}^i AltFORM_k$$

$$\begin{aligned} FORM_{i+1} &= \bigcup_{k=0}^i AltFORM_k \cup \{(\neg\phi) \mid \phi \in \bigcup_{k=0}^i AltFORM_k\} \cup \{(\phi_1 \vee \phi_2) \mid \phi_1, \phi_2 \in \bigcup_{k=0}^i AltFORM_k\} \\ &\supseteq \bigcup_{k=0}^i AltFORM_k \cup \{(\neg\phi) \mid \phi \in AltFORM_i\} \cup \{(\phi_1 \vee \phi_2) \mid \phi_1, \phi_2 \in \bigcup_{k=0}^i AltFORM_k\} \end{aligned}$$

$$\bigcup_{k=0}^{i+1} AltFORM_k \subseteq FORM_{i+1}$$

Now we prove the other way:

$$\forall \phi \in FORM_{i+1}, \phi \in \bigcup_{k=0}^i AltFORM_k \text{ or } \phi = (\neg\psi), \psi \in \bigcup_{k=0}^i AltFORM_k \text{ or } \phi = (\psi_1 \vee \psi_2), \psi_1, \psi_2 \in \bigcup_{k=0}^i AltFORM_k$$

Case 1:

$$\phi \in \bigcup_{k=0}^i AltFORM_k \rightarrow \phi \in \bigcup_{k=0}^{i+1} AltFORM_k$$

Case 2:

$$\begin{aligned} \psi \in \bigcup_{k=0}^i AltFORM_k &\rightarrow \exists j \leq i, \psi \in AltFORM_j \\ \phi = (\neg\psi), \psi \in AltFORM_j, \phi &\in AltFORM_{j+1} \\ &\rightarrow \phi \in \bigcup_{k=0}^{i+1} AltFORM_k \end{aligned}$$

Case 3:

$$\begin{aligned} \psi_1, \psi_2 \in \bigcup_{k=0}^i AltFORM_k &\rightarrow \exists j, l \leq i, \psi_1 \in AltFORM_j, \psi_2 \in AltFORM_l \\ \phi = (\psi_1 \vee \psi_2), \psi_1 \in AltFORM_j, \psi_2 &\in AltFORM_l, \phi \in AltFORM_{j+1} \\ &\rightarrow \phi \in \bigcup_{k=0}^{i+1} AltFORM_k \\ FORM_{i+1} &\subseteq \bigcup_{k=0}^{i+1} AltFORM_k \rightarrow FORM_{i+1} = \bigcup_{k=0}^{i+1} AltFORM_k \end{aligned}$$

□

Formation Sequences

Definition 10.18 (Formation Sequence). A (P, C) -formation sequence is a nonempty finite sequence $\rho = w_1, \dots, w_k$ such that for each i :

- $w_i \in P$, or
- $w_i = (\neg w_j)$ for some $j < i$, or
- $w_i = (w_j \vee w_k)$ for some $j, k < i$.

The sequence belongs to w if $w_k = w$.

Q9

Give two distinct formation sequences that belong to

$$w = ((\neg p_1) \vee ((\neg p_3) \vee (\neg(p_2 \vee (\neg p_1)))))$$

Two distinct formation sequences ρ_1, ρ_2 are:

ρ_1 :

$$\begin{aligned} w_1 &= p_1 \\ w_2 &= (\neg w_1) \\ w_3 &= p_2 \\ w_4 &= (w_3 \vee w_2) \\ w_5 &= (\neg w_4) \\ w_6 &= (p_3) \\ w_7 &= (\neg w_6) \\ w_8 &= (w_7 \vee w_5) \\ w_9 &= (w_2 \vee w_8) \end{aligned}$$

ρ_2 :

$$\begin{aligned} w_1 &= p_1 \\ w_2 &= p_2 \\ w_3 &= p_3 \\ w_4 &= (\neg w_3) \\ w_5 &= (\neg w_1) \\ w_6 &= (w_2 \vee w_5) \\ w_7 &= (\neg w_6) \\ w_8 &= (w_4 \vee w_7) \\ w_9 &= (w_5 \vee w_8) \end{aligned}$$

These are distinct because the components at index $i = 2$ differ $((\neg p_1)$ vs p_2).

Lemma 10.19. $INDFORM = \{w \in \Sigma^* \mid w \text{ is formation-sequence-well-formed}\}$.

Q10

Prove the lemma by showing:

1. Every formation-sequence-well-formed string is in $INDFORM$.

2. Every $w \in \text{INDFORM}$ has a formation sequence.

Before attempting Q10 from the above definition of formation-sequence-well-formed, I will prove the following lemma using the definition below first:

Definition 10.20. Define $\text{ExFORM}_0 = P$

$$\begin{aligned} \text{ExFORM}_{i+1} = & \{(\neg w) \mid w \in \text{ExFORM}_i\} \\ & \cup \{(w_1 \vee w_2) \mid w_1 \in \text{ExFORM}_i, \exists j \leq i, w_2 \in \text{ExFORM}_j\} \\ & \cup \{(w_1 \vee w_2) \mid w_2 \in \text{ExFORM}_i, \exists j \leq i, w_1 \in \text{ExFORM}_j\} \\ &) \setminus \bigcup_{j=0}^i \text{ExFORM}_j \end{aligned}$$

Lemma 10.21.

$$\bigcup_{j=0}^i \text{ExFORM}_j = \text{FORM}_i$$

Proof. Base case:

$$\text{ExFORM}_0 = P = \text{FORM}_0$$

Inductive Hypothesis: Suppose $\forall n \geq 0$,

$$\bigcup_{i=0}^n \text{ExFORM}_i = \text{FORM}_n$$

From definition 2.19,

$$\begin{aligned} \text{FORM}_{n+1} = & \bigcup_{i=0}^n \text{ExFORM}_i \cup \{(\neg w) \mid w \in \bigcup_{i=0}^n \text{ExFORM}_i\} \\ & \cup \{(w_1 \vee w_2) \mid w_1 \in \bigcup_{i=0}^n \text{ExFORM}_i, \exists j \leq i, w_2 \in \bigcup_{i=0}^n \text{ExFORM}_j\} \\ & \cup \{(w_1 \vee w_2) \mid w_2 \in \bigcup_{i=0}^n \text{ExFORM}_i, \exists j \leq i, w_1 \in \bigcup_{i=0}^n \text{ExFORM}_j\} \end{aligned}$$

Definition 10.22.

$$\begin{aligned} \text{FromFORM} : n \mapsto & \{(\neg w) \mid w \in \bigcup_{i=0}^n \text{ExFORM}_i\} \\ & \cup \{(w_1 \vee w_2) \mid w_1 \in \bigcup_{i=0}^n \text{ExFORM}_i, \exists j \leq i, w_2 \in \bigcup_{i=0}^n \text{ExFORM}_j\} \\ & \cup \{(w_1 \vee w_2) \mid w_2 \in \bigcup_{i=0}^n \text{ExFORM}_i, \exists j \leq i, w_1 \in \bigcup_{i=0}^n \text{ExFORM}_j\} \end{aligned}$$

Lemma 10.23.

$$\text{FORM}_{n+1} = \bigcup_{i=0}^n \text{ExFORM}_i \cup \text{FromFORM}(n)$$

Lemma 10.24.

$$ExFORM_{n+1} = \left(\bigcup_{i=0}^n ExFORM_i \cup FromFORM(n) \right) \setminus \bigcup_{i=0}^n ExFORM_i$$

$$ExFORM_{n+1} = FromFORM(n) \setminus \bigcup_{i=0}^n ExFORM_i$$

From Lemma 10.23 and Lemma 10.24, we have

$$\begin{aligned} \bigcup_{i=0}^{n+1} ExFORM_i &= \bigcup_{i=0}^n ExFORM_i \cup ExFORM_{n+1} \\ &= \bigcup_{i=0}^n ExFORM_i \cup (FromFORM(n) \setminus \bigcup_{i=0}^n ExFORM_i) \\ &= \bigcup_{i=0}^n ExFORM_i \cup FromFORM(n) \\ &= FORM_{n+1} \end{aligned}$$

□

Lemma 10.25.

$$\bigcup_{i \geq 0} ExFORM_i = \bigcup_{i \geq 0} FORM_i$$

Proof. $\bigcup_{i \geq 0} ExFORM_i \subseteq \bigcup_{i \geq 0} FORM_i$

$\forall n \geq 0$, From Lemma 2.28

$$\phi \in \bigcup_{i=0}^n ExFORM_i \rightarrow \phi \in FORM_n \rightarrow \phi \in \bigcup_{i=0}^n FORM_i$$

$$\bigcup_{i \geq 0} ExFORM_i \supseteq \bigcup_{i \geq 0} FORM_i$$

Corollary from Lemma 2.28

□

Corollary 10.26.

$$\bigcup_{i \geq 0} ExFORM_i = INDFORM$$

$\forall \phi \in ExFORM_i \rightarrow \phi \text{ is well-formed formula}$

10.1, 2

Claim:

$$FORMSeq = \{w \in \Sigma^* \mid w \text{ is formation-sequence-well-formed}\} = INDFORM$$

Proof. Let ρ_i denotes the set of ρ that contains a sequence of i strings $w_1, w_2, \dots, w_i$

Define

$$\begin{aligned} str : \rho &\mapsto w, str(\rho = w_1, w_2, \dots, w_k) = w_k \\ str : \rho_i &\mapsto \{w\}, str(\rho = w_1, w_2, \dots, w_k) = \{w \mid w = str(\rho), \rho \in \rho_i\} \end{aligned}$$

Claim:

$$\bigcup_{k \geq 1} str(\rho_k) = \bigcup_{i \geq 0} ExFORM_i$$

Proof. $\bigcup_{k \geq 1} \rho_k \subseteq \bigcup_{i \geq 0} ExFORM_i$

$\forall w \in FORMSeq, \exists \rho = w_1, w_2, \dots, w_k, w = w_k$

Base case:

$$\bigcup_{k=1}^1 str(\rho_1) = \{w \mid w \in P\} = \bigcup_{i=0}^0 ExFORM_i$$

Inductive hypothesis: Suppose

$$\bigcup_{j=1}^{k+1} str(\rho_j) \subseteq \bigcup_{i=0}^k ExFORM_i$$

$$\bigcup_{j=1}^{k+2} str(\rho_j) = \bigcup_{j=1}^{k+1} str(\rho_j) \cup \{w \mid \exists \rho \in \rho_{k+2}, w = str(\rho)\}$$

By definition, $w_{k+2} = str(\rho)$ for some $\rho \in \rho_{k+2}$, one of the 3 below holds:

1. $w_{k+2} \in P \rightarrow w_{k+2} \in \bigcup_{i \geq 0} ExFORM_i$
2. $w_{k+2} = (\neg w_n), n < k+2 \rightarrow w_n \in \bigcup_{j=1}^n str(\rho_j) \subseteq \bigcup_{i=0}^k ExFORM_i$
3. $w_{k+2} = (w_m \vee w_n), m, n < k+2 \rightarrow w_m, w_n \in \bigcup_{j=1}^{\max(m,n)} str(\rho_j) \subseteq \bigcup_{i=0}^k ExFORM_i$

$$\bigcup_{k \geq 1} \rho_k \supseteq \bigcup_{i \geq 0} ExFORM_i$$

Base case is the same as above.

Inductive hypothesis: (this will be used in Q11)

Lemma 10.27.

$$\bigcup_{i=0}^k ExFORM_i \subseteq \bigcup_{j=1}^{2(k+1)} str(\rho_j)$$

$\forall w \in ExFORM_{k+1}$, one of the 3 below holds: 1.

$$w \in P \rightarrow w \in ExFORM_0 \rightarrow w \in \bigcup_{i=0}^k ExFORM_i \subseteq \bigcup_{j=1}^{k+1} str(\rho_j)$$

2.

$$\begin{aligned} w = (\neg w_n), w_n \in ExFORM_k &\rightarrow \exists \rho = w_1, w_2, \dots, w_k \in \bigcup_{j=1}^{k+1} \rho_j, w_n = str(\rho) \\ &\rightarrow \exists \rho' \in \rho_{k+2}, \rho' = w_1, w_2, \dots, w_k, (\neg w_n) \\ &\rightarrow w \in str(\rho_{k+2}) \rightarrow w \in \bigcup_{j=1}^{k+2} str(\rho_j) \\ &\rightarrow w \in \bigcup_{j=1}^{2(k+1)} str(\rho_j) \end{aligned}$$

3. (Concat finite sequences that yield formula composing disjunctions gives finite sequence)

$$\begin{aligned}
w &= (w_m \vee w_n), WLOG, w_m \in ExFORM_k, \exists q \leq k, w_n \in ExFORM_q \\
&\rightarrow \exists \rho^m \in \rho_{k+1} = w_1^m, w_2^m, \dots, w_k^m, \\
&\quad \rho^n \in \rho_{q+1} = w_1^n, w_2^n, \dots, w_q^n \in \bigcup_{j=1}^{k+1} \rho_j, w_m = str(\rho^m), w_n = str(\rho^n) \\
&\rightarrow \exists \rho' = w_1^m, w_2^m, \dots, w_k^m, w_1^n, w_2^n, \dots, w_q^n, (w_k^m \vee w_q^n) \in \rho_{k+q+2} \\
&\rightarrow w \in \bigcup_{j=1}^{k+q+1} str(\rho_j) \\
&\rightarrow w \in \bigcup_{j=1}^{2(k+1)} str(\rho_j)
\end{aligned}$$

Therefore,

$$\bigcup_{k \geq 1} str(\rho_k) = \bigcup_{i \geq 0} ExFORM_i$$

□

By Corollary 10.26,

$$\bigcup_{k \geq 1} str(\rho_k) = INDFORM$$

□

Q11

Lemma 10.28. *Every well-formed formula ϕ is of one of the following forms:*

1. $\phi \in P$,
2. $\phi = (\neg\psi)$,
3. $\phi = (\psi_1 \vee \psi_2)$.

Proof. From Corollary 10.26,

$$\forall \phi \in INDFORM, \exists k \geq 0, \phi \in ExFORM_k$$

Case $x = 0$:

$$x = 0 \rightarrow \phi \in ExFORM_0 = P$$

Case $x > 0$:

$$\begin{aligned}
&\phi \in ExFORM_x \text{ and } \forall y \in [0, x), \phi \notin ExFORM_y \\
&\forall \phi \in ExFORM_x, \phi \in P \text{ or} \\
&\phi \in (\{(\neg\psi) \mid \psi \in ExFORM_{x-1}\} \\
&\quad \cup \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_{x-1}, \exists j \leq x-1, \psi_2 \in ExFORM_j\})
\end{aligned}$$

Since we define $ExFORM_i$ to be pair wise disjoint, we guarantee that

$$\phi \in ExFORM_k \iff k \text{ is the minimum depth needed to form } \phi \text{ in the syntax tree}$$

Intuitively, x is the minimum depth required to formulate ϕ , thus with depth $d \in [0, x)$, ϕ cannot be formed
Suppose $\phi \in P$,

$$\phi \in ExFORM_0, \phi \in ExFORM_x, x > 0$$

Contradiction. Therefore

$$\phi \in (\{(\neg\psi) \mid \psi \in ExFORM_{x-1}\} \cup \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_{x-1}, \exists j \leq x-1, \psi_2 \in ExFORM_j\})$$

Now we will prove $\forall \phi \in INDFORM$

$$(\phi \in \{(\neg\psi) \mid \psi \in ExFORM_{x-1}\}) \oplus (\phi \in \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_{x-1}, \exists j \leq x-1, \psi_2 \in ExFORM_j\})$$

Suppose we enforce the use of parentheses for all subformula, that is except $\phi \in P$, all formula starts with '('. If for some i , $\phi \in \{(\neg\psi) \mid \psi \in ExFORM_i\}$, ' $\neg$ ' must be at index 1. Likewise, if for some i , $\phi \in \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_i, \exists j \leq i, \psi_2 \in ExFORM_j\}$, ' $\vee$ ' must not be at index 1. Thus $\nexists \phi \in INDFORM, \phi \in \{(\neg\psi) \mid \psi \in ExFORM_{x-1}\}$ and $\phi \in \{(\psi_1 \vee \psi_2) \mid \psi_1 \in ExFORM_i, \exists j \leq i, \psi_2 \in ExFORM_j\}$ $\square$

HW2

Lemma 10.29 (Characterization of Well-Formed Formulae). *Let P be a set of propositions and $C = \{\neg, \vee\}$. Every well-formed formula ϕ over (P, C) is of exactly one of the following forms:*

1. $\phi \in P$,
2. $\phi = (\neg\psi)$ for some well-formed formula ψ ,
3. $\phi = (\psi_1 \vee \psi_2)$ for some well-formed formulae ψ_1, ψ_2 .

Structural Induction Principle. A property $P(\phi)$ holds for all well-formed formulae if:

- $P(p)$ holds for all $p \in P$,
- $P(\psi)$ implies $P(\neg\psi)$,
- $P(\psi_1)$ and $P(\psi_2)$ imply $P(\psi_1 \vee \psi_2)$.

Q1

1.1

Theorem 10.30. *Let α be a well-formed formula. Let $c(\alpha)$ denote the number of occurrences of binary connective symbols ($\wedge, \vee, \rightarrow, \leftrightarrow$) in α , and let $s(\alpha)$ denote the number of occurrences of sentence symbols in α . Then*

$$s(\alpha) = c(\alpha) + 1.$$

Proof. **Base Case:**

Let $\alpha = p$ for some proposition $p \in \mathcal{P}$.

- $s(p) = 1$ and $c(p) = 0$.
- Since $1 = 0 + 1$, $P(p)$ holds for all atomic propositions.

Inductive Case 1 (Negation):

Suppose $P(\psi)$ holds, i.e., $s(\psi) = c(\psi) + 1$. Let $\alpha = (\neg\psi)$.

- $s(\neg\psi) = s(\psi)$ no new sentence symbols are added.
- $c(\neg\psi) = c(\psi)$ $\neg$ not binary connective.
- Thus, $s(\alpha) = s(\psi) = c(\psi) + 1 = c(\alpha) + 1$. $P(\alpha)$ holds.

Inductive Case 2 (Binary Connectives):

Suppose $P(\psi_1)$ and $P(\psi_2)$ hold. Let $\alpha = (\psi_1 \star \psi_2)$ where $\star \in \{\vee, \rightarrow\}$.

- $s(\alpha) = s(\psi_1) + s(\psi_2)$ (sum of occurrences in sub-formulae).
- $c(\alpha) = c(\psi_1) + c(\psi_2) + 1$ (sum of binary connectives plus the new $\star$).
- By the inductive hypothesis: $s(\alpha) = (c(\psi_1) + 1) + (c(\psi_2) + 1)$.
- Rearranging gives $s(\alpha) = (c(\psi_1) + c(\psi_2) + 1) + 1$.
- Therefore, $s(\alpha) = c(\alpha) + 1$. $P(\alpha)$ holds.

□

1.2

We define the property $P(\alpha)$ as follows: the length of α , denoted $L(\alpha)$, is of the form $4k + 1$ and the number of sentence symbols, $s(\alpha)$, is $k + 1$ for some $k \in \mathbb{N}$

Proof. **Base Case:**

Let $\alpha = p$ for some proposition $p \in \mathcal{P}$

- The length $L(p) = 1 = 4 \cdot 0 + 1$
- The number of sentence symbols $s(p) = 1$
- Thus, $P(p)$ holds.

Inductive Case:

Suppose $P(\psi_1)$ and $P(\psi_2)$ hold Then $L(\psi_1) = 4k_1 + 1$ and $s(\psi_1) = k_1 + 1$. Also $L(\psi_2) = 4k_2 + 1$ and $s(\psi_2) = k_2 + 1$. Let $\alpha = (\psi_1 c \psi_2)$ where c is a binary connective

Length of α :

- The string α consists of: one left parenthesis, ψ_1 , one binary connective, ψ_2 , and one right parenthesis
- $L(\alpha) = 1 + L(\psi_1) + 1 + L(\psi_2) + 1$.
- $L(\alpha) = 3 + (4k_1 + 1) + (4k_2 + 1) = 4k_1 + 4k_2 + 5$.
- $L(\alpha) = 4(k_1 + k_2 + 1) + 1$.
- Let $k = k_1 + k_2 + 1$, we see $L(\alpha) = 4k + 1$, odd

Number of sentence symbols:

- $s(\alpha) = s(\psi_1) + s(\psi_2) = (k_1 + 1) + (k_2 + 1) = k_1 + k_2 + 2$.
- Using $k = k_1 + k_2 + 1$ from above, $s(\alpha) = (k_1 + k_2 + 1) + 1 = k + 1$
- $k + 1 = \frac{4k+4}{4} > \frac{4k+1}{4}$

□

Q2

Recall the following derived connectives:

- $\phi_1 \wedge \phi_2 \equiv \neg((\neg\phi_1) \vee (\neg\phi_2))$,
- $\perp \equiv \neg(p \vee \neg p)$ for some $p \in P$,
- $\phi_1 \rightarrow \phi_2 \equiv (\neg\phi_1) \vee \phi_2$.

Using structural induction, show that for every formula ϕ over $(P, \{\neg, \vee\})$ with $P \neq \emptyset$, there exists a formula ψ using only connectives $\{\perp, \rightarrow\}$ such that ϕ and ψ are logically equivalent.

Proof. We will show that every formula formed over $\{\neg, \vee\}$ can be formed using $\{\perp, \rightarrow\}$, and denote the logically equivalent formula using the later connector set from ϕ as ψ Define property $P(\phi)$ as follows: there exists a logically equivalent formula over P using the connector set $\{\perp, \rightarrow\}$.

Base Case:

- $\forall \phi = p \in P, \psi = p$

Inductive Case 1:

Suppose $P(\phi)$ for some formula, $\exists \psi \equiv \phi$

- $(\neg\phi) = (\neg\phi) \vee ((\neg\phi) \vee (\neg\phi)) = \phi \rightarrow \perp = \psi \rightarrow \perp$

Inductive Case 2:

Suppose $P(\phi_1), P(\phi_2)$ for some formula, $\exists \psi_1 \equiv \phi_1, \psi_2 \equiv \phi_2$

- $(\phi_1 \vee \phi_2) = ((\neg(\neg\phi_1)) \vee \phi_2) = ((\neg\phi_1) \rightarrow \phi_2)$
- From inductive case 1, $((\neg\phi_1) \rightarrow \phi_2) = ((\phi_1 \rightarrow \perp) \rightarrow \phi_2) = ((\psi_1 \rightarrow \perp) \rightarrow \psi_2)$

□

Q3

Definition 10.31 (Conjunctive Normal Form). *A formula ϕ over $(P, \{\neg, \vee, \wedge\})$ is in conjunctive normal form (CNF) if*

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m \quad (m \geq 1),$$

where each clause C_i has the form

$$C_i = \ell_{i,1} \vee \ell_{i,2} \vee \cdots \vee \ell_{i,k_i},$$

and each literal $\ell_{i,j}$ is either p or $\neg p$ for some $p \in P$.

Using structural induction, show that for every well-formed formula ϕ , there exists a logically equivalent formula ψ in CNF.

Proof. Define property $P(\phi)$ to be there exists a CNF ψ that is logically equivalent to ϕ

Base Case:

$\phi = p$. An atomic proposition p is a clause containing a single literal. A single clause is, by definition, a CNF. Thus, $P(p)$ holds.

Inductive Case 1 $\neg$:

Assume $P(\phi)$ holds. If ϕ has a CNF, we can first convert $\neg\phi$ into NNF https://en.wikipedia.org/wiki/Negation_normal_form

- $\neg(A \wedge B) \equiv \neg A \vee \neg B$
- $\neg(A \vee B) \equiv \neg A \wedge \neg B$
- $\neg\neg A \equiv A$

Inductive Case 2 $\vee$:

Assume $P(\phi_1)$ and $P(\phi_2)$ hold. Let $\psi_1 = \bigwedge_i C_i$ and $\psi_2 = \bigwedge_j D_j$ be CNFs.

- $\phi_1 \vee \phi_2 \equiv (\bigwedge_i C_i) \vee (\bigwedge_j D_j)$.
- By the Distributive Law $(A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C))$, we distribute the $\vee$ over the $\wedge$:
- $\phi_1 \vee \phi_2 \equiv \bigwedge_i \bigwedge_j (C_i \vee D_j)$.
- Since the disjunction of two clauses $C_i \vee D_j$ is also a clause, the final expression is a CNF.

□

Q4

Definition 10.32 (Occurs). *Define $OCCUR : \text{FORM} \rightarrow \mathcal{P}(P)$ inductively by:*

- If $\phi = p \in P$, then $OCCUR(\phi) = \{p\}$,
- If $\phi = \neg\psi$, then $OCCUR(\phi) = OCCUR(\psi)$,
- If $\phi = \psi_1 \oplus \psi_2$ with $\oplus \in \{\wedge, \vee\}$, then $OCCUR(\phi) = OCCUR(\psi_1) \cup OCCUR(\psi_2)$.

Lemma 10.33 (Relevance Lemma). *Let ϕ be a propositional formula and let $v_1, v_2 : P \rightarrow \{\text{true}, \text{false}\}$ be valuations such that $v_1(p) = v_2(p)$ for all $p \in \text{OCCUR}(\phi)$. Then $v_1 \models \phi$ if and only if $v_2 \models \phi$.*

Prove this lemma using structural induction.

Proof. We define the property $P(\phi)$ as: for any v_1, v_2 , if $v_1(p) = v_2(p) \forall p \in \text{OCCUR}(\phi)$, $v_1 \models \phi \iff v_2 \models \phi$.

Base Case:

$\phi = p$ for some $p \in P$

- $\text{OCCUR}(p) = \{p\}$, $v_1(p) = v_2(p)$
- $v_1 \models p \iff v_1(p) = \text{true}$ and $v_2 \models p \iff v_2(p) = \text{true}$
- $v_1(p) = v_2(p)$, $v_1 \models p \iff v_2 \models p$

Inductive Step 1 (negation):

Assume $P(\psi)$ holds. Let $\phi = \neg\psi$.

- $\text{OCCUR}(\phi) = \text{OCCUR}(\psi)$, $v_1 \models \psi \iff v_2 \models \psi$
- $v_1 \models \neg\psi \iff v_1 \not\models \psi$ and $v_2 \models \neg\psi \iff v_2 \not\models \psi$
- Since the truth values of ψ are the same, so do their negation.

Inductive Step 2 (Binary connective):

Assume $P(\psi_1)$ and $P(\psi_2)$ hold.

- Let $\phi = \psi_1 \oplus \psi_2$ for $\oplus \in \{\wedge, \vee\}$
- $\text{OCCUR}(\phi) = \text{OCCUR}(\psi_1) \cup \text{OCCUR}(\psi_2)$, if v_1, v_2 agree on $\text{OCCUR}(\phi)$, they agree on both $\text{OCCUR}(\psi_1)$ and $\text{OCCUR}(\psi_2)$
- $v_1 \models \psi_1 \iff v_2 \models \psi_1$ and $v_1 \models \psi_2 \iff v_2 \models \psi_2$
- the satisfaction of $\psi_1 \oplus \psi_2$ depends only on the truth values of ψ_1 and ψ_2 , and those values are the same for v_1 and v_2 , thus $v_1 \models \phi \iff v_2 \models \phi$

□

Q5

Lemma 10.34 (Generalized Relevance Lemma). *Let ϕ be a propositional formula and let v_1, v_2 be valuations such that $v_1(p) = v_2(p)$ for all p in some set $S \supseteq \text{OCCUR}(\phi)$. Then $v_1 \models \phi$ if and only if $v_2 \models \phi$.*

Prove this lemma.

Proof. Let ϕ be a formula and S be a set such that $S \supseteq \text{OCCUR}(\phi)$. Suppose $v_1(p) = v_2(p) \forall p \in S$. Since S contains all elements of $\text{OCCUR}(\phi)$, the condition $v_1(p) = v_2(p) \forall p \in S \implies v_1(p) = v_2(p) \forall p \in \text{OCCUR}(\phi)$. By the Relevance lemma, if v_1 and v_2 agree on $\forall p \in \text{OCCUR}(\phi)$, then $v_1 \models \phi \iff v_2 \models \phi$. □

Q6

Lemma 10.35 (Relevance Lemma for Logical Equivalence). *Let ϕ_1, ϕ_2 be propositional formulae. The following are equivalent:*

- For all valuations v_1, v_2 agreeing on $OCCUR(\phi_1) \cup OCCUR(\phi_2)$, we have $v_1 \models \phi_1$ iff $v_2 \models \phi_2$.
- ϕ_1 and ϕ_2 are logically equivalent.

Prove this lemma.

Proof. We prove the equivalence of the two statements by showing both implications.

($\Leftarrow$) Suppose ϕ_1 and ϕ_2 are logically equivalent. $\forall v, v \models \phi_1 \iff v \models \phi_2$. Consider v_1, v_2 that agree on $OCCUR(\phi_1) \cup OCCUR(\phi_2)$, we have:

- $v_1 \models \phi_1 \iff v_1 \models \phi_2$
- $v_2 \models \phi_1 \iff v_2 \models \phi_2$

Trivial

($\Rightarrow$) Suppose $\forall v_1, v_2$ agreeing on $S = OCCUR(\phi_1) \cup OCCUR(\phi_2)$, we have $v_1 \models \phi_1 \iff v_2 \models \phi_2$. Let v be any valuation. We will show that $v \models \phi_1 \iff v \models \phi_2$. Define $v_1 = v$ and $v_2 = v$. These valuations trivially agree on S .

- $v_1 \models \phi_1 \iff v_2 \models \phi_2$
- $v \models \phi_1 \iff v \models \phi_2$
- ϕ_1 and ϕ_2 are logically equivalent

□

Q7

Let $G = (V, E)$ be an undirected graph and $k \in \mathbb{N}$. Construct a propositional formula ϕ_G such that:

1. $|\phi|$ is polynomial in $k + |V| + |E|$,
2. ϕ is satisfiable if and only if G has a vertex cover of size at most k .

Formally prove correctness of the encoding.

Proof. Define the proposition set P :

$$\begin{aligned} P &= P_{in} \cup P_{out} \cup P_{count} \\ P_{in} &= \{p_{v,in} \mid v \in V\} \\ P_{out} &= \{p_{v,out} \mid v \in V\} \\ P_{count} &= \{c_{i,j} \mid i, j \in [0, |V|]\} \\ |P| &\in O(|V|^2) \end{aligned}$$

There are constraint we need to consider:

$$\forall v \in V, v \in V' \text{ or } v \in V \setminus V'$$

$$\phi_{in} = \bigwedge_{v \in V} (p_{v,in} \vee p_{v,out})$$

$$|\phi_{in}| \in O(|V|)$$

$$\forall v \in V, v \notin V' \text{ or } v \in V \setminus V'$$

$$\phi_{out} = \bigwedge_{v \in V} \neg(p_{v,in} \wedge p_{v,out})$$

$$|\phi_{out}| \in O(|V|)$$

All edges covered

$$\phi_e = \bigwedge_{\{u,v\} \in E} (p_{u,in} \vee (p_{v,in}))$$

$$|\phi_e| \in O(|V|^2)$$

$$|V'| \leq k$$

We use the following facts to define inductive rules:

- The number of true proposition is non decreasing as i increases in the sequence $p_1, \dots, p_i$. $\rightarrow$ if the $c_{i,j}$ is true, $\forall \ell \geq i, c_{\ell,j}$ is true.
- if the $c_{i,j}$ is false, $\forall \ell \geq j, c_{i,\ell}$ is false

Inductively define $c_{i,j}$:

$$[c_{i-1,j} \vee (c_{i-1,j-1} \wedge p_{i,in})] \leftrightarrow c_{i,j}$$

We use bidirectional implication to prevent vacuous truthity of $c_{i,j}$ when conditions do not hold. And the last constraint can be represented by $\neg(c_{|V|,k+1})$

$$|\{c_{i,j} \mid i \in [1, |V|], j \in [1, k+1]\}| \in O(|V| \cdot k)$$

Putting everything together we can encode k -vertex cover using

$$\phi = \phi_{in} \wedge \phi_{out} \wedge \phi_e \wedge (\neg c_{|V|,k+1})$$

□

Correctness of encoding

To prove the correctness of the encoding, we will show:

Theorem 10.36. ϕ is satisfiable $\rightarrow \exists k$ -vertex cover on G

Proof. ϕ is satisfiable $\rightarrow \exists v, v \models \phi, \rightarrow$

$$\begin{aligned} v &\models \phi_{in} \\ v &\models \phi_{out} \\ v &\models \phi_e \\ v &\models (\neg c_{|V|,k+1}) \end{aligned}$$

As per construction, this guarentees $\forall v \in V, v$ is either in the vertex cover or exclusively; $\forall e = \{u, v\} \in E$, either u or v inclusively are in the vertex cover, and the size of the cover is at most k □

Theorem 10.37. $\exists k$ -vertex cover on $G \rightarrow \phi$ is satisfiable

Proof. Suppose $G = (V, E)$ has a vertex cover $V' \subseteq V$ such that $|V'| \leq k$. We construct a valuation v as follows:

1. For each $u \in V$, set $v(p_{u,in}) = \text{true}$ if $u \in V'$, and $v(p_{u,in}) = \text{false}$ otherwise
2. Set $v(p_{u,out}) = \neg v(p_{u,in}) \implies \phi_{in} \wedge \phi_{out}$ is satisfiable
3. Since V' is a vertex cover, for every edge $\{u, v\} \in E$, at least one of u or v must be in V' . Thus, $v(p_{u,in}) \vee v(p_{v,in})$ is true for all edges, satisfying ϕ_e .
4. Define $v(c_{i,j})$ to be true if and only if $|\{u_1, \dots, u_i\} \cap V'| \geq j$.
 - the recurrence $[c_{i-1,j} \vee (c_{i-1,j-1} \wedge p_{i,in})] \leftrightarrow c_{i,j}$ is satisfied because the count of vertices in the cover up to i increases only if the i -th vertex is in the cover.
 - Since $|V'| \leq k$, the total count of vertices in V' after considering all $|V|$ vertices cannot be $k+1$. Thus, $v(c_{|V|,k+1}) = \text{false}$, which satisfies $\neg c_{|V|,k+1}$.

Since all components are satisfied, ϕ is satisfiable. □

Q8

Given a partial $N^2 \times N^2$ Sudoku grid S , construct a propositional formula ϕ_S of size polynomial in N^2 such that ϕ_S is satisfiable if and only if S has a completion to a valid Sudoku grid. Prove correctness.

Proof. Define proposition set $P = \{p_{i,j,k} \mid i, j, k \in [1, N]\}$, representing cell (i, j) is assigned the number k . We need to satisfy the following constraints:

Cell: At most 1 value assigned

$$\phi_{cell, \leq 1} = \bigwedge_{i=1}^N \bigwedge_{j=1}^N \bigwedge_{1 \leq k < m \leq N} (\neg p_{i,j,k} \vee \neg p_{i,j,m})$$

Cell: At least 1 value assigned

$$\phi_{cell, \geq 1} = \bigwedge_{i=1}^N \bigwedge_{j=1}^N \bigvee_{k=1}^N p_{i,j,k}$$

Row: At most 1 value assigned

$$\phi_{row, \leq 1} = \bigwedge_{i=1}^N \bigwedge_{k=1}^N \bigwedge_{1 \leq j < m \leq N} (\neg p_{i,j,k} \vee \neg p_{i,m,k})$$

Column: At most 1 value assigned

$$\phi_{col, \leq 1} = \bigwedge_{j=1}^N \bigwedge_{k=1}^N \bigwedge_{1 \leq i < m \leq N} (\neg p_{i,j,k} \vee \neg p_{m,j,k})$$

Block: At most 1 value assigned

Let $n = \sqrt{N}$. For each block starting at row $r \in \{0, \dots, n-1\}$ and column $c \in \{0, \dots, n-1\}$:

$$\phi_{block, \leq 1} = \bigwedge_{r=0}^{n-1} \bigwedge_{c=0}^{n-1} \bigwedge_{k=1}^N \bigwedge_{(i,j) \neq (i',j')} (\neg p_{rn+i, cn+j, k} \vee \neg p_{rn+i', cn+j', k})$$

where $1 \leq i, j, i', j' \leq n$.

Initial Setup

Let S be the set of triples (i, j, k) such that cell (i, j) is pre-assigned value k :

$$\phi_{clues} = \bigwedge_{(i,j,k) \in S} p_{i,j,k}$$

The complete formula is:

$$\phi_{Sudoku} = \phi_{cell, \leq 1} \wedge \phi_{cell, \geq 1} \wedge \phi_{row, \leq 1} \wedge \phi_{col, \leq 1} \wedge \phi_{block, \leq 1} \wedge \phi_{clues}$$

□

Correctness of Encoding

Theorem 10.38. *The formula ϕ_{Sudoku} is satisfiable $\iff$ the Sudoku puzzle has a valid solution*

Proof. ($\Rightarrow$) Suppose ϕ_{Sudoku} is satisfiable under valuation v . We construct a Sudoku assignment by setting cell $(i, j) = k$ if $v(p_{i,j,k}) = \text{true}$.

- By $\phi_{cell, \geq 1}$ and $\phi_{cell, \leq 1}$, every cell is assigned exactly one value.
- By $\phi_{row, \leq 1}$, no value k is repeated in a row. Since there are N cells in a row and N possible values, each value must appear exactly once.

- The same logic applies to $\phi_{col, \leq 1}$ and $\phi_{block, \leq 1}$ for columns and $n \times n$ subgrids.

- ϕ_{clues} ensures the solution respects the starting digits.

($\Leftarrow$) Suppose a valid Sudoku solution exists. Define a valuation v such that $v(p_{i,j,k}) = \text{true}$ if cell (i, j) contains k , and false otherwise.

- Because the solution is valid, each cell has one number (satisfying cell constraints).
- Because rows, columns, and blocks contain no duplicates, the "at most one" (≤ 1) constraints for each are satisfied.
- Because the solution contains the initial clues, ϕ_{clues} is satisfied.

Therefore, $v \models \phi_{Sudoku}$. □

Background: Compactness

Definition 10.39 (Satisfiable). *A set Γ of propositional formulae is satisfiable if there exists a valuation v such that $v \models \Gamma$.*

Definition 10.40 (Finitely Satisfiable). *A set Γ is finitely satisfiable if every finite $\Gamma' \subseteq \Gamma$ is satisfiable.*

Theorem 10.41 (Compactness Theorem). *A set Γ of propositional formulae is satisfiable if and only if it is finitely satisfiable.*

Q11

Lemma 10.42. *If Γ is satisfiable, then Γ is finitely satisfiable.*

Proof. Γ is satisfiable $\implies \exists v, \forall \phi \in \Gamma, v \models \phi$

$\forall$ finite subset $\Gamma' \subseteq \Gamma, \forall \psi \in \Gamma', \psi \in \Gamma \implies \forall \Gamma' \subseteq \Gamma, \forall \psi \in \Gamma', v \models \psi$ □

Q12

Definition 10.43 (P -complete). Γ is P -complete if for every $p \in P$, either $p \in \Gamma$ or $\neg p \in \Gamma$.

Definition 10.44 (P -consistent). Γ is P -consistent if for no $p \in P$ do both p and $\neg p$ belong to Γ .

Lemma 10.45. *If Γ is finitely satisfiable and P -complete, then Γ is P -consistent.*

1. Prove the lemma.

Proof. Suppose otherwise, Γ is P -complete, finitely satisfiable and NOT P -consistent, $\exists$ finite set $\Gamma' \subseteq \Gamma, \exists p \in P, p \in \Gamma', (\neg p) \in \Gamma' \implies \Gamma'$ is not satisfiable, a contradiction. □

12.2

- $P = \{p, \}$
- $\Gamma = \emptyset$

12.3

- $P = \{p\}$
- $\Gamma = \{(\neg(p \vee (\neg p)))\}$

12.4

I cannot

Q13

Let Γ be finitely satisfiable and let ϕ be a formula.

1. Prove that at least one of $\Gamma \cup \{\phi\}$ or $\Gamma \cup \{\neg\phi\}$ is finitely satisfiable.
2. Give examples as requested.

Lemma 10.46. *at least one of $\Gamma \cup \{\phi\}$ or $\Gamma \cup \{\neg\phi\}$ is finitely satisfiable.*

Proof. We will prove the contrapositive of this statement: if neither $\Gamma \cup \{\phi\}$ nor $\Gamma \cup \{\neg\phi\}$ is satisfiable, Γ is not finitely satisfiable. Consider finite subset $\Gamma_1, \Gamma_2 \subseteq \Gamma$, and $\phi \in FORM$, $\Gamma_1 \cup \{\phi\}$ and $\Gamma_2 \cup \{\neg\phi\}$ are both not satisfiable, $\forall v, v \models (\Gamma_1 \cup \{\phi\}) \implies v \models \Gamma_1$ or $v \models \phi$, and $\forall v, v \models (\Gamma_2 \cup \{\neg\phi\}) \implies v \models \Gamma_2$ or $v \models (\neg\phi)$

Rewrite with respect to the set of all valuations, define

- $V_1 = \{v : P \mapsto \{T, F\} \mid v \models \Gamma_1\}$
- $V_2 = \{v : P \mapsto \{T, F\} \mid v \models \Gamma_2\}$
- $V_\phi = \{v : P \mapsto \{T, F\} \mid v \models \{\phi\}\}$
- $V_{(\neg\phi)} = \{v : P \mapsto \{T, F\} \mid v \models \{\neg\phi\}\}$

$$V_1 \cap V_\phi = \emptyset \implies \forall v \in V_1, v \not\models \phi \implies v \models (\neg\phi) \implies V_1 \subseteq V_{(\neg\phi)}$$

$$V_2 \cap V_{(\neg\phi)} = \emptyset \implies \forall v \in V_2, v \not\models (\neg\phi) \implies v \models \phi \implies V_2 \subseteq V_\phi$$

Now consider also a finite subset to $\Gamma, \Gamma_1 \cup \Gamma_2$, define

- $V_U = \{v : P \mapsto \{T, F\} \mid v \models (\Gamma_1 \cup \Gamma_2)\}$

$$V_U = (V_1 \cap V_2) \subseteq (V_\phi \cup V_{(\neg\phi)}) = \emptyset \implies \nexists v, v \models (\Gamma_1 \cup \Gamma_2) \implies \Gamma \text{ is not finitely satisfiable.} \quad \square$$

13.2

- $\Gamma = \{p\}, \phi = q$

13.3

- $\Gamma = \{p, (\neg q)\}, \phi = q$

Q14, 15

Let $P = \{p_1, p_2, \dots\}$ and define $\Gamma_0, \Gamma_1, \dots$

Lemma 10.47. $\Gamma_U = \bigcup_{i \geq 0} \Gamma_i$ is finitely satisfiable.

Lemma 10.48.

$\forall i \geq 0, \Gamma_i$ is finitely satisfiable

Proof. Base case: $i = 0, \Gamma_i = \Gamma_0 = \Gamma$ is finitely satisfiable by definition.

Inductive Hypothesis: Γ_n is finitely satisfiable. By Lemma 10.46, $\Gamma_n \cup \{p_n\}$ is finitely satisfiable or $\Gamma_n \cup \{\neg p_n\}$ is finitely satisfiable. $\square$

Proof. Consider finite set $\Gamma' \subseteq \Gamma_U$, since Γ' is finite, and $\forall k \geq 0, \Gamma_k \subseteq \Gamma_{k+1}$ by definition, $\exists j \geq 0, \Gamma' \subseteq \Gamma_j = \bigcup_{i \geq 0}^j \Gamma_i$. By Lemma 10.42, Γ_j is satisfiable $\rightarrow \Gamma'$ is finitely satisfiable. $\square$

Proof. From the construction, $\forall i \geq 0$ at least one of p_i and $(\neg p_i)$ is in $\bigcup_{i \geq 0} \Gamma_i$. Since $\bigcup_{i \geq 0} \Gamma_i$ is finitely satisfiable, $\forall i \geq 0, p_i$ and $(\neg p_i)$ cannot both be in $\bigcup_{i \geq 0} \Gamma_i$. Otherwise $\{p_i\} \cup \{(\neg p_1)\} \subseteq \bigcup_{i \geq 0} \Gamma_i$ is not satisfiable, contradiction. $\square$

Now we have proven that the construction proposed from a finitely satisfiable formula set Γ to $\bigcup_{i \geq 0} \Gamma_i$ is also finitely satisfiable. Now to complete the proof of the compactness theorem, we need to prove that $\exists v, v \models \Gamma$.

Proof. From Lemma 10.47, $\forall i \geq 0, (p_i \in \Gamma_U) \oplus ((\neg p_i) \in \Gamma_U), \exists v, v \models \Gamma_U$

$$v(p) : P \mapsto \{T, F\} = \begin{cases} T, & \text{if } p \in \Gamma_U \\ F, & \text{if } (\neg p) \in \Gamma_U \end{cases}$$

Define the literal set $Literals = \{p_i, (\neg p_i) \mid i \geq 0\}$, and function

$$\tau(p) : P \mapsto Literals = \begin{cases} p, & \text{if } p \in \Gamma_U \\ (\neg p), & \text{if } (\neg p) \in \Gamma_U \end{cases}$$

Lemma 10.49.

$$v \models \Gamma_U \rightarrow v \models \tau(\Gamma)$$

Let $\phi \in \Gamma$, Define literal set $X = \{\tau(p) \mid p \in OCCUR(\phi)\}$ where X is a set containing single proposition literals. By definition we have

$$X \subseteq \Gamma_U \rightarrow v \models X \rightarrow \forall \psi \in X, v(\psi) = T$$

and $OCCUR(X) = OCCUR(\phi)$ Since both X is finite, $X \cup \{\phi\}$ is a finite subset to Γ_U , thus satisfiable. $\rightarrow \exists v', v' \models X \cup \{\phi\} \rightarrow v' \models X, v' \models \phi \rightarrow \forall \psi \in X, v'(\psi) = T$.

$$\forall p \in OCCUR(\phi), v(p) = v'(p), v' \models \phi \rightarrow v \models \phi$$

$\square$

Q16

Let $\top$ and $\perp$ be the 0-place connectives (truth and falsity). For a wff ϕ and sentence symbol A , let $\phi_{\top}^A$ and $\phi_{\perp}^A$ denote the substitution of A with $\top$ and $\perp$ respectively. Define the formula $\phi_A^* = (\phi_{\top}^A \vee \phi_{\perp}^A)$.

Proof. 1. $\phi \models \phi_A^*$.

Let v be a valuation such that $v \models \phi$.

- If $v(A) = T \implies v(\phi) = v(\phi_{\top}^A)$. $v \models \phi \implies v \models \phi_{\top}^A$.
- If $v(A) = F \implies v(\phi) = v(\phi_{\perp}^A)$. $v \models \phi \implies v \models \phi_{\perp}^A$.
- either way, $v \models (\phi_{\top}^A \vee \phi_{\perp}^A)$ Thus, $\phi \models \phi_A^*$.

2. If $\phi \models \psi$ and A does not occur in ψ , then $\phi_A^* \models \psi$.

Let $v \models \phi_A^* \implies v \models \phi_{\top}^A$ or $v \models \phi_{\perp}^A$.

- Case 1: $v \models \phi_{\top}^A$. Let v' be valuation $v[A \mapsto \text{true}] \implies v' \models \phi$. $\phi \models \psi \implies v' \models \psi$. $A \notin OCCUR(\psi) \implies \psi$ is independent of $A \implies v \models \psi$.
- Case 2: similar
- Thus, $\phi_A^* \models \psi$

3. ϕ is satisfiable iff ϕ_A^* is satisfiable.

- $(\Rightarrow)$ ϕ is satisfiable $\Rightarrow \exists v \models \phi$. By part 1, $\phi \models \phi_A^* \Rightarrow v \models \phi_A^*$
- $(\Leftarrow)$ ϕ_A^* is satisfiable $\Rightarrow \exists v, v \models \phi_+^A$ or $v \models \phi_-^A$. $v \models \phi_+^A \Rightarrow v[A \mapsto \text{true}] \models \phi$. $v \models \phi_-^A \Rightarrow v[A \mapsto \text{false}] \models \phi$

□

Theorem 10.50 (Interpolation Theorem). *If $\alpha \models \beta$, then there exists a formula γ such that $\alpha \models \gamma \models \beta$, and $OCCUR(\gamma) \subseteq OCCUR(\alpha) \cap OCCUR(\beta)$.*

Proof. Let $S = OCCUR(\alpha) \setminus OCCUR(\beta)$,

- Let $\alpha_0 = \alpha$.
- $\forall i \in [1, n]$, let $\alpha_i = (\alpha_{i-1})_{A_i}^*$.
- By part 1, $\alpha_{i-1} \models \alpha_i$, by transitivity $\alpha \models \alpha_n$.
- By part 2, $\alpha \models \beta$ and $A_1 \notin OCCUR(\beta) \Rightarrow \alpha_1 \models \beta$. Repeat $\forall A_i \in S, \alpha_n \models \beta$.
- $\gamma = \alpha_n$ contains no propositions from S , meaning $OCCUR(\gamma) \subseteq OCCUR(\alpha) \cap OCCUR(\beta)$.

□

HW3

Hilbert-style Proof System

The Hilbert-style (or Frege) proof system for propositional logic comprises of 3 axiom schema and the modus ponens rule of inference

- **Axiom Schema 1:** $\varphi \rightarrow (\psi \rightarrow \varphi)$
- **Axiom Schema 2:** $(\varphi \rightarrow (\psi \rightarrow \rho)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \rho))$
- **Axiom Schema 3:** $((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$
- **Modus Ponens:** From φ and $\varphi \rightarrow \psi$, infer ψ

Question 1

Let Γ, Γ' be sets of propositional logic formulae and let φ be a formula. Show that if $\Gamma \vdash \varphi$ and if $\Gamma \subseteq \Gamma'$, then $\Gamma' \vdash \varphi$

Proof. $\Gamma \vdash \varphi \implies \exists$ a proof for $\phi, \langle \phi_1, \phi_2, \dots, \phi_n \rangle, \phi_n = \varphi$

Base case: $n = 1$

- Case 1: $\varphi \in \Gamma \implies \varphi \in \Gamma' \implies \Gamma' \vdash \varphi$
- Case 2: φ is from Axiom 1 or 2 or 3, $\vdash \varphi \implies \Gamma' \vdash \varphi$

Inductive case: Suppose $\forall i < n, \Gamma \vdash \phi_i \implies \Gamma' \vdash \phi_i$

- Case 1: φ is from axiom 1, 2, 3, trivial
- Case 2: φ is from Modus Ponens, $\exists i, j < n, \phi_i \equiv \phi_j \rightarrow \varphi, \Gamma \vdash \phi_i, \Gamma \vdash \phi_j$

$$\frac{\phi_j \rightarrow \varphi \quad \phi_j}{\varphi}$$

From inductive hypothesis, $\Gamma' \vdash \phi_i, \Gamma' \vdash \phi_j, \Gamma' \vdash \varphi$

□

Question 2

Theorem 10.51 (Deduction Theorem). *Let Γ be a set of propositional logic formulae and φ, ψ be propositional logic formulae. $\Gamma \vdash (\varphi \rightarrow \psi)$ iff $\Gamma \cup \{\varphi\} \vdash \psi$.*

Proof. ($\implies$)

1. $\Gamma \vdash \varphi \rightarrow \psi$ (Given)
2. $\Gamma \cup \{\varphi\} \vdash \varphi \rightarrow \psi$ (Monotonicity)
3. $\Gamma \cup \{\varphi\} \vdash \varphi$ (Assumption)
4. $\Gamma \cup \{\varphi\} \vdash \psi$ (MP 2, 3)

($\impliedby$)

Let $\pi = \psi_1, \dots, \psi_n$ be a proof of ψ from $\Gamma \cup \{\varphi\}$. Assume $\forall j < i, \Gamma \vdash \varphi \rightarrow \psi_j$.

Case 1: $\psi_i \in \Gamma$ or ψ_i is an Axiom

1. $\Gamma \vdash \psi_i$ (Assumption/Axiom)
2. $\Gamma \vdash \psi_i \rightarrow (\varphi \rightarrow \psi_i)$ (Ax 1)
3. $\Gamma \vdash \varphi \rightarrow \psi_i$ (MP 1, 2)

Case 2: $\psi_i = \varphi$

1. $\Gamma \vdash (\varphi \rightarrow ((\varphi \rightarrow \varphi) \rightarrow \varphi)) \rightarrow ((\varphi \rightarrow (\varphi \rightarrow \varphi)) \rightarrow (\varphi \rightarrow \varphi))$ (Ax 2)
2. $\Gamma \vdash \varphi \rightarrow ((\varphi \rightarrow \varphi) \rightarrow \varphi)$ (Ax 1)
3. $\Gamma \vdash (\varphi \rightarrow (\varphi \rightarrow \varphi)) \rightarrow (\varphi \rightarrow \varphi)$ (MP 1, 2)
4. $\Gamma \vdash \varphi \rightarrow (\varphi \rightarrow \varphi)$ (Ax 1)
5. $\Gamma \vdash (\varphi \rightarrow \varphi)$ (MP 3, 4)

Case 3: ψ_i via MP from ψ_j, ψ_k where $\psi_j = \psi_k \rightarrow \psi_i$

1. $\Gamma \vdash \varphi \rightarrow (\psi_k \rightarrow \psi_i)$ (IH)
2. $\Gamma \vdash (\varphi \rightarrow (\psi_k \rightarrow \psi_i)) \rightarrow ((\varphi \rightarrow \psi_k) \rightarrow (\varphi \rightarrow \psi_i))$ (Ax 2)
3. $\Gamma \vdash (\varphi \rightarrow \psi_k) \rightarrow (\varphi \rightarrow \psi_i)$ (MP 1, 2)
4. $\Gamma \vdash \varphi \rightarrow \psi_k$ (IH)
5. $\Gamma \vdash \varphi \rightarrow \psi_i$ (MP 3, 4)

□

Question 3

(a) $\vdash \varphi \rightarrow \varphi$

Done in Q2

(b) $\vdash (\psi \rightarrow \varphi) \rightarrow \varphi$, where $\psi = \varphi \rightarrow \perp$

Proof. $\equiv \{\psi \rightarrow \varphi\} \vdash \varphi$.

1. $(\varphi \rightarrow \perp) \rightarrow \varphi$ (Assumption)
2. $((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$ (Axiom 3)
3. $((\varphi \rightarrow \perp) \rightarrow \varphi) \rightarrow (((\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \perp)) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp))$ (Axiom 2)
4. $((\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \perp)) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)$ (MP 1, 3)
5. $(\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \perp)$ (Part a)
6. $(\varphi \rightarrow \perp) \rightarrow \perp$ (MP 4, 5)
7. φ (MP 2, 6)

(Wikipedia: Peirce's law).

□

(c) $\{\perp\} \vdash \varphi$

Proof. 1. $\perp$ (Assumption)

2. $\perp \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)$ (Axiom 1)
3. $(\varphi \rightarrow \perp) \rightarrow \perp$ (MP 1, 2)
4. $((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$ (Axiom 3)
5. φ (MP 3, 4)

□

(d) $\vdash \psi \rightarrow (\varphi \rightarrow \rho)$ **where** $\psi = \varphi \rightarrow \perp$

Proof. $\equiv \{\varphi \rightarrow \perp, \varphi\} \vdash \rho$.

1. $\varphi \rightarrow \perp$ (Assumption)
2. φ (Assumption)
3. $\perp$ (MP 1, 2)
4. $\perp \rightarrow ((\rho \rightarrow \perp) \rightarrow \perp)$ (Axiom 1)
5. $(\rho \rightarrow \perp) \rightarrow \perp$ (MP 3, 4)
6. $((\rho \rightarrow \perp) \rightarrow \perp) \rightarrow \rho$ (Axiom 3)
7. ρ (MP 5, 6)

□

Theorem 2 (Soundness of Hilbert-style Proof System)

Let Γ be a set of propositional logic formulae and φ be a propositional logic formula. If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$

Question 4

Prove Theorem 2. *Hint: Given a proof, inductively establish that every formula in the proof is a logical consequence of the assumptions.*

Theorem 3 (Completeness of Hilbert-style Proof System)

Let Γ be a set of propositional logic formulae and φ be a propositional logic formula. If $\Gamma \models \varphi$, then $\Gamma \vdash \varphi$

Theorem 4 (Weak Completeness of Hilbert-style Proof System)

Let φ be a propositional logic formula. If $\models \varphi$, then $\vdash \varphi$.

Definition 10.52 (Inconsistent set). Γ is inconsistent $\iff$

$$\exists \varphi, \Gamma \vdash \varphi, \Gamma \vdash (\varphi \rightarrow \perp)$$

Question 5

Prove that two notions of a set being inconsistent are identical. That is, prove that the following two statements are equivalent

1. There is a formula φ such that $\Gamma \vdash \varphi$ and $\Gamma \vdash (\varphi \rightarrow \perp)$
2. For every formula ρ , $\Gamma \vdash \rho$

Lemma 1

Let Γ be a set of formulae and φ be a formula. If $\Gamma \cup \{\varphi \rightarrow \perp\}$ is inconsistent, then $\Gamma \vdash \varphi$

Lemma 2

Let Γ be a set of formulae and φ be a formula. If $\Gamma \cup \{\varphi\}$ is inconsistent, then $\Gamma \vdash (\varphi \rightarrow \perp)$

Lemma 3

Let Γ be a set of formulae. If Γ is inconsistent, then there is a finite subset $\Gamma' \subseteq_{fin} \Gamma$ of Γ such that Γ' is also inconsistent

Question 6

- (a) Prove Lemma 1

Proof. From deduction thm, $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow \psi$ and $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow (\psi \rightarrow \perp)$.

1. $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow (\psi \rightarrow \perp)$
2. $\Gamma \vdash ((\varphi \rightarrow \perp) \rightarrow (\psi \rightarrow \perp)) \rightarrow (((\varphi \rightarrow \perp) \rightarrow \psi) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp))$ (Ax 2)
3. $\Gamma \vdash ((\varphi \rightarrow \perp) \rightarrow \psi) \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)$ (MP 1, 2)
4. $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow \psi$
5. $\Gamma \vdash (\varphi \rightarrow \perp) \rightarrow \perp$ (MP 3, 4)
6. $\Gamma \vdash ((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi$ (Ax 3)
7. $\Gamma \vdash \varphi$ (MP 5, 6)

□

- (b) Prove Lemma 2

Proof. By definition, $\Gamma \cup \{\varphi\} \vdash \psi$ and $\Gamma \cup \{\varphi\} \vdash (\psi \rightarrow \perp)$ for some ψ . By the Deduction:

1. $\Gamma \vdash \varphi \rightarrow \psi$
2. $\Gamma \vdash \varphi \rightarrow (\psi \rightarrow \perp)$
3. $\Gamma \vdash (\varphi \rightarrow (\psi \rightarrow \perp)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \perp))$ (Ax 2)
4. $\Gamma \vdash (\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \perp)$ (MP 2, 3)
5. $\Gamma \vdash \varphi \rightarrow \perp$ (MP 1, 4)

□

- (c) Let Γ be a set of formulae and φ be a formula. Show that, if Γ is consistent, then at least one of $\Gamma \cup \{\varphi\}$ or $\Gamma \cup \{\varphi \rightarrow \perp\}$ is consistent

Proof. Suppos otherwise, Γ is consistent, $\Gamma \cup \{\varphi\}$ and $\Gamma \cup \{\varphi \rightarrow \perp\}$ are inconsistent. Since $\Gamma \cup \{\varphi\}$ is inconsistent, by Lemma 2

$$\Gamma \vdash \varphi \rightarrow \perp .$$

Since $\Gamma \cup \{\varphi \rightarrow \perp\}$ is inconsistent, by Lemma 1 we have

$$\Gamma \vdash \varphi .$$

$$\Gamma \vdash \perp .$$

By explosion, pick an arbitrary formula ψ , $\Gamma \vdash \perp \vdash \psi$, $\Gamma \vdash \perp \vdash \psi \rightarrow \perp$, Γ is inconsistent, contradiction. □

- (d) Let Γ, Γ' be sets of formulae such that $\Gamma \subseteq \Gamma'$. If Γ' is consistent, then so is Γ

Proof. Suppose othrewise, Γ is inconsistent,

$$\Gamma \vdash \perp .$$

$\Gamma \subseteq \Gamma'$, by monotonicity of derivability we have

$$\Gamma' \vdash \perp .$$

Contradiction. □

- (e)

Proof. Γ is inconsistent, $\exists \varphi, \Gamma \vdash \varphi, \Gamma \vdash (\varphi \rightarrow \perp)$ there exists a finite proof for both φ and $\varphi \rightarrow \perp$. Consider

- $\pi_1 = \langle \phi_1, \phi_2, \dots, \phi_n \rangle, \phi_n = \varphi$
- $\pi_2 = \langle \psi_1, \psi_2, \dots, \psi_m \rangle, \psi_m = \varphi \rightarrow \perp$

We construct Γ' as such: $\forall i < n, j < m$, if $\phi_i \in \Gamma, \phi_i \in \Gamma'$; if $\psi_j \in \Gamma, \psi_j \in \Gamma'$. Therefore $\forall \eta \in \Gamma', \eta \in \Gamma, \Gamma' \subseteq \Gamma$; $|\Gamma'| \leq n + m$, Γ' is a finite subset of Γ . Since π_1 is a proof for ϕ , $\forall \phi \in \pi_1, \phi \in \Gamma \implies \phi \in \Gamma'$; We can easily show that same proof can be constructed from Γ' by induction. Same case for $\Gamma' \vdash \varphi \rightarrow \perp$. Therefore Γ' is inconsistent. $\square$

Definition 10.53. • $\Delta_0 = \{\varphi \rightarrow \perp\}$

- $\Delta_{i+1} = \begin{cases} \Delta_i \cup \{\varphi_i\} & \text{if } \Delta_i \cup \{\varphi_i\} \text{ is consistent} \\ \Delta_i \cup \{\varphi_i \rightarrow \perp\} & \text{otherwise} \end{cases}$

$$\Delta = \bigcup_{i \geq 0} \Delta_i$$

Lemma 10.54. Γ is consistent $\implies \Gamma \cup \{\varphi\}$ or $\Gamma \cup \{\varphi \rightarrow \perp\}$ is consistent.

Question 7

- (a) Prove, using induction, that Δ_i is consistent, for every i

Proof. **Base case:** Δ_0

Suppose otherwise, $\Delta_0 = \{\varphi \rightarrow \perp\}$ is inconsistent. Then by Lemma 1,

$$\vdash \varphi.$$

Contradiction.

Inductive case: Suppose $\forall j \leq i, \Delta_j$ is consistent

By lemma above, Δ_i is consistent $\implies \Delta_i \cup \{\varphi_i\}$ is consistent or $\Delta_i \cup \{\varphi_i \rightarrow \perp\}$ is consistent $\implies \Delta_{i+1}$ is consistent $\square$

- (b) Prove that Δ is consistent

Proof. Suppose otherwise, Δ is inconsistent, by Lemma 3, $\exists \Delta' \subseteq_{fin} \Delta, \Delta'$ is inconsistent. Since Δ' is finite, let the highest ranked formula in Δ' be $n, \forall \varphi \in \Delta', \varphi \in \Delta_n$. From b), Δ_n is consistent, by 6d), Δ' is consistent, contradiction. $\square$

- (c) Prove that Δ is negation-complete, i.e., for every formula ψ , we have $(\psi \rightarrow \perp) \in \Delta$ iff $(\psi \notin \Delta)$

Proof. $\implies$

Let $\psi = \varphi_{n+1}, \Delta_n$ is consistent. $\Delta_{n+1} = \Delta_n \cup \{\psi\}$ if $\Delta_n \cup \{\psi\}$ is consistent. Suppose $\psi \rightarrow \perp \in \Delta$, $\Delta \vdash \psi, \Delta \vdash \psi \rightarrow \perp \implies \Delta$ is inconsistent, contradiction. Therefore $\psi \in \Delta \implies \psi \rightarrow \perp \notin \Delta$

$\Leftarrow$

Suppose otherwise, $\psi \rightarrow \perp \in \Delta, \psi \in \Delta, \Delta \vdash \psi, \Delta \vdash \psi \rightarrow \perp \implies \Delta$ is inconsistent, contradiction. $\square$

- (d) Use the fact that Δ is negation-complete to show that for every formula ψ , $\psi \in \Delta$ iff $\Delta \vdash \psi$

Proof. $\implies$

trivial by definition

$\longleftarrow$

Consider $\Delta \vdash \psi$. Suppose $\psi \notin \Delta, \psi \rightarrow \perp \in \Delta, \Delta \vdash \psi \rightarrow \perp$. Since $\Delta \vdash \psi$, by MP, $\Delta \vdash \perp$, Δ is inconsistent, contradiction. Therefore $\psi \in \Delta$. $\square$

Question 8

- (a) Show that for any formula ψ , $v \models \psi$ iff $\psi \in \Delta$

Proof. **Base case:**

1. $\psi \in P$: by definition, $v(p) = \text{true} \iff p \in \Delta$
2. $\psi = \perp$: always $v(\perp) = \text{false}, \perp \notin \Delta \implies v \models \perp \iff \perp \in \Delta$ vacuously

Inductive case: $\psi = a \rightarrow b$

Suppose

- $v \models a \iff a \in \Delta$
- $v \models b \iff b \in \Delta$

Case $\implies$

Suppose $v \models a \rightarrow b$, $a \rightarrow b \notin \Delta, (a \rightarrow b) \rightarrow \perp \in \Delta, \Delta \vdash (a \rightarrow b) \rightarrow \perp$.

Now we show that $b \notin \Delta$

- Suppose otherwise, $b \in \Delta, \Delta \vdash b$, by ax 1, $\Delta \vdash b \rightarrow (a \rightarrow b) \implies \Delta \vdash a \rightarrow b$ by MP, contradiction.

Now we show that $a \in \Delta$

- Suppose otherwise, $a \notin \Delta \implies a \rightarrow \perp \in \Delta$, from

$$\vdash (\varphi \rightarrow \perp) \rightarrow (\varphi \rightarrow \rho)$$

$\Delta \vdash (a \rightarrow \perp) \rightarrow (a \rightarrow b)$. By MP, $\Delta \vdash a \rightarrow b$, contradiction.

Now with $b \notin \Delta$ and $a \in \Delta$, by IH, $v \models a$ and $v \not\models b \implies v \not\models a \rightarrow b$, contradiction.

Case $\longleftarrow$

Suppose otherwise, $a \rightarrow b \in \Delta, v \models a$ and $v \not\models b$. By IH, $a \in \Delta$ and $b \notin \Delta, b \rightarrow \perp \in \Delta \implies \Delta \vdash a, \Delta \vdash a \rightarrow b$, by MP, $\Delta \vdash b \implies b \in \Delta$, contradiction. $\square$

- (b) Show that $v \models (\varphi \rightarrow \perp)$ and thus $\not\models \varphi$

Proof. $\{\varphi \rightarrow \perp\} \in \Delta \implies v \models (\varphi \rightarrow \perp) \implies v \not\models \varphi \implies \not\models \varphi$. $\square$

Question 9

Prove Theorem 3 using Theorem 4, the Compactness theorem and the Deduction theorem

Proof. $\Gamma \models \varphi \implies \Gamma \cup \{\varphi \rightarrow \perp\}$ is unsat, so by Compactness $\exists \Gamma_0 = \{\psi_1, \dots, \psi_n\} \subseteq_{\text{fin}} \Gamma$, such that $\Gamma_0 \cup \{\varphi \rightarrow \perp\}$ is unsat,

$$\{\psi_1, \dots, \psi_n, (\varphi \rightarrow \perp)\} \text{ is unsat } \implies (\psi_1 \wedge \dots \wedge \psi_n \wedge (\varphi \rightarrow \perp)) \models \perp$$

$$\models \psi_1 \rightarrow (\psi_2 \rightarrow \dots (\psi_n \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)) \dots)$$

([https://en.wikipedia.org/wiki/Exportation_\(logic\)](https://en.wikipedia.org/wiki/Exportation_(logic))) by Theorem 4

$$\emptyset \vdash \psi_1 \rightarrow (\psi_2 \rightarrow \dots (\psi_n \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)) \dots)$$

Apply deduction thm

$$\{\psi_1\} \vdash (\psi_2 \rightarrow \dots (\psi_n \rightarrow ((\varphi \rightarrow \perp) \rightarrow \perp)) \dots)$$

until we get

$$\Gamma_0 \vdash (\varphi \rightarrow \perp) \rightarrow \perp.$$

Using ax 3:

$$\vdash ((\varphi \rightarrow \perp) \rightarrow \perp) \rightarrow \varphi,$$

By MP

$$\Gamma_0 \vdash \varphi.$$

Since $\Gamma_0 \subseteq \Gamma$, by monotonicity,

$$\Gamma \vdash \varphi.$$

□

Theorem 5 (Soundness of Resolution Proof System)

Let Γ be a set of clauses (over propositions P). If there is a resolution proof of Γ , then Γ is unsatisfiable.

Question 10

Prove Theorem 5, *Hint: Consider a resolution proof $\pi = C_1, C_2, \dots, C_n$ of Γ . Use induction to establish that for every $1 \leq i \leq n$ if a valuation v is such that $v \models \Gamma$, then $v \models C_i$.*

Proof. Let $\pi = C_1, C_2, \dots, C_n$ be a resolution proof of Γ , where $C_n = \emptyset$.

We prove by induction on i that for every $1 \leq i \leq n$, $v \models \Gamma$, $v \models C_i$.

$i \in \{1, \dots, n\}$ and suppose $v \models \Gamma$.

1. $C_i \in \Gamma$, $v \models C_i$.
2. C_i is a resolvent of two earlier clauses C_j and C_k wrt some proposition p , where

$$C_j = C \uplus \{p\}, \quad C_k = D \uplus \{\neg p\},$$

and $C_i = C \cup D$. By the IH, $j, k < i$, $v \models C_j$ and $v \models C_k$. We show that $v \models C_i$. Suppose, $v \not\models C_i = C \cup D$. Then v satisfies neither any literal in C nor any literal in D . Since $v \models C_j = C \uplus \{p\}$ and no literal in C is satisfied by v , it must be that $v \models p$, i.e. $v(p) = \text{true}$. Since $v \models C_k = D \uplus \{\neg p\}$ and no literal in D is satisfied by v , it must be that $v \models \neg p$, i.e. $v(p) = \text{false}$, Contradiction

□

Theorem 6 (Completeness of Resolution Proof System)

Let Γ be a set of clauses (over propositions P). If Γ is unsatisfiable, then there is a resolution proof of Γ

Question 11

Let Γ be an unsatisfiable set of clauses Show that, if Γ is unsatisfiable, then there is a finite set $\Gamma_0 \subseteq_{fin} \Gamma$ such that Γ_0 is unsatisfiable.

Proof. Suppose Γ is unsat, by Compactness, $\exists \Gamma' \subseteq_{fin} \Gamma, \Gamma'$ is unsat (?) □

Question 12

Let Γ_1, Γ_2 be sets of clauses such that $\Gamma_1 \subseteq \Gamma_2$. Show that, if there is a resolution proof of Γ_1 , then there is also a resolution proof of Γ_2

Proof. Let $\pi = C_1, C_2, \dots, C_n$ be a resolution proof of Γ_1

$\forall 1 \leq i \leq n, \pi$ is a resolution proof of Γ_1 , either:

1. $C_i \in \Gamma_1$, or
2. $\exists j, k < i, C_i$ is a resolvent of C_j and C_k

If $C_i \in \Gamma_1, C_i \in \Gamma_2$ Otherwise, this fact is independent of the choice of the initial clause set, so the same derivation is valid in a proof from Γ_2 . □

Question 13

Next, we constructed a sequence of sets as follows. Here, we assume that the set of propositions occurring in Γ_0 can be enumerated as $p_1, p_2, \dots, p_k$.

$$\Delta_0 = \Gamma_0$$

$$\Delta_{i+1} = \Delta_i^\emptyset \cup \{C \cup D \mid C \uplus \{p_{i+1}\} \in \Delta_i^+, D \uplus \{\neg p_{i+1}\} \in \Delta_i^-\}$$

where

- $\Delta_i^\emptyset$ is the set of clauses in Δ_i that contain neither the literal p_{i+1} nor $\neg p_{i+1}$,
- Δ_i^+ is the set of clauses in Δ_i that contain the literal p_{i+1} but not $\neg p_{i+1}$,
- Δ_i^- is the set of clauses in Δ_i that contain the literal $\neg p_{i+1}$ but not p_{i+1} .

Inductively show that Δ_i is unsat for every $0 \leq i \leq k$. Also show that $\Delta_i \neq \emptyset$ for every $0 \leq i \leq k$.

Proof. Base case:

$i = 0$. By construction, $\Delta_0 = \Gamma_0$. From 11, Γ_0 is a finite unsat subset of Γ

Inductive step:

Suppose Δ_i is unsat and $\Delta_i \neq \emptyset$. We show Δ_{i+1} is unsat and nonempty. Suppose that $\exists v, v \models \Delta_{i+1}$, consider $v(p_{i+1})$

- If $v(p_{i+1}) = \text{true}$, then every clause in Δ_i^+ is sat by v due to p_{i+1} , and $\Delta_i^\emptyset \subseteq \Delta_{i+1}$ thus sat by v .
For every resolvent $C \cup D \in \Delta_{i+1}$ with

$$C \uplus \{p_{i+1}\} \in \Delta_i^+, \quad D \uplus \{\neg p_{i+1}\} \in \Delta_i^-,$$

$v \models C \cup D$. Since $v(p_{i+1}) = \text{true}$, the clause $D \uplus \{\neg p_{i+1}\}$ must be satisfied by some literal in D , hence $v \models D$. Thus v satisfies every clause in Δ_i^- . Contradiction

- case $v(p_{i+1}) = \text{false}$: similar

□

Question 14

Inductively show that Δ_i does not contain the literals $\{p_1, \neg p_1, \dots, p_i, \neg p_i\}$ for every $1 \leq i \leq k$

Proof. We prove by induction on i that Δ_i does not contain any of the literals

$$\{p_1, \neg p_1, \dots, p_i, \neg p_i\}$$

for every $1 \leq i \leq k$.

Base case:

$i = 1$.

By definition,

$$\Delta_1 = \Delta_0^\emptyset \cup \{C \cup D \mid C \uplus \{p_1\} \in \Delta_0^+, D \uplus \{\neg p_1\} \in \Delta_0^-\}.$$

Every clause in $\Delta_0^\emptyset$ contains neither p_1 nor $\neg p_1$, and every resolvent $C \cup D$ is obtained by removing p_1 and $\neg p_1$. Hence no clause in Δ_1 contains p_1 or $\neg p_1$.

Inductive step:

□

Question 15

Finish the proof of Theorem 6 by explicitly constructing a resolution proof for Γ and establishing its correctness

Proof. Let Γ be unsat. By 11, $\exists \Gamma_0 \subseteq_{fin} \Gamma$ such that Γ_0 is unsat. Let $p_1, \dots, p_k$ be all propositions occurred in Γ_0 , and $\Delta_0, \Delta_1, \dots, \Delta_k$ as in the construction. Δ_k is unsat, and by 14, Δ_k does not contain $p_1, \neg p_1, \dots, p_k, \neg p_k$. therefore every clause in Δ_k contains no literal, thus every clause in Δ_k is $\emptyset$. Since $\Delta_k \neq \emptyset, \emptyset \in \Delta_k$

Base case:

$i = 0, \Delta_0 = \Gamma_0$

Inductive step:

Suppose every clause in Δ_i has a finite resolution proof from Γ_0 . Let $E \in \Delta_{i+1}$. By definition ,

1. $E \in \Delta_i^\emptyset$, hence $E \in \Delta_i$, so by IH E has a finite resolution proof from Γ_0 ; or
2. $E = C \cup D$, where

$$C \uplus \{p_{i+1}\} \in \Delta_i^+ \quad \text{and} \quad D \uplus \{\neg p_{i+1}\} \in \Delta_i^-.$$

Then E is the resolvent of these 2

$$C \uplus \{p_{i+1}\} \quad \text{and} \quad D \uplus \{\neg p_{i+1}\}.$$

By IH, both clauses have a finite resolution proof from Γ_0 , thus the resolution proof for E is also finite

$\Gamma_0 \subseteq \Gamma$, by 12 the same proof is also a resolution proof for Γ .

Therefore Γ is unsat, then there is a resolution proof for Γ .

□

We will prove the contrapositive of compactness theorem for the next 2 qn.

Question 16

Assume that the Hilbert-style Proof System is sound and complete for propositional logic, Use these facts to establish that the Compactness Theorem holds true for propositional logic

Proof. Γ is unsat $\implies \Gamma \models \perp$, so by completeness $\Gamma \vdash \perp$ since proofs are finite, only finitely many assumptions from Γ appear in the proof, yielding a finite $\Gamma_0 \subseteq_{fin} \Gamma$ with $\Gamma_0 \vdash \perp$, and by soundness $\Gamma_0 \models \perp$, hence Γ_0 is unsat □

Question 17

Assume that the Resolution Proof System is sound and complete for propositional logic. Use these facts to establish that the Compactness Theorem holds true for propositional logic

Proof. If Γ is an unsatisfiable set of clauses, by completeness there is a finite resolution proof for Γ , and since this proof uses only finitely many clauses from Γ , there is a finite subset $\Gamma_0 \subseteq_{\text{fin}} \Gamma$ from which the same proof is a resolution proof. By soundness, Γ_0 is unsat □

Question 1

Lemma 10.55 (Relevance Lemma for FOL). *Let ϕ be a FOL formula, let $\mathfrak{A}$ be a structure and let α_1, α_2 be variable assignments such that for all $x \in FVar(\phi)$,*

$$\alpha_1(x) = \alpha_2(x).$$

Then

$$\mathfrak{A}, \alpha_1 \models \phi \iff \mathfrak{A}, \alpha_2 \models \phi.$$

Question. Prove Lemma 1.

Proof. By symmetry, we only need to prove if for all $x \in FVar(\phi)$, $\alpha_1(x) = \alpha_2(x)$ then

$$\mathfrak{A}, \alpha_1 \models \phi \implies \mathfrak{A}, \alpha_2 \models \phi.$$

Lemma 10.56. *If*

$$\forall x \in Var(t), \alpha_1(x) = \alpha_2(x)$$

then $Val_{\mathfrak{A}, \alpha_1}(t) = Val_{\mathfrak{A}, \alpha_2}(t)$

Base case:

1. $t \equiv c$: $I(c) = I(c) \implies Val_{\mathfrak{A}, \alpha_1}(c) = Val_{\mathfrak{A}, \alpha_2}(c)$
2. $t \equiv x$: By definition, $Val_{\mathfrak{A}, \alpha_1}(x) = \alpha_1(x) = \alpha_2(x) = Val_{\mathfrak{A}, \alpha_2}(x)$

Inductive steps: Suppose this holds for $t_1, t_2, \dots$

$t' \equiv f(t_1, t_2, \dots)$: from definition, $Val_{\mathfrak{A}, \alpha_1}(f(t_1, t_2, \dots)) = I(f)(Val_{\mathfrak{A}, \alpha_1}(t_1), Val_{\mathfrak{A}, \alpha_1}(t_2), \dots)$. By IH, tuple $(Val_{\mathfrak{A}, \alpha_1}(t_1), Val_{\mathfrak{A}, \alpha_1}(t_2), \dots) = (Val_{\mathfrak{A}, \alpha_2}(t_1), Val_{\mathfrak{A}, \alpha_2}(t_2), \dots) \implies I_{\mathfrak{A}}(f)(Val_{\mathfrak{A}, \alpha_1}(t_1), Val_{\mathfrak{A}, \alpha_1}(t_2), \dots) = I_{\mathfrak{A}}(f)(Val_{\mathfrak{A}, \alpha_2}(t_1), Val_{\mathfrak{A}, \alpha_2}(t_2), \dots) \implies Val_{\mathfrak{A}, \alpha_1}(t') = Val_{\mathfrak{A}, \alpha_2}(t')$ Now we can use the lemma to prove the previous lemma:

Base cases

1. $\phi \equiv t_1 = t_2$. Suppose

$$\mathfrak{A}, \alpha_1 \models t_1 = t_2.$$

Then we have

$$Val_{\mathfrak{A}, \alpha_1}(t_1) = Val_{\mathfrak{A}, \alpha_1}(t_2).$$

$$FVar(t_1 = t_2) = Var(t_1) \cup Var(t_2),$$

$$Val_{\mathfrak{A}, \alpha_1}(t_1) = Val_{\mathfrak{A}, \alpha_2}(t_1) \quad \text{and} \quad Val_{\mathfrak{A}, \alpha_1}(t_2) = Val_{\mathfrak{A}, \alpha_2}(t_2).$$

$$Val_{\mathfrak{A}, \alpha_2}(t_1) = Val_{\mathfrak{A}, \alpha_2}(t_2) \implies \mathfrak{A}, \alpha_2 \models t_1 = t_2.$$

2. $\phi \equiv R(t_1, \dots, t_k)$.

Suppose

$$\mathfrak{A}, \alpha_1 \models R(t_1, \dots, t_k).$$

Then

$$(Val_{\mathfrak{A}, \alpha_1}(t_1), \dots, Val_{\mathfrak{A}, \alpha_1}(t_k)) \in I(R).$$

$$FVar(R(t_1, \dots, t_k)) = Var(t_1) \cup \dots \cup Var(t_k),$$

by the term lemma,

$$\forall i, Val_{\mathfrak{A}, \alpha_1}(t_i) = Val_{\mathfrak{A}, \alpha_2}(t_i).$$

$$(Val_{\mathfrak{A}, \alpha_1}(t_1), \dots, Val_{\mathfrak{A}, \alpha_1}(t_k)) = (Val_{\mathfrak{A}, \alpha_2}(t_1), \dots, Val_{\mathfrak{A}, \alpha_2}(t_k)) \in I(R)$$

$$\mathfrak{A}, \alpha_2 \models R(t_1, \dots, t_k).$$

Inductive cases

1. $\phi \equiv \neg\psi$.

Suppose

$$\mathfrak{A}, \alpha_1 \models \neg\psi.$$

Then

$$\mathfrak{A}, \alpha_1 \not\models \psi.$$

$$FVar(\neg\psi) = FVar(\psi),$$

$$\mathfrak{A}, \alpha_2 \models \psi \implies \mathfrak{A}, \alpha_1 \models \psi.$$

$$\mathfrak{A}, \alpha_2 \not\models \psi,$$

$$\mathfrak{A}, \alpha_2 \models \neg\psi.$$

2. $\phi \equiv \psi_1 \vee \psi_2$.

$$\mathfrak{A}, \alpha_1 \models \psi_1 \vee \psi_2.$$

then either

$$\mathfrak{A}, \alpha_1 \models \psi_1 \quad \text{or} \quad \mathfrak{A}, \alpha_1 \models \psi_2.$$

Also,

$$FVar(\psi_1 \vee \psi_2) = FVar(\psi_1) \cup FVar(\psi_2).$$

by IH

$$\mathfrak{A}, \alpha_2 \models \psi_1 \quad \text{or} \quad \mathfrak{A}, \alpha_2 \models \psi_2.$$

Therefore,

$$\mathfrak{A}, \alpha_2 \models \psi_1 \vee \psi_2.$$

3. $\phi \equiv \exists x \psi$.

Suppose

$$\mathfrak{A}, \alpha_1 \models \exists x \psi.$$

Then there exists some $u \in U$ such that

$$\mathfrak{A}, \alpha_1[x \mapsto u] \models \psi.$$

w can show that $\alpha_1[x \mapsto u]$ and $\alpha_2[x \mapsto u]$ agree on all variables in $FVar(\psi)$.

fix a $y \in FVar(\psi)$,

- If $y = x$, then

$$\alpha_1[x \mapsto u](y) = u = \alpha_2[x \mapsto u](y).$$

- If $y \neq x$, then $y \in FVar(\psi) \setminus \{x\} = FVar(\exists x, \psi)$

$$\alpha_1[x \mapsto u](y) = \alpha_1(y) = \alpha_2(y) = \alpha_2[x \mapsto u](y).$$

Thus, $\forall y \in FVar(\psi)$,

$$\alpha_1[x \mapsto u](y) = \alpha_2[x \mapsto u](y).$$

By IH

$$\mathfrak{A}, \alpha_2[x \mapsto u] \models \psi.$$

Hence

$$\mathfrak{A}, \alpha_2 \models \exists x \psi.$$

4. $\phi \equiv \forall x \psi$.

Suppose

$$\mathfrak{A}, \alpha_1 \models \forall x \psi.$$

then for every $u \in U$,

$$\mathfrak{A}, \alpha_1[x \mapsto u] \models \psi.$$

Fix a $u \in U$. We can show that $\alpha_1[x \mapsto u]$ and $\alpha_2[x \mapsto u]$ agree on all variables in $FVar(\psi)$.

fix a $y \in FVar(\psi)$.

- If $y = x$, then

$$\alpha_1[x \mapsto u](y) = u = \alpha_2[x \mapsto u](y).$$

- If $y \neq x$, then $y \in FVar(\psi) \setminus \{x\} = FVar(\exists x, \psi)$

$$\alpha_1[x \mapsto u](y) = \alpha_1(y) = \alpha_2(y) = \alpha_2[x \mapsto u](y).$$

by IH,

$$\mathfrak{A}, \alpha_2[x \mapsto u] \models \psi.$$

Since $u \in U$ is arbitrary,

$$\mathfrak{A}, \alpha_2 \models \forall x \psi.$$

□

Question 2

Definition 10.57 (Ground Terms, Atoms and Formulae). *A term over a signature Σ is said to be a ground term if it does not contain any occurrence of a variable. The set of all ground terms over Σ is denoted by*

$$\text{Terms}_\Sigma.$$

A formula is said to be ground if it contains no variables (and therefore no quantifiers).

A formula is said to be a ground atom if it is both ground and atomic.

We denote by

$$\text{Atoms}_\Sigma$$

the set of all ground atoms, and by

$$\text{FORM}_\Sigma$$

the set of all ground formulae over Σ .

Definition 10.58 (Ground Instantiation). *Let*

$$\phi \equiv \forall x_1 \forall x_2 \cdots \forall x_n \psi$$

be a universally quantified FOL sentence in prenex normal form.

A ground instantiation of ϕ is obtained by replacing each occurrence of x_i in ψ by a ground term t_i .

Formally, there exists a mapping

$$T : V \rightarrow \text{Terms}_\Sigma$$

such that

$$\varphi \equiv \psi[x_1 \rightarrow T(x_1)] \cdots [x_n \rightarrow T(x_n)].$$

The set of all ground instantiations of ϕ is denoted

$$\text{Ground}(\phi).$$

For a set Γ of universally quantified sentences,

$$\text{Ground}(\Gamma) = \bigcup_{\phi \in \Gamma} \text{Ground}(\phi).$$

Definition 10.59 (Propositional Satisfiability). *A ground atomic valuation over Σ is a mapping*

$$\tau : \text{Atoms}_\Sigma \rightarrow \{\text{true}, \text{false}\}.$$

This valuation is extended to ground formulae using the relation $\models_{Gr}$ defined by:

$$\tau \models_{Gr} \psi \quad \text{iff} \quad \tau(\psi) = \text{true} \quad \text{if } \psi \in \text{Atoms}_\Sigma$$

$$\tau \models_{Gr} \neg \phi \quad \text{iff} \quad \tau \not\models_{Gr} \phi$$

$$\tau \models_{Gr} (\phi_1 \vee \phi_2) \quad \text{iff} \quad \tau \models_{Gr} \phi_1 \text{ or } \tau \models_{Gr} \phi_2.$$

A set of ground FOL formulae Γ is propositionally satisfiable if there exists a ground atomic valuation τ such that

$$\tau \models_{Gr} \phi \quad \text{for every } \phi \in \Gamma.$$

Lemma 10.60. *If Γ is satisfiable then $\text{Ground}(\Gamma)$ is propositionally satisfiable.*

Question. Prove Lemma 2.

Proof. Γ is satisfiable, there exists a structure $\mathfrak{A}$ such that

$$\mathfrak{A} \models \Gamma.$$

Define ground atom valuation

$$\tau_{\mathfrak{A}} : \text{Atoms}_{\Sigma} \rightarrow \{\text{true}, \text{false}\}$$

$$\tau_{\mathfrak{A}}(a) = \text{true} \iff \mathfrak{A} \models a$$

for every ground atom a over Σ . Now we show that for every ground formula ϕ ,

$$\mathfrak{A} \models \phi \implies \tau_{\mathfrak{A}} \models_{Gr} \phi.$$

Induction on the structure of ϕ

Base case: χ is a ground atom

trivial

Inductive cases

1. $\psi \equiv \neg\phi$. Suppose

$$\mathfrak{A} \models \neg\phi \implies \mathfrak{A} \not\models \phi$$

By IH,

$$\mathfrak{A} \models \phi \implies \tau_{\mathfrak{A}} \models_{Gr} \phi$$

$$\tau_{\mathfrak{A}} \not\models_{Gr} \phi$$

$$\tau_{\mathfrak{A}} \models_{Gr} \neg\phi$$

2. $\chi \equiv \phi_1 \vee \phi_2$.

Suppose

$$\mathfrak{A} \models \phi_1 \vee \phi_2$$

Then

$$\mathfrak{A} \models \phi_1 \quad \text{or} \quad \mathfrak{A} \models \phi_2$$

by IH

$$\tau_{\mathfrak{A}} \models_{Gr} \phi_1 \quad \text{or} \quad \tau_{\mathfrak{A}} \models_{Gr} \phi_2$$

$$\tau_{\mathfrak{A}} \models_{Gr} \phi_1 \vee \phi_2$$

Fix any $\chi \in \text{Ground}(\Gamma)$. Then there exists

$$\varphi = \forall x_1 \dots \forall x_n \psi \in \Gamma$$

such that

$$\chi = \psi[x_1 \mapsto t_1] \dots [x_n \mapsto t_n]$$

for ground terms $t_1, \dots, t_n$

$$\mathfrak{A} \models \Gamma \implies \mathfrak{A} \models \varphi$$

For $1 \leq i \leq n$

$$u_i = \text{Val}_{\mathfrak{A}}(t_i)$$

$$\mathfrak{A}, \alpha[x_1 \mapsto u_1] \dots [x_n \mapsto u_n] \models \psi.$$

By substitution lemma, which we will prove later

$$\mathfrak{A} \models \psi[x_1 \mapsto t_1] \cdots [x_n \mapsto t_n].$$

$$\mathfrak{A} \models \chi.$$

$$\tau_{\mathfrak{A}} \models_{Gr} \chi.$$

Since $\chi \in \text{Ground}(\Gamma)$ is arbitrarily chosen, we have

$$\forall \chi \in \text{Ground}(\Gamma), \tau_{\mathfrak{A}} \models_{Gr} \chi$$

$$\tau_{\mathfrak{A}} \models_{Gr} \text{Ground}(\Gamma).$$

Therefore $\text{Ground}(\Gamma)$ is propositionally sat. □

Lemma 10.61 (Substitution lemma).

$$\mathfrak{A} \models \psi[x \mapsto t] \iff \mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models \psi$$

Lemma 10.62.

$$\text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(t)$$

Proof. **Base case:**

1. $t = c \in C$:

$$\text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(c) = I(c)$$

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(t) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(c) = I(c)$$

2. $t = y \in V$:

Case 1: $x = y$

$$\implies y[x \mapsto t'] = t'$$

$$\text{Val}_{\mathfrak{A}, \alpha}(y[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(t')$$

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(y) = \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')](y) = \text{Val}_{\mathfrak{A}, \alpha}(t')$$

Case 2: $x \neq y$

$$\implies y[x \mapsto t'] = y$$

$$\text{Val}_{\mathfrak{A}, \alpha}(y[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(y) = \alpha(y)$$

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(y) = \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')](y) = \alpha(y)$$

Inductive step: $t = f(t_1, \dots)$

$$t[x \mapsto t'] = f(t_1[x \mapsto t'], t_2[x \mapsto t'], \dots)$$

$$\text{Val}_{\mathfrak{A}, \alpha}(t[x \mapsto t']) = \text{Val}_{\mathfrak{A}, \alpha}(f(t_1[x \mapsto t'], t_2[x \mapsto t'], \dots)) =$$

$$I(f)(\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t']), \text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t']), \dots)$$

The other side

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t')]}(f(t_1, t_2, \dots)) = I(f)(\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t']), \text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t']), \dots)$$

□

Proof. We prove the substitution lemma for formulae by structural induction on ψ .

Base case: $\psi \equiv t_1 = t_2$

$$\psi[x \mapsto t] \equiv t_1[x \mapsto t] = t_2[x \mapsto t]$$

Thus

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff \text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t]).$$

By the substitution lemma for terms,

$$\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_1)$$

and

$$\text{Val}_{\mathfrak{A}, \alpha}(t_2[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_2).$$

Hence

$$\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_1) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_2),$$

that is,

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models t_1 = t_2.$$

Inductive steps

1. $\psi \equiv R(t_1, \dots, t_n)$.

$$\psi[x \mapsto t] \equiv R(t_1[x \mapsto t], \dots, t_n[x \mapsto t]).$$

Then

$$\mathfrak{A}, \alpha \models \psi[x \mapsto t] \iff (\text{Val}_{\mathfrak{A}, \alpha}(t_1[x \mapsto t]), \dots, \text{Val}_{\mathfrak{A}, \alpha}(t_n[x \mapsto t])) \in I(R).$$

By the substitution lemma for terms, for each i ,

$$\text{Val}_{\mathfrak{A}, \alpha}(t_i[x \mapsto t]) = \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_i).$$

$$(\text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_1), \dots, \text{Val}_{\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)]}(t_n)) \in I(R),$$

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models R(t_1, \dots, t_n).$$

2. $\psi \equiv \neg\phi$.

$$\mathfrak{A}, \alpha \models (\neg\phi)[x \mapsto t] \iff \mathfrak{A}, \alpha \models \neg(\phi[x \mapsto t])$$

$$\iff \mathfrak{A}, \alpha \not\models \phi[x \mapsto t].$$

By IH,

$$\mathfrak{A}, \alpha \models \phi[x \mapsto t] \iff \mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models \phi.$$

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \not\models \phi,$$

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models \neg\phi.$$

3. $\psi \equiv \phi_1 \vee \phi_2$.

$$\mathfrak{A}, \alpha \models (\phi_1 \vee \phi_2)[x \mapsto t] \iff \mathfrak{A}, \alpha \models \phi_1[x \mapsto t] \vee \phi_2[x \mapsto t].$$

$$\mathfrak{A}, \alpha \models \phi_1[x \mapsto t] \quad \text{or} \quad \mathfrak{A}, \alpha \models \phi_2[x \mapsto t].$$

By IH

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models \phi_1 \quad \text{or} \quad \mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models \phi_2.$$

$$\mathfrak{A}, \alpha[x \mapsto \text{Val}_{\mathfrak{A}, \alpha}(t)] \models \phi_1 \vee \phi_2.$$

□

Question 3

Definition 10.63 (Herbrand Structure). *Let $\Sigma = (C, F, R)$ be a signature with $C \neq \emptyset$.*

The Σ -Herbrand structure is

$$\mathcal{A}_\Sigma = (U_\Sigma, I_\Sigma)$$

defined as follows.

- The universe is

$$U_\Sigma = \text{Terms}_\Sigma$$

(the set of all ground terms).

- Constants are interpreted naturally:

$$I_\Sigma(c) = c.$$

- For every function symbol f of arity r ,

$$I_\Sigma(f)(t_1, \dots, t_r) = f(t_1, \dots, t_r).$$

- Relation symbols may be interpreted arbitrarily.

Lemma 10.64. *If $\text{Ground}(\Gamma)$ is propositionally satisfiable then Γ is satisfiable.*

Question. Prove Lemma 3.

Proof. Suppose $\text{Ground}(\Gamma)$ is propositionally sat, there is a ground atomic valuation τ

$$\forall \phi \in \text{Ground}(\Gamma), \tau \models_{Gr} \phi$$

Construct a Herbrand structure as per definition. For $I(R)$, we define as such:

$$I_\Sigma^\tau(R) = \{(t_1, t_2, \dots, t_{\text{arity}(R)}) \mid \tau \models_{Gr} R(t_1, t_2, \dots, t_{\text{arity}(R)})\}$$

Now we show that for every ground formula ψ ,

$$\mathfrak{A}_\Sigma^\tau \models \psi \iff \tau \models_{Gr} \psi.$$

Lemma 10.65. *For every ground term t ,*

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t) = t.$$

Proof. By structural induction on t . If $t \equiv c$ for some constant c , then

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(c) = I_\Sigma^\tau(c) = c.$$

If $t \equiv f(t_1, \dots, t_r)$, then by IH

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_i) = t_i \quad \text{for each } i.$$

$$\text{Val}_{\mathfrak{A}_\Sigma^\tau}(f(t_1, \dots, t_r)) = I_\Sigma^\tau(f)(\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_1), \dots, \text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_r)) = I_\Sigma^\tau(f)(t_1, \dots, t_r) = f(t_1, \dots, t_r).$$

□

Base cases

1. $\psi \equiv R(t_1, \dots, t_r)$.

By definition,

$$\mathfrak{A}_\Sigma^\tau \models R(t_1, \dots, t_r) \iff (\text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_1), \dots, \text{Val}_{\mathfrak{A}_\Sigma^\tau}(t_r)) \in I_\Sigma^\tau(R).$$

$$(t_1, \dots, t_r) \in I_\Sigma^\tau(R).$$

By definition of $I_\Sigma^\tau(R)$,

$$\tau \models_{Gr} R(t_1, \dots, t_r).$$

Inductive cases

1. $\psi \equiv \neg\phi$.

$$\mathfrak{A}_\Sigma^\tau \models \neg\phi \iff \mathfrak{A}_\Sigma^\tau \not\models \phi.$$

By IH

$$\mathfrak{A}_\Sigma^\tau \not\models \phi \iff \tau \not\models_{Gr} \phi.$$

$$\mathfrak{A}_\Sigma^\tau \models \neg\phi \iff \tau \models_{Gr} \neg\phi.$$

2. $\psi \equiv \phi_1 \vee \phi_2$.

$$\mathfrak{A}_\Sigma^\tau \models \phi_1 \vee \phi_2 \iff \mathfrak{A}_\Sigma^\tau \models \phi_1 \text{ or } \mathfrak{A}_\Sigma^\tau \models \phi_2.$$

By IH

$$\tau \models_{Gr} \phi_1 \text{ or } \tau \models_{Gr} \phi_2.$$

$$\mathfrak{A}_\Sigma^\tau \models \phi_1 \vee \phi_2 \iff \tau \models_{Gr} \phi_1 \vee \phi_2.$$

Now fix a formula

$$\varphi = \forall x_1 \dots \forall x_n \psi \in \Gamma.$$

We show that

$$\mathfrak{A}_\Sigma^\tau \models \varphi$$

fix arbitrary $u_1, \dots, u_n \in U_\Sigma = Terms_\Sigma$ each u_i is a ground term

$$\chi \equiv \psi[x_1 \mapsto u_1] \dots [x_n \mapsto u_n]$$

is a ground instantiation of φ , thus

$$\chi \in Ground(\Gamma).$$

as τ propositionally satisfies $Ground(\Gamma)$

$$\tau \models_{Gr} \chi.$$

$$\mathfrak{A}_\Sigma^\tau \models \chi.$$

$$\mathfrak{A}_\Sigma^\tau \models \psi[x_1 \mapsto u_1] \dots [x_n \mapsto u_n].$$

Since $u_1, \dots, u_n \in U_\Sigma$ are arbitrarily chosen, by the semantics of universal quantifier,

$$\mathfrak{A}_\Sigma^\tau \models \forall x_1 \dots \forall x_n \psi.$$

$$\mathfrak{A}_\Sigma^\tau \models \varphi.$$

Since $\varphi \in \Gamma$ is also arbitrarily chosen, we have

$$\mathfrak{A}_\Sigma^\tau \models \Gamma.$$

□

Question 4

Theorem 10.66 (Downward Löwenheim–Skolem Theorem). *Let Σ be a countable signature and let Γ be a set of universally quantified FOL sentences without equality.*

If Γ is satisfiable, then there exists a countable structure $\mathcal{A}$ such that

$$\mathcal{A} \models \Gamma.$$

Question. Prove Theorem 2.

Proof. Suppose Γ is satisfiable, $Ground(\Gamma)$ is propositionally sat, there exists a Herbrand structure

$$\mathfrak{A}_\Sigma^\tau = (U_\Sigma, I_\Sigma^\tau)$$

such that

$$\mathfrak{A}_\Sigma^\tau \models \Gamma.$$

Easy to show that $\mathfrak{A}_\Sigma^\tau$ is countable

$$U_\Sigma = Terms_\Sigma,$$

the set of all ground terms over the signature Σ . Since Σ is countable, and every ground term is a finite string over the symbols of Σ , the set $Terms_\Sigma$ is countable.

Therefore $\mathfrak{A}_\Sigma^\tau$ is a countable structure satisfying Γ .

□

Question 5

Theorem 10.67. Γ is satisfiable if and only if $\text{Ground}(\Gamma)$ is satisfiable.

Two sets of sentences Γ_1, Γ_2 are said to be *equivalent* if for every structure $\mathcal{A}$,

$$\mathcal{A} \models \Gamma_1 \iff \mathcal{A} \models \Gamma_2.$$

Question.

Describe:

- a signature Σ ,
- a set Γ of universally quantified FOL sentences without equality over Σ ,
- a Σ -structure $\mathcal{A}$,

such that

$$\mathcal{A} \models \text{Ground}(\Gamma) \quad \text{but} \quad \mathcal{A} \not\models \Gamma.$$

Proof. Consider signature

$$\Sigma = (\{0\}, \emptyset, \{R^1\})$$

structure

$$\mathfrak{A} = (\{0, 1\}, I(0) = 0, I(R) = \{0\})$$

singleton set

$$\Gamma = \{\forall x R(x)\}$$

$$\text{Ground}(\Gamma) = \{R(0)\}$$

$$\mathfrak{A} \models \text{Ground}(\Gamma)$$

but

$$\mathfrak{A} \not\models \Gamma$$

as $R(1) \notin I(R)$. Basically grounding loses information about elements in the universe that no ground terms represent $\square$

Question 6

Let

$$\Sigma = (\{c, d, e\}, \{f\}, \{R\})$$

where f is unary and R is unary.

Define the formulas

$$\psi_1 \equiv \neg(c = c)$$

$$\psi_2 \equiv (c = d) \wedge \neg(d = c)$$

$$\psi_3 \equiv (c = d) \wedge (d = e) \wedge \neg(c = e)$$

$$\psi_4 \equiv (c = d) \wedge \neg(f(c) = f(d))$$

$$\psi_5 \equiv (c = d) \wedge R(c) \wedge \neg R(d)$$

Let

$$\Gamma_i = \{\psi_i\}$$

for $i \in \{1, 2, 3, 4, 5\}$.

Question.

Show that for each i ,

$$\text{Ground}(\Gamma_i)$$

is propositionally satisfiable but

$$\Gamma_i$$

is not satisfiable.

1. $\psi_1 \equiv \neg(c = c)$

- Propositionally satisfiable: let $\tau(c = c) = \text{false}$.
- Not satisfiable in FOL: equality is reflexive, so $c = c$ must be true.

2. $\psi_2 \equiv (c = d) \wedge \neg(d = c)$

- Propositionally satisfiable: let $\tau(c = d) = \text{true}$ and $\tau(d = c) = \text{false}$.
- Not satisfiable in FOL: equality is symmetric, so $c = d \Rightarrow d = c$.

3. $\psi_3 \equiv (c = d) \wedge (d = e) \wedge \neg(c = e)$

- Propositionally satisfiable: let $\tau(c = d) = \tau(d = e) = \text{true}$ and $\tau(c = e) = \text{false}$.
- Not satisfiable in FOL: equality is transitive, so $c = d$ and $d = e$ imply $c = e$.

4. $\psi_4 \equiv (c = d) \wedge \neg(f(c) = f(d))$

- Propositionally satisfiable: let $\tau(c = d) = \text{true}$ and $\tau(f(c) = f(d)) = \text{false}$.
- Not satisfiable in FOL: functions preserve equality, so $c = d \Rightarrow f(c) = f(d)$.

5. $\psi_5 \equiv (c = d) \wedge R(c) \wedge \neg R(d)$

- Propositionally satisfiable: let $\tau(c = d) = \text{true}$, $\tau(R(c)) = \text{true}$, $\tau(R(d)) = \text{false}$.
- Not satisfiable in FOL: relations respect equality, so $c = d \Rightarrow (R(c) \leftrightarrow R(d))$.

Question 7

Let

$$\Gamma^* = eq(\Gamma) \cup E$$

where E is the set of equality axioms.

Lemma 10.68. *If Γ is satisfiable then Γ^* is satisfiable.*

Question. Prove Lemma 4.

Proof. Intuition: $\Gamma \mapsto \Gamma^*$ just constraints the interpretation of relation EQ, therefore under any structure, if a formula $\phi \in \Gamma$ is true, corresponding $eq(\phi) \in \Gamma^*$ is true

Suppose Γ is satisfiable. Then there exists a structure $\mathfrak{A} = (U, I)$ such that $\mathfrak{A} \models \Gamma$.

Define a new structure $\mathfrak{A}' = (U, I')$ based on $\mathfrak{A}$ as such

- For all symbols in Σ , interpretation remains the same
- Interpret EQ as

$$I'(EQ) = \{(u, u) \mid u \in U\}.$$

Then $\mathfrak{A}'$ satisfies all equality axioms E , since identity is an equivalence relation and is preserved by functions and relations. also, since $eq(\Gamma)$ is obtained from Γ by replacing equality with EQ , and EQ is interpreted as identity, trivially

$$\begin{aligned}\mathfrak{A}' &\models eq(\Gamma). \\ \mathfrak{A}' &\models eq(\Gamma) \cup E = \Gamma^*.\end{aligned}$$

□

Question 8

Definition 10.69 (Equivalence Relation). *A binary relation $R \subseteq U \times U$ is an equivalence relation if it is*

- *reflexive*
- *symmetric*
- *transitive*

For $u \in U$ the equivalence class is

$$[u]_R = \{v \mid (u, v) \in R\}.$$

The quotient set is

$$U/R = \{[u]_R \mid u \in U\}.$$

Definition 10.70 (Congruence Relation). *Let U be a set and $E \subseteq U \times U$. E is a congruence relation with respect to functions F and relations R if*

1. *E is an equivalence relation.*
2. *For every function f of arity r ,*

$$E(u_1, v_1), \dots, E(u_r, v_r) \Rightarrow E(f(u_1, \dots, u_r), f(v_1, \dots, v_r)).$$

3. *For every relation R of arity r ,*

$$E(u_1, v_1), \dots, E(u_r, v_r)$$

implies the tuples either both belong to R or both do not.

Question.

Let $\mathcal{A} = (U, I)$ be a structure for the signature

$$\Sigma^* = (C, F, R \cup \{EQ\})$$

such that

$$\mathcal{A} \models E.$$

Show that $I(EQ)$ is a congruence relation with respect to every function symbol $f \in F$ and relation symbol $R \in R$.

Proof. since $\mathcal{A} \models E_1, E_2, E_3$, the relation $I(EQ)$ is reflexive, symmetric, and transitive, therefore $I(EQ)$ is an equivalence relation.

let $f \in F$ be of arity r , and let $u_1, \dots, u_r, v_1, \dots, v_r \in U$ satisfy

$$(u_i, v_i) \in I(EQ) \quad (1 \leq i \leq r).$$

Since $\mathcal{A} \models E_4$, it follows that

$$(f(u_1, \dots, u_r), f(v_1, \dots, v_r)) \in I(EQ).$$

Thus $I(EQ)$ is preserved by every function symbol.

Similarly for relation symbols we have

$$R(u_1, \dots, u_r) \iff R(v_1, \dots, v_r).$$

Hence $(u_1, \dots, u_r) \in I(R)$ iff $(v_1, \dots, v_r) \in I(R)$, so $I(EQ)$ is consistent with every relation symbol.

Therefore $I(EQ)$ is a congruence relation with respect to every function symbol in F and every relation symbol in $\mathcal{R}$. $\square$

Question 9

Lemma 10.71. *If Γ^* is satisfiable then Γ is satisfiable.*

Question. Prove Lemma 5. Heavy duty at work this 2 weeks, unable to finish, will attempt alongside with hw5

Question 10

Theorem 10.72. *Assume that the signature Σ contains at least one constant.
Let Γ be a set of universally quantified FOL sentences in prenex normal form.
Then*

$$\Gamma \text{ is satisfiable} \iff \text{Ground}(\Gamma^*) \text{ is propositionally satisfiable.}$$

Question. Prove Theorem 5.